

Some Basic Concepts of Chemistry

Question1

x mg of pure HCl was used to make an aqueous solution. 25.0 mL of 0.1M Ba(OH)₂ solution is used when the HCl solution was titrated against it. The numerical value of x is $\text{_____} \times 10^{-1}$. (Nearest integer)

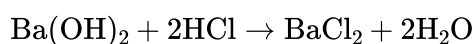
Given : Molar mass of HCl and Ba(OH)₂ are 36.5 and 171.0 g mol⁻¹ respectively.

JEE Main 2026 (Online) 23rd January Morning Shift

Answer: 1825

Solution:

Balanced equation:



Moles of Ba(OH)₂ used:

$$n_{\text{Ba(OH)}_2} = MV = 0.1 \times 25.0 \times 10^{-3} = 2.5 \times 10^{-3} \text{ mol}$$

From the equation, 1 mol Ba(OH)₂ reacts with 2 mol HCl, so

$$n_{\text{HCl}} = 2 \times 2.5 \times 10^{-3} = 5.0 \times 10^{-3} \text{ mol}$$

Mass of HCl:

$$m_{\text{HCl}} = n \times M = 5.0 \times 10^{-3} \times 36.5 = 0.1825 \text{ g}$$

$$0.1825 \text{ g} = 182.5 \text{ mg}$$

$$\text{So } x = 182.5 \text{ mg} = 1825 \times 10^{-1} \text{ mg.}$$

$$\text{Nearest integer} = \boxed{1825}.$$

Question2

0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of CaCl_2 tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is

____ %. (Nearest integer)

(Given: molar mass in gmol^{-1} H : 1, C : 12, O : 16)

JEE Main 2026 (Online) 24th January Evening Shift

Answer: 73

Solution:

From combustion:

- Increase in mass of CaCl_2 tube = mass of H_2O formed

$$m(\text{H}_2\text{O}) = 0.15 \text{ g}, \quad n(\text{H}_2\text{O}) = \frac{0.15}{18} = 0.008333 \text{ mol}$$

Moles of H atoms:

$$n(\text{H}) = 2n(\text{H}_2\text{O}) = 0.016667 \text{ mol}$$

Mass of H:

$$m(\text{H}) = 0.016667 \times 1 = 0.016667 \text{ g}$$

- Increase in mass of potash tube = mass of CO_2 formed

$$m(\text{CO}_2) = 0.1837 \text{ g}, \quad n(\text{CO}_2) = \frac{0.1837}{44} = 0.004175 \text{ mol}$$

Moles of C atoms = moles of CO_2 :

$$n(\text{C}) = 0.004175 \text{ mol}$$

Mass of C:

$$m(\text{C}) = 0.004175 \times 12 = 0.05010 \text{ g}$$

Sample mass = 0.25 g, so mass of oxygen in the compound:

$$m(\text{O}) = 0.25 - (0.016667 + 0.05010) = 0.18323 \text{ g}$$

Percentage of oxygen:

$$\%O = \frac{0.18323}{0.25} \times 100 = 73.29\% \approx \boxed{73\%}$$

Question3

14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K . Which of the following statements is incorrect?

[Given : Molar mass in gmol^{-1} of Ca – 40, Cl – 35.5, H – 1]

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

The limiting reagent is calcium metal.

B.

33.3 g of CaCl_2 is produced.

C.

7.84 L of H_2 gas is evolved.

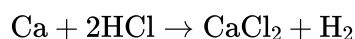
D.

0.35 mol of H_2 gas is evolved.

Answer: B

Solution:

Balanced reaction (NCERT method):



Step 1: Moles of calcium

Given mass of Ca = 14.0 g, molar mass of Ca = 40 g mol^{-1}

$$n(\text{Ca}) = \frac{14.0}{40} = 0.35 \text{ mol}$$

Since HCl is in excess, the **limiting reagent is Ca**.

So **Option A is correct**.

Step 2: Amount of products formed (from stoichiometry)

From the balanced equation:

- $1 \text{ mol Ca} \rightarrow 1 \text{ mol CaCl}_2$
- $1 \text{ mol Ca} \rightarrow 1 \text{ mol H}_2$

Hence,

$$n(\text{CaCl}_2) = 0.35 \text{ mol}$$

$$n(\text{H}_2) = 0.35 \text{ mol}$$

So **Option D is correct.**

Step 3: Volume of H_2 at 1 atm and 273 K (STP)

At STP, 1 mol gas occupies 22.4 L.

$$V(\text{H}_2) = 0.35 \times 22.4 = 7.84 \text{ L}$$

So **Option C is correct.**

Step 4: Mass of CaCl_2

Molar mass of CaCl_2 :

$$= 40 + 2(35.5) = 40 + 71 = 111 \text{ g mol}^{-1}$$

Mass produced:

$$m(\text{CaCl}_2) = 0.35 \times 111 = 38.85 \text{ g}$$

But Option B says 33.3 g, which is **wrong**.

Incorrect statement: Option B.

Question4

Aqueous HCl reacts with $\text{MnO}_2(\text{s})$ to form $\text{MnCl}_2(\text{aq})$, $\text{Cl}_2(\text{g})$, and $\text{H}_2\text{O}(\text{l})$. What is the weight (in g) of Cl_2 liberated when 8.7 g of $\text{MnO}_2(\text{s})$ is reacted with excess aqueous HCl solution? (Given Molar mass in g mol^{-1} $\text{Mn} = 55$, $\text{Cl} = 35.5$, $\text{O} = 16$, $\text{H} = 1$)

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

B.

21.3

C.

7.1

D.

71

Answer: C

Solution:

Balance the reaction first (NCERT method):



Step 1: Calculate moles of MnO_2

Molar mass of MnO_2 :

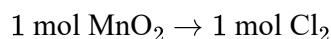
$$M(\text{MnO}_2) = 55 + 2(16) = 55 + 32 = 87 \text{ g mol}^{-1}$$

Moles of MnO_2 in 8.7 g:

$$n(\text{MnO}_2) = \frac{8.7}{87} = 0.1 \text{ mol}$$

Step 2: Use mole ratio to find moles of Cl_2

From the balanced equation:



So,

$$n(\text{Cl}_2) = 0.1 \text{ mol}$$

Step 3: Convert moles of Cl_2 to mass

Molar mass of Cl_2 :

$$M(\text{Cl}_2) = 2 \times 35.5 = 71 \text{ g mol}^{-1}$$

Mass of Cl_2 :

$$m(\text{Cl}_2) = n \times M = 0.1 \times 71 = 7.1 \text{ g}$$

Answer: 7.1 g (Option C)

Question5

By usual analysis, 1.00 g of compound (X) gave 1.79 g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is : (nearest integer)

(Given, molar mass in g mol^{-1} : O = 16, Mg = 24, P = 31)

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

40

B.

30

C.

20

D.

50

Answer: D

Solution:

The chemical formula for magnesium pyrophosphate is $\text{Mg}_2\text{P}_2\text{O}_7$.

Using the given molar masses (Mg = 24, P = 31, O = 16):

Molar mass of $\text{Mg}_2\text{P}_2\text{O}_7 = (2 \times 24) + (2 \times 31) + (7 \times 16)$

Molar mass = $48 + 62 + 112 = 222 \text{ g mol}^{-1}$

One mole of $\text{Mg}_2\text{P}_2\text{O}_7$ contains 2 moles of phosphorus atoms.

Mass of P = $2 \times 31 = 62 \text{ g}$

This means that 222 g of $\text{Mg}_2\text{P}_2\text{O}_7$ contains 62 g of phosphorus.

The mass of $\text{Mg}_2\text{P}_2\text{O}_7$ obtained in the experiment is 1.79 g.

Applying the unitary method:

Mass of P in 1.79 g of $\text{Mg}_2\text{P}_2\text{O}_7 = \frac{62}{222} \times 1.79 \text{ g}$

Mass of P = $\frac{110.98}{222} \approx 0.4999 \text{ g}$

The percentage of an element in a given compound is calculated as:

$$\% \text{ of P} = \frac{\text{Mass of P}}{\text{Total mass of compound (X)}} \times 100$$

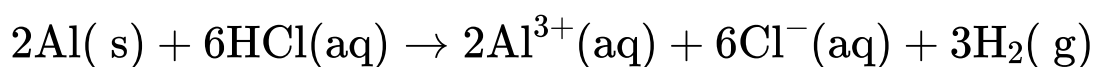
$$\% \text{ of P} = \frac{0.4999 \text{ g}}{1.00 \text{ g}} \times 100 = 49.99\%$$

Rounding off to the nearest integer, we get 50%.

Option D is correct.

Question6

In the reaction,



JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

12 L HCl(aq) is consumed for every 6 L H₂(g) produced.

B.

11.2 L H₂(g) at STP is produced for every mole of HCl consumed.

C.

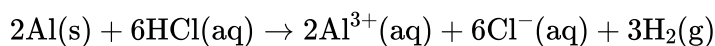
33.6 L H₂(g) is produced regardless of temperature and pressure for every mole of Al that reacts.

D.

67.2 L H₂(g) at STP is produced for every mole of Al that reacts.

Answer: B

Solution:



From the balanced equation, 6 moles of HCl give 3 moles of H_2 .

So, moles of H_2 produced

$$= \frac{3}{6} \times \text{moles of HCl used}$$

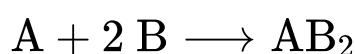
$$= \frac{1}{2} \times \text{moles of HCl used}$$

Also, 2 moles of Al give 3 moles of H_2 .

So, moles of H_2 produced

$$= \frac{3}{2} \times \text{moles of Al used}$$

Question 7



36.0 g of 'A' (Molar mass : 60 g mol^{-1}) and 56.0 g of 'B' (Molar mass : 80 g mol^{-1}) are allowed to react. Which of the following statements are correct ?

A. 'A' is the limiting reagent.

B. 77.0 g of AB_2 is formed.

C. Molar mass of AB_2 is 140 g mol^{-1} .

D. 15.0 g of A is left unreacted after the completion of reaction.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

A and C Only

B.

A and B Only

C.

C and D Only

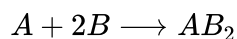
D.

B and D Only

Answer: D

Solution:

Balanced reaction:



Step 1: Calculate moles of reactants

For A :

$$n_A = \frac{36.0}{60} = 0.60 \text{ mol}$$

For B :

$$n_B = \frac{56.0}{80} = 0.70 \text{ mol}$$

Step 2: Find the limiting reagent (NCERT method: compare required moles)

For 0.60 mol of A , required moles of B :

$$n_{B(\text{required})} = 2 \times 0.60 = 1.20 \text{ mol}$$

But available $B = 0.70 \text{ mol}$, which is less than 1.20 mol.

So, **B is the limiting reagent.**

Hence statement **A** is false.

Step 3: Moles and mass of product formed

From the reaction, 2 mol B gives 1 mol AB_2 .

So moles of AB_2 formed:

$$n_{AB_2} = \frac{0.70}{2} = 0.35 \text{ mol}$$

Molar mass of AB_2 :

$$M_{AB_2} = 60 + 2(80) = 220 \text{ g mol}^{-1}$$

So statement **C** is false.

Mass of AB_2 formed:

$$m_{AB_2} = 0.35 \times 220 = 77.0 \text{ g}$$

So statement **B** is true.

Step 4: Unreacted A

Moles of A consumed = moles of AB_2 formed (ratio 1 : 1):

$$n_A(\text{consumed}) = 0.35 \text{ mol}$$

Mass of A consumed:

$$m_A(\text{consumed}) = 0.35 \times 60 = 21.0 \text{ g}$$

Unreacted A :

$$m_A(\text{left}) = 36.0 - 21.0 = 15.0 \text{ g}$$

So statement **D** is true.

Correct statements: B and D only

☒ Option D

Question8

Identify the correct statements :

- A. Hydrated salts can be used as primary standard.**
- B. Primary standard should not undergo any reaction with air.**
- C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.**
- D. Primary standard should not be soluble in water.**
- E. Primary standard should have low relative molar mass.**

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

D and E Only

B.

A, B and E Only

C.

A, B, C and E Only

D.

A, B and C Only

Answer: D

Solution:

Primary standard (as per NCERT idea) should be a substance which is **pure, stable in air, non-hygroscopic, readily soluble in water**, and reacts **completely** with the titrant in a **stoichiometric** way.

Now check each statement:

A. Hydrated salts can be used as primary standard.

Some hydrated salts are sufficiently **stable** and can be used (example often cited: Mohr's salt). So this can be **correct**.

B. Primary standard should not undergo any reaction with air.

Correct. It should not react with O_2 , CO_2 or moisture of air.

C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.

Correct. The reaction should be **fast** and have a **definite stoichiometry** so the end point is sharp.

D. Primary standard should not be soluble in water.

Incorrect. It should be **readily soluble** so that a standard solution can be prepared.

E. Primary standard should have low relative molar mass.

Incorrect. It should preferably have **high molar mass** to minimise weighing error.

So, correct statements are **A, B and C only**.

Correct option: Option D (A, B and C Only).

Question9

For the given reaction;



If 90 g $CaCO_3$ is added to 300 mL of HCl which contains 38.55% HCl by mass and has density 1.13 g mL^{-1} , then which of the following option is correct?

Given molar mass of H, Cl, Ca and O are 1, 35.5, 40 and 16 g mol⁻¹ respectively.

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

64.97 g of HCl remains unreacted

B.

60.32 g of HCl remains unreacted

C.

32.85 g of CaCO₃ remains unreacted

D.

97.30 g of HCl reacted

Answer: A

Solution:

Density of HCl solution (d) = 1.13 g/ml

V = 300ml

Wt. of HCl solution = density × volume = 1.13 × 300 = 339 g

Wt. of HCl = $339 \times \frac{38.55}{100} = 130.68$ g (LR)

$\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$

Molar mass of CaCO₃ = 40 + 12 + (3 × 16) = 100 g mol⁻¹

So, moles of CaCO₃ = $\frac{90}{100} = 0.90$ mole

Molar mass of HCl = 1 + 35.5 = 36.5 g mol⁻¹

So, moles of HCl = $\frac{130.68}{36.5} = 3.58$ mole

From the equation, 1 mol CaCO₃ needs 2 mol HCl

So, 0.90 mol CaCO₃ needs 2 × 0.90 = 1.80 mol HCl

Since available HCl = 3.58 mol and required HCl = 1.80 mol, HCl is in excess and CaCO₃ is the limiting reagent (LR).

Moles of HCl remained = 3.58 – 1.80 = 1.78 mole.

Mass of HCl remained = 1.78 × 36.5 = 64.97 g.

Question 10

The mass of iron converted into Fe_3O_4 by the action of 18 g of steam is:

(Given: Molar mass of H, O and Fe are 1, 16 and 56 g mol^{-1} respectively)

Assume iron is present in excess:

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

2.1 g

B.

4.2 g

C.

21 g

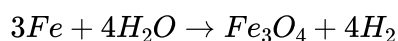
D.

42 g

Answer: D

Solution:

The reaction of iron with steam is:



We are given:

Mass of steam = 18 g

Molar mass of water:

$$\text{H}_2\text{O} = 2(1) + 16 = 18 \text{ g mol}^{-1}$$

So, moles of steam are:

$$\text{Moles of } H_2O = \frac{18}{18} = 1 \text{ mol}$$

From the balanced equation:

4 mol H_2O react with 3 mol Fe

Therefore, for 1 mol H_2O :

$$\text{Moles of Fe required} = \frac{3}{4}$$

Now, molar mass of iron is:

$$56 \text{ g mol}^{-1}$$

So, mass of iron reacted:

$$\text{Mass of Fe} = \frac{3}{4} \times 56 = 42 \text{ g}$$

Hence, the mass of iron converted into Fe_3O_4 is:

42 g

So, the correct option is:

Option D

Question11

The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of $CO_2(g)$ formed when 3.38 g of this hydrocarbon is completely burnt in oxygen is :

(Given : Molar mass in $g \text{ mol}^{-1}$ C : 12, H : 1, O : 16)

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

5.68

B.

11.44

C.

22.74

D.

17.05

Answer: B

Solution:

Let the hydrocarbon be C_2H_x since it has two carbon atoms.

Given, the mass percentage ratio of $C : H$ is 12 : 1.

So, in the hydrocarbon,

- mass of carbon = $2 \times 12 = 24$
- mass of hydrogen = $x \times 1 = x$

Now,

$$24 : x = 12 : 1$$

So,

$$\frac{24}{x} = \frac{12}{1}$$

$$24 = 12x$$

$$x = 2$$

Hence, the hydrocarbon is



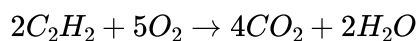
Its molar mass is

$$(2 \times 12) + (2 \times 1) = 24 + 2 = 26 \text{ g mol}^{-1}$$

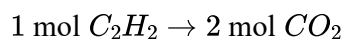
Number of moles of C_2H_2 in 3.38 g:

$$\text{moles of } C_2H_2 = \frac{3.38}{26} = 0.13 \text{ mol}$$

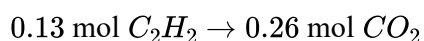
Complete combustion of acetylene:



From the equation,



Therefore,



Molar mass of CO_2 :

$$12 + (2 \times 16) = 44 \text{ g mol}^{-1}$$

Mass of CO_2 formed:

$$0.26 \times 44 = 11.44 \text{ g}$$

So, the correct answer is

11.44 g

Option B.

Question 12

Number of moles and number of molecules in 1.4187 L of SO₂ at STP respectively are

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

$$0.1266; 3.812 \times 10^{22}$$

B.

$$0.0633; 3.812 \times 10^{22}$$

C.

$$0.1266; 7.6238 \times 10^{22}$$

D.

$$0.0633; 7.6238 \times 10^{22}$$

Answer: B

Solution:

At STP, according to Avogadro's law, 1 mole of any gas occupies 22.4 L.

So, number of moles of SO₂ is

$$n = \frac{V}{22.4} = \frac{1.4187}{22.4}$$

$$n = 0.0633 \text{ mol}$$

Now, number of molecules

$$= n \times N_A$$

where $N_A = 6.022 \times 10^{23}$

So,

Number of molecules = $0.0633 \times 6.022 \times 10^{23}$

= 3.812×10^{22}

Hence, the correct answer is:

0.0633 mol ; 3.812×10^{22} molecules

So, Option B is correct.

Question13

The correct order of total number of atoms present in

(A) 2 moles of cyclohexane

(B) 684 g of sucrose

(C) 90.8 L of dihydrogen at STP

is :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$C > A > B$

B.

$C > B > A$

C.

$B > C > A$

D.

$B > A > C$

Answer: D

Solution:

Let us find the **total number of atoms** in each case.

We only need to compare, so first calculate the number of **moles of atoms** in each sample.

(A) 2 moles of cyclohexane

Cyclohexane has formula:



So, one molecule of cyclohexane contains:

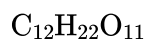
$$6 + 12 = 18 \text{ atoms}$$

Therefore, in 2 moles of cyclohexane, moles of atoms are:

$$2 \times 18 = 36 \text{ moles of atoms}$$

(B) 684 g of sucrose

Sucrose has formula:



Total atoms in one molecule:

$$12 + 22 + 11 = 45$$

Molar mass of sucrose:

$$12 \times 12 + 22 \times 1 + 11 \times 16$$

$$= 144 + 22 + 176 = 342 \text{ g mol}^{-1}$$

Given mass = 684 g, so moles of sucrose:

$$\frac{684}{342} = 2 \text{ mol}$$

Hence, moles of atoms:

$$2 \times 45 = 90 \text{ moles of atoms}$$

(C) 90.8 L of dihydrogen at STP

At STP, 1 mole of any gas occupies 22.4 L.

So moles of H_2 :

$$\frac{90.8}{22.4} \approx 4.05 \text{ mol}$$

Each molecule of H_2 has 2 atoms.

Therefore, moles of atoms:

$$4.05 \times 2 = 8.10 \text{ moles of atoms}$$

Comparison

- (A) = 36 moles of atoms

- (B) = 90 moles of atoms
- (C) = 8.10 moles of atoms

So the correct order is:

$$B > A > C$$

Therefore, the correct option is:

Option D

Question 14

What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution?

[Given: Molar mass of H, O, C and N are 1, 16, 12 and 14 g mol⁻¹ respectively.]

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

0.825

B.

0.032

C.

0.867

D.

0.967

Answer: D

Solution:

For a 10% by weight (w/w) aqueous urea solution, let us take 100 g of solution.

Then:

- Mass of urea = 10 g
- Mass of water = 90 g

Now find the molar masses.

Urea has formula NH_2CONH_2 or $CO(NH_2)_2$

$$M(\text{urea}) = 12 + 16 + 2(14) + 4(1) = 60 \text{ g mol}^{-1}$$

Moles of urea:

$$n_{\text{urea}} = \frac{10}{60} = \frac{1}{6} \approx 0.167$$

Molar mass of water:

$$M(\text{water}) = 2(1) + 16 = 18 \text{ g mol}^{-1}$$

Moles of water:

$$n_{\text{water}} = \frac{90}{18} = 5$$

Now, mole fraction of water is

$$X_{\text{water}} = \frac{n_{\text{water}}}{n_{\text{water}} + n_{\text{urea}}}$$

$$X_{\text{water}} = \frac{5}{5 + \frac{1}{6}} = \frac{5}{\frac{31}{6}} = \frac{30}{31}$$

$$X_{\text{water}} \approx 0.967$$

So, the correct answer is:

0.967

Option D

Question15

How many grams of residue is obtained by heating 2.76 g of silver carbonate?

(Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol⁻¹ respectively)

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

1.08 g

B.

2.16 g

C.

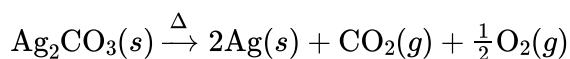
3.24 g

D.

4.32 g

Answer: B

Solution:



From the chemical equation, we can see that carbon dioxide (CO_2) and oxygen (O_2) are gases and will easily escape. The only solid residue left behind is silver (Ag).

Next, let us calculate the molar mass of silver carbonate (Ag_2CO_3) by adding the molar masses of its constituent elements:

$$\text{Molar mass of Ag}_2\text{CO}_3 = (2 \times 108) + 12 + (3 \times 16)$$

$$\text{Molar mass of Ag}_2\text{CO}_3 = 216 + 12 + 48 = 276 \text{ g mol}^{-1}$$

Now, we can find the number of moles of silver carbonate provided in the question. The given mass is 2.76 g.

$$\text{Number of moles} = \frac{\text{Given mass}}{\text{Molar mass}}$$

$$\text{Moles of Ag}_2\text{CO}_3 = \frac{2.76 \text{ g}}{276 \text{ g mol}^{-1}} = 0.01 \text{ mol}$$

Using the stoichiometry from our balanced chemical equation, we know that heating 1 mol of Ag_2CO_3 yields 2 mol of solid Ag residue.

Therefore, if we decompose 0.01 mol of Ag_2CO_3 , the number of moles of silver residue produced will be:

$$\text{Moles of residue (Ag)} = 2 \times 0.01 \text{ mol} = 0.02 \text{ mol}$$

Finally, we convert these moles of silver back to mass in grams. We know the given molar mass of Ag is 108 g mol^{-1} .

$$\text{Mass of Ag} = \text{Moles of Ag} \times \text{Molar mass of Ag}$$

$$\text{Mass of Ag} = 0.02 \text{ mol} \times 108 \text{ g mol}^{-1} = 2.16 \text{ g}$$

The mass of the solid residue obtained is 2.16 g.

Correct Option: Option B

Question16

What volume of hydrogen gas at STP would be liberated by action of 50 mL of H_2SO_4 of 50% purity (density = 1.3 g mL^{-1}) on 20 g of zinc?

Given : Molar mass of H, O, S, Zn are 1, 16, 32, 65 g mol^{-1} respectively.

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

5.824 L

B.

7.428 L

C.

6.892 L

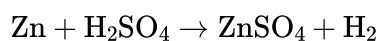
D.

8.375 L

Answer: C

Solution:

The reaction is:



From the balanced equation,

1 mol Zn reacts with 1 mol H_2SO_4 and gives 1 mol H_2

Now let us find the number of moles of each reactant.

1. Amount of H_2SO_4 present

Volume of acid solution = 50 mL

Density = 1.3 g mL^{-1}

So, mass of acid solution is

Mass = Volume $\times$ Density = $50 \times 1.3 = 65 \text{ g}$

$$\text{Purity} = 50\%$$

Hence, mass of pure H_2SO_4 is

$$\frac{50}{100} \times 65 = 32.5 \text{ g}$$

Molar mass of H_2SO_4 :

$$2(1) + 32 + 4(16) = 98 \text{ g mol}^{-1}$$

So, moles of H_2SO_4 are

$$\frac{32.5}{98} = 0.3316 \text{ mol}$$

2. Amount of zinc

Mass of Zn = 20 g

Molar mass of Zn = 65 g mol^{-1}

So, moles of Zn are

$$\frac{20}{65} = 0.3077 \text{ mol}$$

3. Limiting reagent

Since the reaction ratio is 1 : 1:

- H_2SO_4 available = 0.3316 mol
- Zn available = 0.3077 mol

Zn is smaller in amount, so **zinc is the limiting reagent**.

Therefore, moles of H_2 formed are

$$0.3077 \text{ mol}$$

4. Volume of hydrogen at STP

At STP, 1 mol gas occupies 22.4 L.

So, volume of H_2 is

$$0.3077 \times 22.4 = 6.892 \text{ L}$$

$$\boxed{6.892 \text{ L}}$$

So, the correct option is:

$$\boxed{\text{Option C}}$$

Question 17

An oxide of iron contains 69.9% iron, its empirical formula, is :

(Given : Molar mass of Fe and O are 56 and 16 g mol⁻¹ respectively.)

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

FeO

B.

Fe₂O₃

C.

Fe₃O₄

D.

FeO₃

Answer: B

Solution:

Let the oxide of iron contain 100 g of the compound.

Then,

Mass of Fe = 69.9 g

Mass of O = 100 – 69.9 = 30.1 g

Calculate moles of each element

Moles of Fe = $\frac{69.9}{56} = 1.248$

Moles of O = $\frac{30.1}{16} = 1.881$

Find the simplest whole-number ratio

Divide each by the smaller number (1.248):

Ratio of Fe:O = $\frac{1.248}{1.248} : \frac{1.881}{1.248} = 1 : 1.51$

This ratio is approximately **1 : 1.5**.

To remove the fraction, multiply both by 2 : Fe₂O₃

In Fe₂O₃,

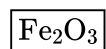
Mass of Fe = 2 × 56 = 112 g

$$\text{Mass of O} = 3 \times 16 = 48 \text{ g}$$

$$\text{Total mass} = 160 \text{ g}$$

$$\text{Percentage of Fe} = \frac{112}{160} \times 100 = 70\% \text{ (approximately 69.9\%)}$$

Therefore, the empirical formula of the oxide is



Correct Option: B.

Question 18

Which of the following contain the same number of atoms?

(Given : Molar mass in gmol^{-1} of H, He, O and S are 1, 4, 16 and 32 respectively)

A. 2 g of O_2 gas

B. 4 g of SO_2 gas

C. 1400 mL of O_2 at STP

D. 0.05 L of He at STP

E. 0.0625 mol of H_2 gas

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

A and B only

B.

B and C only

C.

C and D only

D.

A, C and E only

Answer: D

Solution:

Total number of atoms = (Number of moles) $\times N_A \times$ (Atomicity)

Let's calculate this for each option.

A. 2 g of O₂ gas

1. **Molar mass of O₂:** The atomic mass of O is 16 u. So, the molar mass of O₂ is $2 \times 16 = 32 \text{ g mol}^{-1}$.

2. **Number of moles (n):**

$$n = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{2 \text{ g}}{32 \text{ g mol}^{-1}} = \frac{1}{16} \text{ mol}$$

3. **Atomicity of O₂:** One molecule of O₂ contains 2 oxygen atoms. So, the atomicity is 2.

4. **Total number of atoms:**

$$\text{Atoms} = n \times N_A \times \text{Atomicity} = \frac{1}{16} \times N_A \times 2 = \frac{2}{16} N_A = \frac{1}{8} N_A$$

B. 4 g of SO₂ gas

1. **Molar mass of SO₂:** Atomic mass of S is 32 u and O is 16 u. So, the molar mass of SO₂ is $32 + (2 \times 16) = 32 + 32 = 64 \text{ g mol}^{-1}$.

2. **Number of moles (n):**

$$n = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{4 \text{ g}}{64 \text{ g mol}^{-1}} = \frac{1}{16} \text{ mol}$$

3. **Atomicity of SO₂:** One molecule of SO₂ contains 1 sulfur atom and 2 oxygen atoms, making a total of $1 + 2 = 3$ atoms. The atomicity is 3.

4. **Total number of atoms:**

$$\text{Atoms} = n \times N_A \times \text{Atomicity} = \frac{1}{16} \times N_A \times 3 = \frac{3}{16} N_A$$

C. 1400 mL of O₂ at STP

1. **Molar volume at STP:** We know that 1 mole of any gas at STP occupies 22400 mL (or 22.4 L).

2. **Number of moles (n):**

$$n = \frac{\text{Given volume}}{\text{Molar volume at STP}} = \frac{1400 \text{ mL}}{22400 \text{ mL mol}^{-1}} = \frac{14}{224} = \frac{1}{16} \text{ mol}$$

3. **Atomicity of O₂:** As we found earlier, the atomicity is 2.

4. **Total number of atoms:**

$$\text{Atoms} = n \times N_A \times \text{Atomicity} = \frac{1}{16} \times N_A \times 2 = \frac{2}{16} N_A = \frac{1}{8} N_A$$

D. 0.05 L of He at STP

1. **Molar volume at STP:** 1 mole of any gas at STP occupies 22.4 L.

2. **Number of moles (n):**

$$n = \frac{\text{Given volume}}{\text{Molar volume at STP}} = \frac{0.05 \text{ L}}{22.4 \text{ L mol}^{-1}} \approx 0.0022 \text{ mol}$$

3. **Atomicity of He:** Helium (He) is a noble gas and exists as single atoms. So, its atomicity is 1.

4. **Total number of atoms:**

$$\text{Atoms} = n \times N_A \times \text{Atomicity} = \frac{0.05}{22.4} \times N_A \times 1 = \frac{0.05}{22.4} N_A$$

E. 0.0625 mol of H_2 gas

1. **Number of moles (n):** This is given directly as 0.0625 mol. Let's convert this decimal to a fraction to make comparison easier: $0.0625 = \frac{625}{10000} = \frac{1}{16}$. So, $n = \frac{1}{16}$ mol.

2. **Atomicity of H_2 :** One molecule of H_2 contains 2 hydrogen atoms. The atomicity is 2.

3. **Total number of atoms:**

$$\text{Atoms} = n \times N_A \times \text{Atomicity} = \frac{1}{16} \times N_A \times 2 = \frac{2}{16} N_A = \frac{1}{8} N_A$$

Conclusion

Let's summarise the total number of atoms we calculated for each option:

- A: $\frac{1}{8} N_A$ atoms
- B: $\frac{3}{16} N_A$ atoms
- C: $\frac{1}{8} N_A$ atoms
- D: $\frac{0.05}{22.4} N_A$ atoms
- E: $\frac{1}{8} N_A$ atoms

By comparing these results, we can clearly see that options A, C, and E contain the same number of atoms.

Therefore, the correct choice is the one that lists A, C, and E.

The correct answer from the options given below is **Option D**.

Question 19

Match List - I with List - II.

List - I Mass of substance		List - II Number of atoms	
A.	1.8 mg water	I.	$2 \times 10^{-4} \times N_A$
B.	9.8 mg sulphuric acid	II.	$1.5 \times 10^{-4} \times N_A$

List - I Mass of substance		List - II Number of atoms	
C.	1.8 mg carbon	III.	$3 \times 10^{-4} \times N_A$
D.	5.85 mg salt (NaCl)	IV.	$7 \times 10^{-4} \times N_A$

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

A-IV, B-III, C-I, D-II

B.

A-III, B-II, C-IV, D-I

C.

A-III, B-IV, C-II, D-I

D.

A-III, B-IV, C-I, D-II

Answer: C

Solution:

We have the following masses:

- A: 1.8 mg H_2O
- B: 9.8 mg H_2SO_4
- C: 1.8 mg C
- D: 5.85 mg NaCl

Since $1 \text{ mg} = 10^{-3} \text{ g}$:

A: $1.8 \text{ mg} = 1.8 \times 10^{-3} \text{ g}$

B: $9.8 \text{ mg} = 9.8 \times 10^{-3} \text{ g}$

C: $1.8 \text{ mg} = 1.8 \times 10^{-3} \text{ g}$

D: $5.85 \text{ mg} = 5.85 \times 10^{-3} \text{ g}$

Number of atoms = (number of moles of molecules) $\times$ (atoms per molecule) $\times N_A$

i.e.

$$\text{Number of atoms} = \frac{\text{Given mass}}{\text{Molar mass}} \times (\text{atoms per molecule}) \times N_A$$

(A) Water (H₂O)

$$\text{Molar mass of H}_2\text{O} = 18 \text{ g/mol}$$

$$\text{Atoms per molecule} = 3 \text{ (2 H + 1 O)}$$

$$\text{Moles of H}_2\text{O} = \frac{1.8 \times 10^{-3}}{18} = 1 \times 10^{-4}$$

$$\text{Number of atoms} = 1 \times 10^{-4} \times 3N_A = 3 \times 10^{-4}N_A$$

So, **A** → **III**

(B) Sulphuric acid (H₂SO₄)

$$\text{Molar mass of H}_2\text{SO}_4 = 98 \text{ g/mol}$$

$$\text{Atoms per molecule} = 7 \text{ (2 H + 1 S + 4 O)}$$

$$\text{Moles of H}_2\text{SO}_4 = \frac{9.8 \times 10^{-3}}{98} = 1 \times 10^{-4}$$

$$\text{Number of atoms} = 1 \times 10^{-4} \times 7N_A = 7 \times 10^{-4}N_A$$

So, **B** → **IV**

(C) Carbon (C)

$$\text{Molar mass of carbon} = 12 \text{ g/mol}$$

$$\text{Each atom} = 1 \text{ atom per molecule}$$

$$\text{Moles of C} = \frac{1.8 \times 10^{-3}}{12} = 1.5 \times 10^{-4}$$

$$\text{Number of atoms} = 1.5 \times 10^{-4}N_A$$

So, **C** → **II**

(D) Sodium chloride (NaCl)

$$\text{Molar mass} = 58.5 \text{ g/mol}$$

$$\text{Atoms per formula unit} = 2 \text{ (Na + Cl)}$$

$$\text{Moles of NaCl} = \frac{5.85 \times 10^{-3}}{58.5} = 1 \times 10^{-4}$$

$$\text{Number of atoms} = 1 \times 10^{-4} \times 2N_A = 2 \times 10^{-4}N_A$$

So, **D** → **I**

$A - III, \quad B - IV, \quad C - II, \quad D - I$
--

Option C

Question20

20 mL of 2 M NaOH solution is added to 400 mL of 0.5 M NaOH solution. The final concentration of the solution is _____ $\times 10^{-2}$ M. (Nearest integer)

JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 57

Solution:

To determine the final concentration of the NaOH solution, we use the formula for mixing solutions :

$$M_F = \frac{M_1 \times V_1 + M_2 \times V_2}{V_1 + V_2}$$

Where :

$M_1 = 2$ M and $V_1 = 20$ mL: Concentration and volume of the first solution.

$M_2 = 0.5$ M and $V_2 = 400$ mL: Concentration and volume of the second solution.

Substitute the values into the equation:

$$M_F = \frac{2 \times 20 + 0.5 \times 400}{420}$$

Calculating each term:

$$2 \times 20 = 40$$

$$0.5 \times 400 = 200$$

Add these results:

$$M_F = \frac{40 + 200}{420} = \frac{240}{420} \approx 0.571 \text{ M}$$

Convert this to scientific notation as the problem specifies the answer should be in $\times 10^{-2}$ form:

$$M_F = 57.1 \times 10^{-2} \text{ M}$$

Rounded to the nearest integer, the final concentration is:

57

Question21

During " S " estimation, 160 mg of an organic compound gives 466 mg of barium sulphate. The percentage of Sulphur in the given compound is _____ %.

(Given molar mass in g mol^{-1} of Ba : 137, S : 32, O : 16)

JEE Main 2025 (Online) 23rd January Morning Shift

Answer: 40

Solution:

$$\text{Millimoles of BaSO}_4 = \frac{466}{233} = 2 \text{ m mol}$$

$$\% \text{ S} = \frac{\frac{466}{233} \times 32}{160} \times 100 = 40\%$$

Question22

When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is _____.
(Nearest integer)

Given :

Molar mass of Al is 27.0 g mol^{-1}

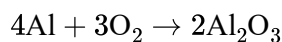
Molar mass of O is 16.0 g mol^{-1}

JEE Main 2025 (Online) 23rd January Evening Shift

Answer: 153

Solution:

To determine the mass of aluminum oxide produced, we first consider the chemical reaction between aluminum and oxygen:



Given:

Molar mass of Al = 27.0 g/mol

Molar mass of O = 16.0 g/mol

81.0 g of Al and 128.0 g of O₂

Calculating moles of reactants:

For aluminum (Al), using its molar mass:

$$\frac{81 \text{ g}}{27 \text{ g/mol}} = 3 \text{ moles of Al}$$

For oxygen (O₂), using its molar mass (O₂ is composed of two oxygen atoms):

$$\frac{128 \text{ g}}{32 \text{ g/mol}} = 4 \text{ moles of O}_2$$

Identifying the limiting reagent:

From the stoichiometry of the reaction:

4 moles of Al react with 3 moles of O₂

3 moles of Al require $\frac{3}{4} \times 3$ moles of O₂ = 2.25 moles of O₂

Since only 2.25 moles of O₂ are needed by 3 moles of Al, and we have 4 moles of O₂, aluminum is the limiting reagent.

Calculating moles of Al₂O₃ formed:

From the reaction, 4 moles of Al yield 2 moles of Al₂O₃. Thus:

$$\frac{3}{4} \times 2 = \frac{3}{2} \text{ moles of Al}_2\text{O}_3$$

Calculating mass of Al₂O₃:

The molar mass of Al₂O₃ is:

$$(2 \times 27) + (3 \times 16) = 102 \text{ g/mol}$$

So, the mass of $\frac{3}{2}$ moles of Al₂O₃ is:

$$\frac{3}{2} \times 102 = 153 \text{ grams}$$

Thus, the mass of aluminum oxide produced is 153 grams.

Question23

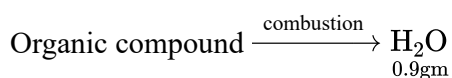
0.01 mole of an organic compound (X) containing 10% hydrogen, on complete combustion produced 0.9 gH₂O. Molar mass of (X) is

_____ g mol⁻¹.

JEE Main 2025 (Online) 23rd January Evening Shift

Answer: 100

Solution:



$$\therefore \text{mole of H}_2\text{O} = \frac{0.9}{18} = 0.05 \text{ mole}$$

$$\begin{aligned}\therefore \text{mole of H in H}_2\text{O} &= 0.05 \times 2 = 0.1 \text{ mole} \\ &= \text{mole of H in } 0.01 \text{ mole} \\ &\quad \text{Organic compound}\end{aligned}$$

$$\begin{aligned}\therefore \text{wt of H atom in } 0.01 \text{ mole compound} &= 0.1 \times 1 \\ &= 0.1\text{gm}\end{aligned}$$

$$\therefore \text{wt of H atom in one mole compound}$$

$$= \frac{0.1}{0.01} = 10\text{gm}$$

$$\therefore \text{wt. \% of H} = \frac{\text{wt. of H in one mole compound}}{\text{Molar mass of compound}} \times 100$$

$$10 = \frac{10}{M} \times 100$$

$$\therefore M = 100$$

Question24

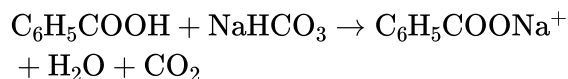
Xg of benzoic acid on reaction with aq NaHCO₃ released CO₂ that occupied 11.2 L volume at STP.

X is _____ g.

JEE Main 2025 (Online) 24th January Morning Shift

Answer: 61

Solution:



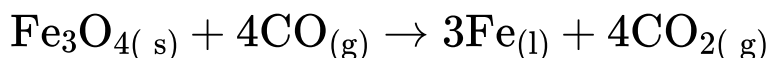
x gm 11.2 L

$$\text{mole of C}_6\text{H}_5\text{COOH} = \text{mole of CO}_2 = \frac{11.2}{22.4} = 0.5$$

$$\text{mass of C}_6\text{H}_5\text{COOH} = x = 0.5 \times 122 = 61\text{gm}$$

Question25

Consider the following reaction occurring in the blast furnace:



' x ' kg of iron is produced when 2.32×10^3 kg Fe_3O_4 and 2.8×10^2 kg CO are brought together in the furnace. The value of ' x ' is _____ . (nearest integer)

{Given:molar mass of $\text{Fe}_3\text{O}_4 = 232 \text{ g mol}^{-1}$

molar mass of $\text{CO} = 28 \text{ g mol}^{-1}$

molar mass of $\text{Fe} = 56 \text{ g mol}^{-1}$ }

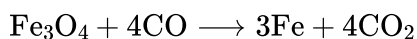
JEE Main 2025 (Online) 24th January Morning Shift

Answer: 420

Solution:

$$\text{moles of Fe}_3\text{O}_4 = \frac{2.32 \times 10^3 \times 10^3}{232} = 10000 \text{ mol}$$

$$\text{moles of CO} = \frac{2.8 \times 10^2 \times 10^3}{28} = 10000 \text{ mol}$$



$$10^4 \text{ mol} \quad 10^4 \text{ mol}$$

CO is L.R.

$$\text{mole of Fe} = \frac{3}{4} \times 10^4$$

$$\text{mass of Fe} = \frac{3}{4} \times \frac{10^4 \times 56}{1000} \text{ kg} = 420 \text{ kg}$$

Question 26

Quantitative analysis of an organic compound (X) shows following % composition.

C : 14.5%

Cl : 64.46%

H: 1.8 %

(Empirical formula mass of the compound (X) is _____ $\times 10^{-1}$

(Given molar mass in g mol^{-1} of C : 12, H : 1, O : 16, Cl : 35.5)

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 1655

Solution:

Identify the given weight percentages from a 100 g sample:

Carbon (C): 14.5 g

Chlorine (Cl): 64.46 g

Hydrogen (H): 1.8 g

Since the sample must sum to 100 g, the remaining mass is from oxygen (O):

$$\text{O: } 100 - (14.5 + 64.46 + 1.8) = 100 - 80.76 = 19.24 \text{ g.}$$

Next, convert these masses to moles using the given atomic masses:

Moles of C:

$$\frac{14.5 \text{ g}}{12 \text{ g/mol}} \approx 1.2083 \text{ mol.}$$

Moles of Cl:

$$\frac{64.46 \text{ g}}{35.5 \text{ g/mol}} \approx 1.8171 \text{ mol.}$$

Moles of H:

$$\frac{1.8 \text{ g}}{1 \text{ g/mol}} = 1.8 \text{ mol.}$$

Moles of O:

$$\frac{19.24 \text{ g}}{16 \text{ g/mol}} \approx 1.2025 \text{ mol.}$$

Now, find the simplest whole-number ratio by dividing each by the smallest number of moles (approximately 1.2025 mol):

Ratio for C:

$$1.2083/1.2025 \approx 1.00$$

Ratio for O:

$$1.2025/1.2025 = 1.00$$

Ratio for H:

$$1.8/1.2025 \approx 1.50$$

Ratio for Cl:

$$1.8171/1.2025 \approx 1.51 \approx 1.50$$

Since the ratios for H and Cl are about 1.5, multiplying all ratios by 2 will give whole numbers:

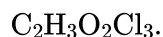
$$\text{C: } 1 \times 2 = 2$$

$$\text{O: } 1 \times 2 = 2$$

$$\text{H: } 1.5 \times 2 = 3$$

$$\text{Cl: } 1.5 \times 2 = 3$$

Thus, the empirical formula of the compound is:



Calculate the empirical formula mass by summing the contributions:

$$\text{C: } 2 \times 12 = 24 \text{ g/mol}$$

$$\text{H: } 3 \times 1 = 3 \text{ g/mol}$$

$$\text{O: } 2 \times 16 = 32 \text{ g/mol}$$

$$\text{Cl: } 3 \times 35.5 = 106.5 \text{ g/mol}$$

Total mass:

$$24 + 3 + 32 + 106.5 = 165.5 \text{ g/mol.}$$

The problem asks for the empirical formula mass in the form “ $__ \times 10^{-1}$ ”. We can express 165.5 g/mol as:

$$165.5 = 1655 \times 10^{-1} \text{ g/mol.}$$

Thus, the answer is: $1655 \times 10^{-1} \text{ g/mol.}$

Question27

The molarity of a 70% (mass/mass) aqueous solution of a monobasic acid (X) is _____ $\times 10^{-1}$ M (Nearest integer)

[Given: Density of aqueous solution of (X) is 1.25 g mL^{-1}

Molar mass of the acid is 70 g mol^{-1}]

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 125

Solution:

Assuming 100 gm solution contain 70 gm solute.

Volume of 100 gm solution will be $\frac{100}{1.25}$ ml.

Molarity = $\frac{70/70}{100/1.25} \times 1000 = 12.5$ or 125×10^{-1}

Question28

0.5 g of an organic compound on combustion gave 1.46 g of CO_2 and 0.9 g of H_2O . The percentage of carbon in the compound is _____. (Nearest integer)

[Given : Molar mass (in gmol^{-1}) C : 12, H : 1, O : 16]

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 80

Solution:

To determine the percentage of carbon in the organic compound, follow these steps:

Calculate Moles of CO_2 :

Given: 1.46 g of CO_2 is produced.

Molar mass of $\text{CO}_2 = 44 \text{ g/mol}$.

$$\text{Moles of } \text{CO}_2 = \frac{1.46 \text{ g}}{44 \text{ g/mol}}.$$

Determine Moles of Carbon Atoms:

Each molecule of CO_2 contains one atom of carbon.

Thus, moles of carbon = moles of CO_2 .

Calculate Mass of Carbon:

Molar mass of carbon (C) = 12 g/mol.

Mass of carbon = Moles of carbon $\times$ 12 g/mol.

$$\text{Mass of carbon} = \frac{1.46}{44} \times 12.$$

Calculate Percentage of Carbon:

Total mass of the organic compound = 0.5 g.

$$\text{Percentage of carbon} = \left(\frac{\text{Mass of carbon}}{0.5 \text{ g}} \right) \times 100\%.$$

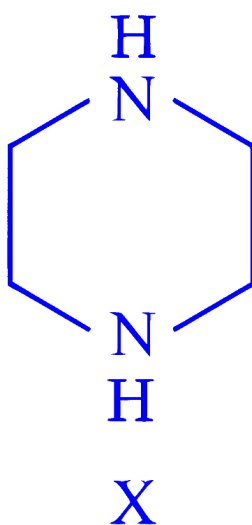
$$\text{Hence, Percentage of carbon} = \frac{1.46}{44} \times \frac{12}{0.5} \times 100.$$

Percentage of carbon = 79.63%.

Therefore, the percentage of carbon in the compound is approximately 80%.

Question29

During estimation of nitrogen by Dumas' method of compound X(0.42 g)



_____mL of N_2 gas will be liberated at STP. (nearest integer)

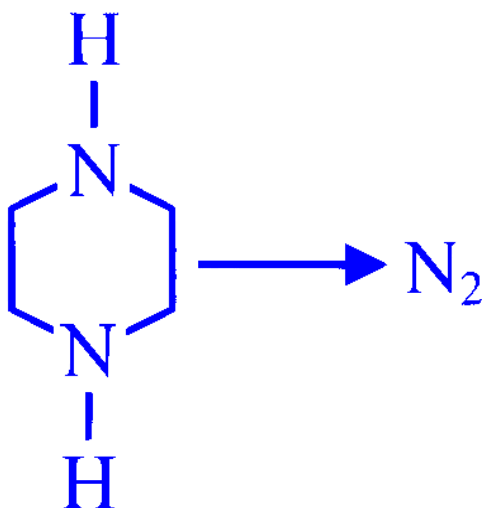
(Given molar mass in g mol^{-1} : C : 12, H : 1, N : 14)

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 109

Solution:

M.wt. of given compound = 86



Applying POAC on ' N '

$$n_X \times 2 = n_{N_2} \times 2$$

$$\frac{0.42}{86} = n_{N_2}$$

$$\Rightarrow (\text{Volume})_{N_2} \text{ at STP} = \frac{0.42}{86} \times 22.4 \text{ L}$$

$$= 0.1108 \text{ L} = 110.8 \text{ mL}$$

Question30

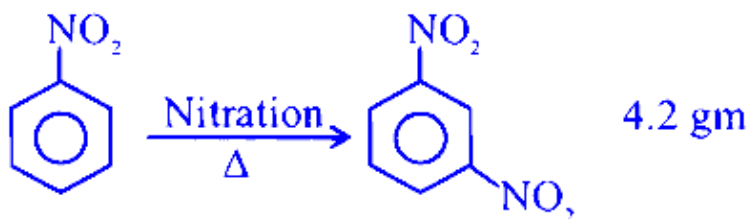
X g of nitrobenzene on nitration gave 4.2 g of m -dinitrobenzene. X = _____g. (nearest integer)

[Given : molar mass (in gmol^{-1}) C : 12, H : 1, O : 16, N : 14]

JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 3

Solution:



$$\begin{aligned} \text{C}_6\text{H}_5\text{NO}_2 \quad \text{MF} &= \text{C}_6\text{H}_4\text{N}_2\text{O}_4 \\ \text{MW} &= 123 \quad \text{MW} = 168 \\ \therefore \frac{4.2}{168} &= 0.025 \text{ mol} \end{aligned}$$

$\therefore$ required gm of nitro benzene

$$\begin{aligned} &= 123 \times 0.025 \\ &= 3.075 \end{aligned}$$

$\therefore$ Nearest integer is 3

Question31

Fortification of food with iron is done using $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$. The mass in grams of the $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ required to achieve 12 ppm of iron in 150 kg of wheat is _____ (Nearest integer)

[Given: Molar mass of Fe, S and O respectively are 56, 32 and 16 g mol^{-1}]

JEE Main 2025 (Online) 4th April Morning Shift

Answer: 9

Solution:

Let mass of iron = w g

$$\begin{aligned} \Rightarrow \frac{w}{150 \times 10^3} \times 10^6 &= 12 \\ \Rightarrow w &= 150 \times 12 \times 10^{-3} = 1.8\text{gm} \end{aligned}$$

Let mass of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ = w_1 gm

$$\Rightarrow \text{Moles of Fe} = \frac{1.8}{56} = \left(\frac{w_1}{56 + 96 + 7 \times 18} \right)$$

$$\Rightarrow w_1 = 8.935 \text{ gm}$$

Question32

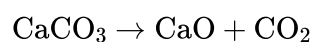
The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is _____ kg. (Nearest integer)

Given: Molar mass (in gmol^{-1}) of Ca-40, O-16, C-12

JEE Main 2025 (Online) 4th April Evening Shift

Answer: 63

Solution:



$$\text{mass of CaCO}_3 = \frac{150 \times 75}{100} = 112.5 \text{ kg}$$

$$= 112500 \text{ g}$$

$$n_{\text{CaCO}_3} = 1125$$

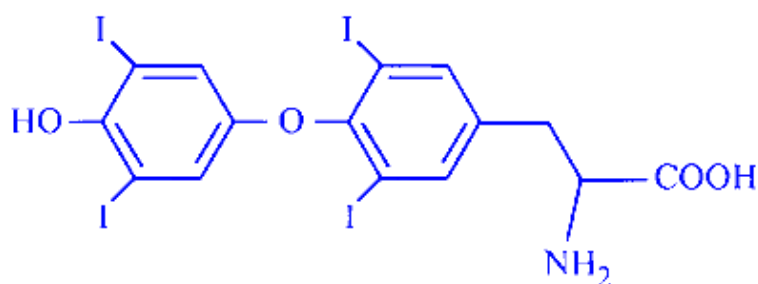
$$\text{So moles of CaO} = 1125$$

$$\text{mass of CaO} = \frac{1125 \times 56}{1000} = 63 \text{ kg}$$

$$\text{Correct answer} \Rightarrow 63$$

Question33

Thyroxine, the hormone has given below structure



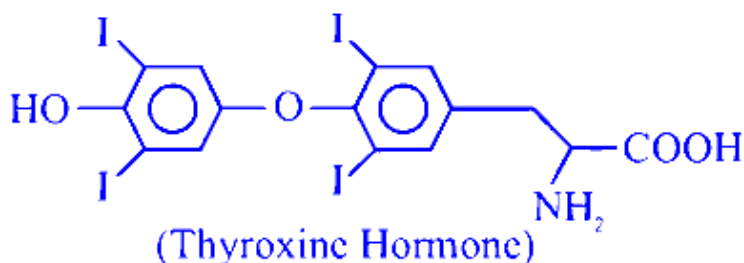
The percentage of iodine in thyroxine is _____ %. (nearest integer)

(Given molar mass in gmol^{-1} C : 12, H : 1, O : 16, N : 14, I : 127)

JEE Main 2025 (Online) 7th April Morning Shift

Answer: 65

Solution:



→ Molecular formula of Thyroxine $\Rightarrow \text{C}_{15}\text{H}_{11}\text{O}_4\text{NI}_4$

→ Molecular mass of Thyroxine -

$$\text{C} \rightarrow 15 \times 12 = 180$$

$$\text{H} \rightarrow 11 \times 1 = 11$$

$$\text{O} \rightarrow 16 \times 4 = 64$$

$$\text{N} \rightarrow 14 \times 1 = 14$$

$$\text{I} \rightarrow 127 \times 4 = 508$$

→ Molecular mass of Thyroxine $\Rightarrow 777$

$$\rightarrow \% \text{ of Iodine} = \frac{508}{777} \times 100$$

$$= 65.38\%$$

$$\text{Nearest integer} = 65$$

Question34

An organic compound weighing 500 mg , produced 220 mg of CO_2 , on complete combustion. The percentage composition of carbon in the compound is _____ %. (nearest integer)

(Given molar mass in gmol^{-1} of C : 12, O : 16)

JEE Main 2025 (Online) 7th April Morning Shift

Answer: 12

Solution:

Convert the mass of CO₂ to moles:

The molar mass of CO₂ is 44 g/mol. Thus, the number of moles of CO₂ produced is calculated as follows:

$$n_{\text{CO}_2} = \frac{220 \times 10^{-3} \text{ g}}{44 \text{ g/mol}} = 5 \times 10^{-3} \text{ moles}$$

Calculate the mass of carbon in CO₂:

Since each mole of CO₂ contains one mole of carbon, the moles of carbon are the same: 5×10^{-3} moles. The molar mass of carbon is 12 g/mol, so the mass of carbon is:

$$m_{\text{C}} = 5 \times 10^{-3} \text{ moles} \times 12 \text{ g/mol} = 60 \times 10^{-3} \text{ g} = 60 \text{ mg}$$

Calculate the percentage composition of carbon:

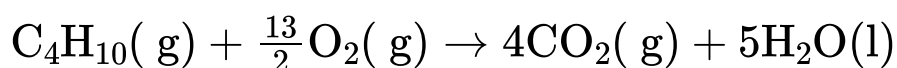
To find the percentage by mass of carbon in the compound, use the formula:

$$\% \text{ Carbon} = \left(\frac{60 \text{ mg}}{500 \text{ mg}} \right) \times 100 = 12\%$$

Thus, the percentage composition of carbon in the compound is approximately 12%.

Question 35

Butane reacts with oxygen to produce carbon dioxide and water following the equation given below.



If 174.0 kg of butane is mixed with 320.0 kg of O₂, the volume of water formed in liters is

_____ . (Nearest integer)

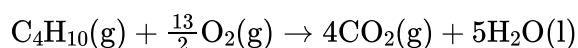
[Given : (a) Molar mass of C, H, O are 12, 1, 16 g mol⁻¹ respectively, (b) Density of water = 1 g mL⁻¹]

JEE Main 2025 (Online) 7th April Evening Shift

Answer: 138

Solution:

The chemical reaction between butane (C_4H_{10}) and oxygen (O_2) to produce carbon dioxide (CO_2) and water (H_2O) can be represented by the following equation:



Given that we have 174.0 kg of butane and 320.0 kg of oxygen, we are tasked with calculating the volume of water produced in liters.

Mole Calculation for Water:

The stoichiometry of the reaction shows that 5 moles of water are produced per $13/2$ moles of oxygen.

Using the moles of oxygen available from input, calculate the moles of water produced:

$$\text{Moles of H}_2\text{O} = 5 \times \frac{2}{13} \times (10 \times 10^3)$$

Mass of Water Produced:

Calculate the mass of water using the molar mass of water (18 g/mol):

$$\text{Mass of H}_2\text{O} = \left(\frac{10^5}{13} \right) \times 18$$

The result will be:

$$= 1.3846 \times 10^5 \text{ g}$$

Volume of Water:

Convert the mass of water to volume, considering the density of water is 1 g/mL (or 1 g/L):

$$\text{Volume of H}_2\text{O} = 138.46 \text{ liters}$$

Therefore, the volume of water formed is approximately 138 liters when rounded to the nearest integer.

Question 36

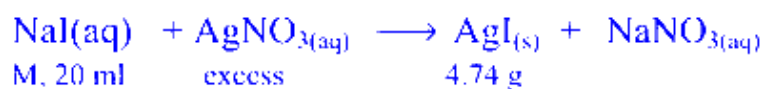
20 mL of sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____ M. (Nearest Integer value)

(Given : Na = 23, I = 127, Ag = 108, N = 14, O = 16 g mol⁻¹)

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 1

Solution:



$$\text{Moles of I}^{-}\text{ in NaI} = \text{Moles of (I}^{-}\text{) in AgI} = \frac{4.74}{235}$$

$$\text{Moles of NaI} = \frac{4.74}{235}$$

$$\text{Molarity [NaI]} = \frac{4.74}{235 \times 0.02} = 1.008$$

Question37

Density of 3 M NaCl solution is 1.25 g/mL. The molality of the solution is :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. 2.79 m

B. 2 m

C. 1.79 m

D. 3 m

Answer: A

Solution:

Molarity of NaCl = 3 M (3 moles in 1 L of solution)

Given the density of the solution is 1.25 g/mL, the total mass of 1 liter (1000 mL) of the solution is:

$$1.25 \text{ g/mL} \times 1000 \text{ mL} = 1250 \text{ g}$$

To find the molality, we need the mass of the solvent (water). First, calculate the mass of NaCl in 1 L of solution. The molar mass of NaCl is approximately:

$$23 \text{ g/mol (Na)} + 35.45 \text{ g/mol (Cl)} \approx 58.45 \text{ g/mol}$$

Thus, the mass of NaCl is:

$$3 \text{ mol} \times 58.45 \text{ g/mol} \approx 175.35 \text{ g}$$

Now, the mass of the solvent (water) is:

$$1250 \text{ g (solution)} - 175.35 \text{ g (NaCl)} \approx 1074.65 \text{ g}$$

Convert the mass of the solvent to kilograms:

$$1074.65 \text{ g} \div 1000 \approx 1.07465 \text{ kg}$$

Molality (m) is defined as the number of moles of solute per kilogram of solvent:

$$m = \frac{3 \text{ mol}}{1.07465 \text{ kg}} \approx 2.79 \text{ mol/kg}$$

Thus, the molality of the solution is approximately 2.79 m.

Question38

2.8×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its ' x ' mg sample. The mass of CO_2 taken initially is Given:

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

- A. 98.3 mg
- B. 196.2 mg
- C. 150.4 mg
- D. 48.2 mg

Answer: B

Solution:

$$\begin{aligned}
 (\text{moles})_{\text{initial}} &= \frac{x \times 10^{-3}}{44} \\
 (\text{moles})_{\text{removal}} &= \frac{10^{21}}{6.02 \times 10^{23}} \\
 (\text{moles})_{\text{left}} &= (\text{moles})_{\text{initial}} - (\text{moles})_{\text{removed}} \\
 2.8 \times 10^{-3} &= \frac{x \times 10^{-3}}{44} - \frac{10^{21}}{6.02 \times 10^{23}} \\
 \Rightarrow x &= 196.2 \text{mg}
 \end{aligned}$$

Question39

The elemental composition of a compound is 54.2%C, 9.2%H and 36.6%O. If the molar mass of the compound is 132 g mol^{-1} , the molecular formula of the compound is : [Given : The relative atomic mass of C : H : O = 12 : 1 : 16]

JEE Main 2025 (Online) 24th January Evening Shift

Options:

- A. $\text{C}_4\text{H}_8\text{O}_2$
- B. $\text{C}_6\text{H}_{12}\text{O}_6$
- C. $\text{C}_4\text{H}_9\text{O}_3$
- D. $\text{C}_6\text{H}_{12}\text{O}_3$

Answer: D

Solution:

C	:	H	:	O
$\frac{54.2}{12}$	:	9.2	:	$\frac{36.6}{16}$
4.516	:	9.2	:	2.287
$\frac{4.516}{2.287}$	:	$\frac{9.2}{2.287}$	:	$\frac{2.287}{2.287}$
1.97	:	4.02	:	1

$\text{C}_2\text{H}_4\text{O} \Rightarrow$ Empirical formula

$$\text{E.F. mass} = 24 + 4 + 16 = 44$$

and molar mass = 132

$$\begin{aligned}\text{Hence molecular formula} &= (\text{C}_2\text{H}_4\text{O})_3 \\ &= \text{C}_6\text{H}_{12}\text{O}_3\end{aligned}$$

Correct Option (3)

Question40

Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given : Density of nitric acid solution is 1.25 g/mL.

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

32

B.

40

C.

55

D.

45

Answer: A

Solution:

$$\%w/w \text{ of HNO}_3 = 75\%$$

means 100 gm of solution containing 75 g of HNO₃

$$\&\left(\frac{\text{gm}}{\text{ml}}\right)_{\text{solution}} = 1.25 = \frac{100\text{gm}}{V}$$

$$V_{\text{ml}} \text{ of 100 gm solution} = \frac{100}{1.25} \text{ml}$$

$$\therefore 75\text{gm of HNO}_3 \text{ present in } \frac{100}{1.25} \text{ml solution}$$

∴ 30gm of HNO_3 present in

$$\frac{100}{1.25 \times 75} \times 30 = 32 \text{ml solution}$$

Question41

Choose the correct statements.

- (A) Weight of a substance is the amount of matter present in it.
- (B) Mass is the force exerted by gravity on an object.
- (C) Volume is the amount of space occupied by a substance.
- (D) Temperatures below 0°C are possible in Celsius scale, but in Kelvin scale negative temperature is not possible.
- (E) Precision refers to the closeness of various measurements for the same quantity.

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

- A.
(B), (C) and (D) Only
- B.
(A), (B) and (C) Only
- C.
(C), (D) and (E) Only
- D.
(A), (D) and (E) Only

Answer: C

Solution:

A) Weight of a substance is the amount of matter present in it.

Not correct.

The amount of matter present in a substance is called its mass, not its weight.

Mass of a substance is the amount of matter present in it.

B) Mass is the force exerted by gravity on an object.

Not correct.

Mass is not the force exerted on an object by gravity. Mass is the amount of matter in an object, while weight is the force of gravity on that object.

C) Volume is the amount of space occupied by a substance.

Correct.

The amount of space occupied by a substance is its volume, which is simply the equivalent three dimensional measurements of spatial geometry (height, width; and (length).

Standard units of volume are cubic centimeters for solids, milliliters or liters for liquids, and liters for gases.

D) Temperatures below 0°C are possible in Celsius scale, but in Kelvin scale negative temperature is not possible.

Correct.

Celsius temp is possible below 0°C (negative temp). But, it is not possible in the Kelvin Scale

$273\text{ K} \rightarrow$ Absolute temperature

$\text{Kelvin temp} = ^{\circ}\text{C temp} + 273.15$

The minimum temperature possible in the Kelvin scale is the absolute zero temperature (OK). OK means -273.15°C . Since absolute zero is the coldest possible temperature, there are no negative values on the Kelvin temperature scale.

E) Precision refers to the closeness of various measurements for the same quantity.

Correct.

A measure of how close repeated measurements are to each other is known as precision. Precision is the reproducibility of a result or measurement.

If weight of a substance is measured five times and get 8.2 kg each time, then the measurement is very precise - Example.

So, the correct statements are (C) , (D) and (E).

Correct answer: Option 3) (C) ,(D) and(E) only

Question 42

On complete combustion 1.0 g of an organic compound (X) gave 1.46 g of CO₂ and 0.567 g of H₂O. The empirical formula mass of compound (X) is _____ g. (Given molar mass in gmol⁻¹ C : 12, H : 1, O : 16)

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. 60

B. 45

C. 30

D. 15

Answer: C

Solution:

Calculate the Moles of Carbon:

Carbon is fully converted to CO₂.

$$\text{Moles of CO}_2 = \frac{1.46 \text{ g}}{44 \text{ g/mol}} = 0.033 \text{ mol.}$$

$$\text{Moles of Carbon (C)} = \text{Moles of CO}_2 = 0.033 \text{ mol.}$$

$$\text{Mass of Carbon} = 0.033 \times 12 \text{ g/mol} = 0.396 \text{ g.}$$

Calculate the Moles of Hydrogen:

Hydrogen is fully converted to H₂O.

$$\text{Moles of H}_2\text{O} = \frac{0.567 \text{ g}}{18 \text{ g/mol}} = 0.0315 \text{ mol.}$$

$$\text{Moles of Hydrogen (H)} = 2 \times 0.0315 = 0.063 \text{ mol.}$$

$$\text{Mass of Hydrogen} = 0.063 \times 1 \text{ g/mol} = 0.063 \text{ g.}$$

Calculate the Mass of Oxygen:

$$\text{Total mass of compound X} = 1.0 \text{ g.}$$

$$\text{Mass of Oxygen} = \text{Total mass} - (\text{Mass of Carbon} + \text{Mass of Hydrogen})$$

$$\text{Mass of Oxygen} = 1.0 \text{ g} - (0.396 \text{ g} + 0.063 \text{ g}) = 0.541 \text{ g.}$$

$$\text{Moles of Oxygen (O)} = \frac{0.541}{16} = 0.0338 \approx 0.033 \text{ mol.}$$

Determine the Empirical Formula:

The ratio of moles: C = 0.033, H = 0.063, O = 0.033.

Simplified, this ratio is approximately 1:2:1.

Empirical formula: CH₂O.

Calculate the Empirical Formula Mass:

Empirical formula mass = $12 \times 1 + 1 \times 2 + 16 \times 1 = 30$ g/mol.

Therefore, the empirical formula mass of compound X is 30 g/mol.

Question43



Consider the above reaction, what mass of CaCl₂ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of CaCO₃ ?

(Given : Molar mass of Ca, C, O, H and Cl are 40, 12, 16, 1 and 35.5 g mol⁻¹, respectively)

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

- A. 3.908 g
- B. 2.636 g
- C. 10.545 g
- D. 5.272 g

Answer: C

Solution:

The reaction is:



Calculate Moles of CaCO₃:

The molar mass of CaCO₃ is calculated as

$$40 + 12 + 3 \times 16 = 100 \text{ g/mol}$$

Given 1000 g of CaCO_3 , the number of moles is

$$\frac{1000}{100} = 10 \text{ moles}$$

Calculate Moles of HCl:

Using the given concentration and volume for HCl, the moles are calculated as follows:

$$0.76 \times \frac{250}{1000} = 0.19 \text{ moles}$$

HCl is the limiting reagent (L.R.) because it has fewer moles than needed to completely react with the CaCO_3 .

Calculate Moles of CaCl_2 Formed:

According to the stoichiometry of the reaction, 2 moles of HCl produce 1 mole of CaCl_2 . Therefore, the moles of CaCl_2 formed are:

$$\frac{0.19}{2} = 0.095 \text{ moles}$$

Calculate Mass of CaCl_2 :

The molar mass of CaCl_2 is:

$$40 + 2 \times 35.5 = 111 \text{ g/mol}$$

Thus, the mass of CaCl_2 is:

$$0.095 \times 111 = 10.545 \text{ g}$$

Hence, the mass of CaCl_2 formed is 10.545 g.

Question44

Among 10^{-9} g (each) of the following elements, which one will have the highest number of atoms?

Element: Pb, Po, Pr and Pt

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. Pr

B. Pt

C. Po

D. Pb

Answer: A

Solution:

To determine which element has the highest number of atoms among 10^{-9} grams of the elements Pb, Po, Pr, and Pt, we use the formula for calculating the number of atoms:

$$\text{Number of atoms} = \frac{\text{Mass}}{\text{Molar Mass (g/mol)}} \times N_A$$

where N_A is Avogadro's number.

From this formula, it's clear that for a given mass, the element with the smallest molar mass will have the greatest number of atoms, as it appears in the denominator.

Below are the molar masses of the elements:

Molar mass of Pr (Praseodymium), $M_{\text{Pr}} = 141 \text{ g/mol}$

Molar mass of Pt (Platinum), $M_{\text{Pt}} = 195 \text{ g/mol}$

Molar mass of Pb (Lead), $M_{\text{Pb}} = 207 \text{ g/mol}$

Molar mass of Po (Polonium), $M_{\text{Po}} = 209 \text{ g/mol}$

Since Praseodymium (Pr) has the smallest molar mass ($M_{\text{Pr}} = 141 \text{ g/mol}$), 10^{-9} grams of Pr will contain the highest number of atoms compared to the other elements listed.

Question45

In Dumas' method for estimation of nitrogen 0.4 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is(Given : Aqueous tension at 300 K = 15 mmHg)

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. 15.71%

B. 17.46%

C. 7.85%

D. 20.95%

Answer: A

Solution:

$$\begin{aligned}\text{Pressure of N}_2 \text{ gas evolved} &= 715 - 15 \\ &= 700 \text{ mmHg} \\ &= \frac{700}{760} \text{ atm.}\end{aligned}$$

$$\therefore \text{Mole of N}_2 \text{ evolved} = \frac{PV}{RT}$$

$$\begin{aligned}&= \frac{700 \times 60 \times 10^{-3}}{760 \times 0.0821 \times 300} \\ &= 0.0022 \text{ mole}\end{aligned}$$

$$\therefore \text{wt. of N}_2 \text{ evolved} = 0.0022 \times 28 = 0.063 \text{ gm}$$

$$\therefore \text{wt. \% of nitrogen in compound}$$

$$\begin{aligned}&= \frac{\text{wt. of nitrogen}}{\text{wt. of compound}} \times 100 \\ &= \frac{0.063}{0.4} \times 100 \\ &= 15.71\%\end{aligned}$$

Question46

10 mL of 2 M NaOH solution is added to 20 mL of 1 M HCl solution kept in a beaker. Now, 10 mL of this mixture is poured into a volumetric flask of 100 mL containing 2 moles of HCl and made the volume upto the mark with distilled water. The solution in this flask is :

JEE Main 2025 (Online) 3rd April Evening Shift

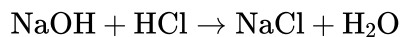
Options:

- A. 0.2 M NaCl solution
- B. 20 M HCl solution
- C. Neutral solution
- D. 10 M HCl solution

Answer: B

Solution:

When 10ml, 2MNaOH solution is added to 20 ml of 1 M HCl solution :



Initial : MV	= 2 × 0.1	MV = 1 × 0.2
	= 0.2 mole	= 0.2 mole
Final	0	0

∴ Resulting solution becomes neutral.

Now when 10 mol of above solution is poured into a flask containing 2 mole HCl and made solution 100 ml will distilled water.

$$\text{Molarity of HCl} = \frac{2}{100} \times 1000 = 20$$

Question47

Mass of magnesium required to produce 220 mL of hydrogen gas at STP on reaction with excess of dil. HCl is

Given: Molar mass of Mg is 24 g mol^{-1} .

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

- A. 0.24 mg
- B. 235.7 g
- C. 2.444 g
- D. 236 mg

Answer: D

Solution:

Given data:

Volume of H_2 : 220 mL

Molar mass of Mg: 24 g/mol

At STP, 1 mole of a gas occupies 22.4 liters (or 22,400 mL).

First, determine the moles of hydrogen gas (H_2) produced:

$$\text{Moles of H}_2 = \frac{220 \times 10^{-3}}{22.4}$$

Because the molar ratio of Mg to H₂ in the reaction is 1:1, the moles of Mg used are equal to the moles of H₂:

Next, calculate the mass of magnesium:

$$\text{Mass of Mg used} = \frac{220 \times 10^{-3}}{22.4} \times 24$$

Simplifying further:

$$= 235.7 \times 10^{-3} \text{ g}$$

Convert the mass from grams to milligrams:

$$= 235.7 \text{ mg}$$

Thus, the mass of magnesium required is 235.7 mg.

Question48

On combustion 0.210 g of an organic compound containing C, H and O gave 0.127 g H₂O and 0.307 g CO₂. The percentages of hydrogen and oxygen in the given organic compound respectively are:

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

7.55, 43.85

B.

6.72, 53.41

C.

6.72, 39.87

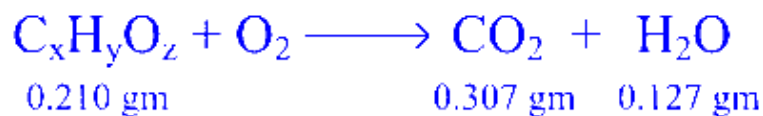
D.

53.41, 39.6

Answer: B

Solution:

In the combustion of organic compound, all "C" in CO₂ and all "H" in H₂O comes from organic compound



$$\text{Weight of "C" in CO}_2 = \frac{12}{44} \times 0.307$$

$$= 0.0837 \text{ gm}$$

$$\text{Weight of "H" in H}_2\text{O} = \frac{2}{18} \times 0.127 = 0.0141 \text{ g}$$

$$\begin{aligned} \% 'H' \text{ in compound} &= \frac{0.0141}{0.21} \times 100 = 6.719\% \\ &= 6.72\% \end{aligned}$$

$$\begin{aligned} \text{Weight of "O" in compound} \\ &= 0.210 - (0.0837 + 0.0141) \\ &= 0.1122 \end{aligned}$$

$$\begin{aligned} \% \text{ of "O" in compound} &= \frac{0.1122}{0.21} \times 100 \\ &= 53.41\% \end{aligned}$$

Question49

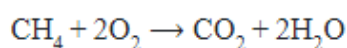
Mass of methane required to produce 22g of CO₂ after complete combustion is g.

(Given Molar mass in g mol⁻¹ C = 12.0 H = 1.0 O = 16.0)

[27-Jan-2024 Shift 1]

Answer: 8

Solution:



$$\text{Moles of CO}_2 = \frac{22}{44} = 0.5$$

$$\text{So, required moles of CH}_4 = 0.5$$

$$\text{Mass} = 0.5 \times 16 = 8 \text{ gm}$$

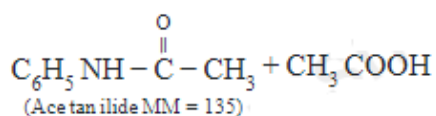
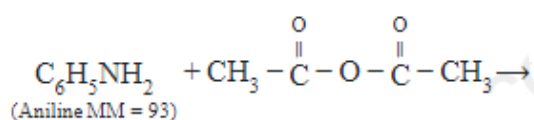
Question50

9.3g of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is $\underline{\hspace{1cm}} \times 10^{-1}$ g. (Given molar mass in gmol^{-1} N : 14, O : 16, C : 12, H : 1)

[27-Jan-2024 Shift 2]

Answer: 13.5

Solution:



$$n_{\text{Acetanilide}} = n_{\text{Aniline}}$$

$$\Rightarrow \frac{m}{135} = \frac{9.3}{93}$$

$$\Rightarrow m = 13.5\text{g}$$

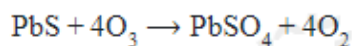
Question51

1 mole of PbS is oxidised by "X" moles of O₃ to get "Y" moles of O₂. X + Y =

[27-Jan-2024 Shift 2]

Answer: 8

Solution:



$$x = 4, y = 4$$

Question52

0.05cm thick coating of silver is deposited on a plate of 0.05m^2 area. The number of silver atoms deposited on plate are _____ $\times 10^{23}$. (At mass Ag = 108, $d = 7.9\text{gcm}^{-3}$)

[30-Jan-2024 Shift 1]

Answer: 11

Solution:

$$\text{Volume of silver coating} = 0.05 \times 0.05 \times 10000$$

$$= 25\text{cm}^3$$

$$\text{Mass of silver deposited} = 25 \times 7.9\text{g}$$

$$\text{Moles of silver atoms} = \frac{25 \times 7.9}{108}$$

$$\text{Number of silver atoms} = \frac{25 \times 7.9}{108} \times 6.023 \times 10^{23}$$

$$= 11.01 \times 10^{23}$$

Ans. 11

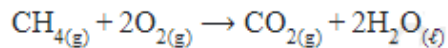
Question53

Number of moles of methane required to produce $22\text{gCO}_{2(\text{g})}$ after combustion is $x \times 10^{-2}$ moles. The value of x is

[31-Jan-2024 Shift 1]

Answer: 50

Solution:



$$n_{\text{CO}_2} = \frac{22}{44} = 0.5 \text{ moles}$$

So moles of CH_4 required = 0.5 moles i.e. 50×10^{-2} mole

$$x = 50$$

Question 54

A sample of CaCO_3 and MgCO_3 weighed 2.21g is ignited to constant weight of 1.152g. The composition of mixture is :

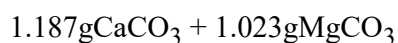
(Given molar mass in gmol^{-1}

CaCO_3 : 100, MgCO_3 : 84)

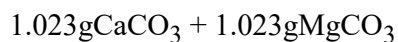
[31-Jan-2024 Shift 2]

Options:

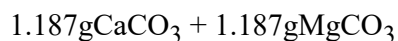
A.



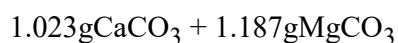
B.



C.



D.



Answer: A

Solution:



Let the weight of CaCO_3 be x gm

$$\therefore \text{weight of MgCO}_3 = (2.21 - x) \text{ gm}$$

Moles of CaCO_3 decomposed = moles of CaO formed

$$\frac{x}{100} = \text{moles of CaO formed}$$

$$\therefore \text{weight of CaO formed} = \frac{x}{100} \times 56$$

Moles of MgCO_3 decomposed = moles of MgO formed

$$\frac{(2.21 - x)}{84} = \text{moles of MgO formed}$$

$$\therefore \text{weight of MgO formed} = \frac{2.21 - x}{84} \times 40$$

$$\Rightarrow \frac{2.21 - x}{84} \times 40 + \frac{x}{100} \times 56 = 1.152$$

$$\therefore x = 1.1886 \text{ g} = \text{weight of CaCO}_3$$

$$\& \text{ weight of MgCO}_3 = 1.0214 \text{ g}$$

Question 55

The molarity of 1L orthophosphoric acid (H_3PO_4) having 70% purity by weight (specific gravity 1.54 g cm^{-3}) is _____ M.

(Molar mass of $\text{H}_3\text{PO}_4 = 98 \text{ g mol}^{-1}$)

[31-Jan-2024 Shift 2]

Answer: 11

Solution:

Specific gravity (density) = 1.54g/cc.

Volume = 1L = 1000ml

Mass of solution = 1.54×1000

= 1540g

% purity of H_2SO_4 is 70%

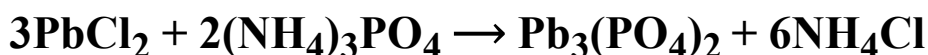
So weight of $\text{H}_3\text{PO}_4 = 0.7 \times 1540 = 1078\text{g}$

Mole of $\text{H}_3\text{PO}_4 = \frac{1078}{98} = 11$

Molarity = $\frac{11}{1\text{L}} = 11$

Question56

Consider the following reaction:



If 72mmol² of PbCl_2 is mixed with 50mmol of $(\text{NH}_4)_3\text{PO}_4$, then amount of $\text{Pb}_3(\text{PO}_4)_2$ formed is _____mmol. (nearest integer)

[1-Feb-2024 Shift 1]

Answer: 24

Solution:

Limiting Reagent is PbCl_2 mmol of $\text{Pb}_3(\text{PO}_4)_2$ formed = $\frac{\text{mmol of PbCl}_2 \text{ reacted}}{3} = 24 \text{ mmol}$

Question57

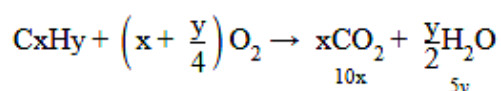
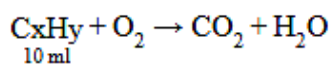
10mL of gaseous hydrocarbon on combustion gives 40mL of $\text{CO}_2(\text{g})$ and 50mL of water vapour. Total number

of carbon and hydrogen atoms in the hydrocarbon is _____

[1-Feb-2024 Shift 2]

Answer: 14

Solution:



$$10x = 40$$

$$x = 4$$

$$5y = 50$$

$$y = 10$$



Question 58

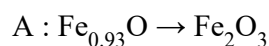
When $\text{Fe}_{0.93}\text{O}$ is heated in presence of oxygen, it converts to Fe_2O_3 . The number of correct statement/s from the following is _____

- A. The equivalent weight of $\text{Fe}_{0.93}\text{O}$ is $\frac{\text{Molecular weight}}{0.79}$.
- B. The number of moles of Fe^{2+} and Fe^{3+} in 1 mole of $\text{Fe}_{0.93}\text{O}$ is 0.79 and 0.14 respectively.
- C. $\text{Fe}_{0.93}\text{O}$ is metal deficient with lattice comprising of cubic closed packed arrangement of O^{2-} ions.
- D. The % composition of Fe^{2+} and Fe^{3+} in $\text{Fe}_{0.93}\text{O}$ is 85% and

15% respectively.
[24-Jan-2023 Shift 1]

Answer: 4

Solution:



$$nf = \left(3 - \frac{200}{93} \right) \times 0.93$$

$$nf = 0.79$$

$$B : 2x + (0.93 - x) \times 3 = 2$$

$$x = 0.79$$

$$\text{Fe}^{2+} = 0.79, \text{Fe}^{3+} = 0.21$$

$$C : \text{Fact}^D : \% \text{Fe}^{2+} = \frac{0.79}{0.93} \times 100 = 85\%; \text{Fe}^{3+} = 15\%$$

Question59

The number of units, which are used to express concentration of solutions from the following is_____

Mass percent, Mole, Mole fraction, Molarity, ppm, Molality.

[24-Jan-2023 Shift 2]

Answer: 5

Solution:

Mass percent, mole fraction, molarity, ppm, molality are used for measuring concentration terms.

Question60

What is the mass ratio of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$, molar mass = 62g/mol) required for making 500g of 0.25 molal aqueous solution and 250 mL of 0.25 molar aqueous solution ?
[25-Jan-2023 Shift 2]

Options:

A. 1 : 1

B. 3 : 1

C. 2 : 1

D. 1 : 2

Answer: C

Solution:

Assume : Mass of solvent $\approx$ Mass of solution

Case I :-

$$0.25 = \frac{W_1}{62} \times \frac{1000}{500}$$

Case II :-

$$0.25 = \frac{W_2}{62} \times \frac{1000}{250}$$

$$\frac{W_1}{W_2} = \frac{2}{1}$$

Question61

Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8% carbon is _____

(Given : Molar mass of A = 84g mol^{-1})

[25-Jan-2023 Shift 2]

Answer: 12

Solution:

Element	Percentage	Mole	Mole ratio
C	85.8	$\frac{85.8}{12} = 7.15$	1
H	14.2	$\frac{14.2}{1} = 14.2$	2

Empirical formula (CH_2)

$$14 \times n = 84$$

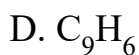
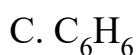
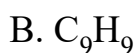
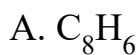
$$n = 6$$

$\therefore$ Molecular formula C_6H_{12}

Question62

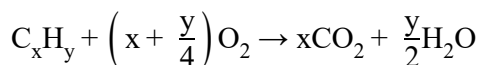
When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A ? [29-Jan-2023 Shift 2]

Options:



Answer: A

Solution:



$$x + \frac{y}{4} = 9.5$$

$$\frac{y}{2} = 3$$

$$\Rightarrow x = 8, y = 6$$

Question63

When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4g of CO_2 was produced. The molar mass of compound is _____ gmol^{-1} (Nearest integer)
[29-Jan-2023 Shift 2]

Answer: 200

Solution:

Let M is the molar mass of the compound (g/mol)

mass of compound = 0.01 Mgm

mass of carbon = $0.01M \times \frac{60}{100}$

moles of carbon = $\frac{0.01M}{12} \times \frac{60}{100}$

moles of CO_2 from combustion = $\frac{4.4}{44}$ = moles of carbon

$$\frac{0.01M}{12} \times \frac{60}{100} = \frac{4.4}{44}$$

$$M = \frac{4.4}{44} \times \frac{100}{60} \times \frac{12}{0.01} = 200 \text{ gm/mol}$$

Question64

Some amount of dichloromethane (CH_2Cl_2) is added to 671.141 mL of chloroform (CHCl_3) to prepare $2.6 \times 10^{-3}\text{M}$ solution of CH_2Cl_2 (DCM). The concentration of DCM is ppm _____ (by mass).

Given: Atomic mass: C = 12; H : 1; Cl = 35.5 density of $\text{CHCl}_3 = 1.49\text{gcm}^{-3}$

[30-Jan-2023 Shift 1]

Answer: 148

Solution:

$$\text{Molarity} = \frac{\text{mole}}{\text{volume}}$$

$$2.6 \times 10^{-3} = \frac{x/85}{0.67141}$$

$$x = 0.148\text{g}$$

$$\text{conc. of DCM in ppm} = \frac{0.148}{1.49 \times 671.141} \times 10^6$$

$$= 148 \text{ ppm}$$

Question 65

Match List I with List II:

List I (Mixture)	List II (Separation Technique)
(A) $\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	I. Steam distillation
(B) $\text{C}_6\text{H}_{14} + \text{C}_5\text{H}_{12}$	II. Differential extraction
(C) $\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2\text{O}$	III. Distillation
(D) Organic compound in H_2O	IV. Fractional distillation

[30-Jan-2023 Shift 2]

Options:

A. A-IV, B-I, C-III, D-II

B. A-III, B-IV, C-I, D-II

C. A-II, B-I, C-III, D-IV

D. A-III, B-I, C-IV, D-II

Answer: B

Solution:

List I (Mixture)	List II (Separation Technique)
$\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	Distillation
$\text{C}_6\text{H}_{14} + \text{C}_5\text{H}_{12}$	Fractional distillation
$\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2\text{O}$	Steam distillation
Organic compound in H_2O	Differential extraction

Question 66

On complete combustion, 0.492g of an organic compound gave 0.792g of CO_2 .

The % of carbon in the organic compound is _____
(Nearest integer)

[31-Jan-2023 Shift 1]

Answer: 44

Solution:

weight of C in 0.792 gm CO_2

$$= \frac{12}{44} \times 0.792 = 0.216$$

$$\begin{aligned}\% \text{ of C in compound} &= \frac{0.216}{0.492} \times 100 \\ &= 43.90\%\end{aligned}$$

Ans: 44

Question 67

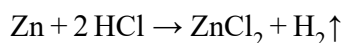
Zinc reacts with hydrochloric acid to give hydrogen and zinc chloride. The volume of hydrogen gas produced at STP from the reaction of 11.5g of zinc with excess HCl is _____ L (Nearest integer)

(Given : Molar mass of Zn is 65.4g mol^{-1} and Molar volume of H_2 at STP = 22.7L)

[31-Jan-2023 Shift 1]

Answer: 4

Solution:



$$\text{Moles of Zn used} = \frac{11.5}{65.4} = \text{Moles of H}_2 \text{ evolved}$$

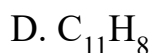
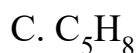
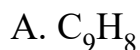
$$\text{Volume of H}_2 = \frac{11.5}{65.4} \times 22.7\text{L} = 3.99\text{L}$$

Question68

When a hydrocarbon A undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water. What is the molecular formula of A ?

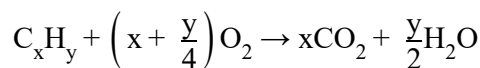
[31-Jan-2023 Shift 2]

Options:



Answer: A

Solution:



$$\frac{y}{2} = 4 \therefore y = 8$$

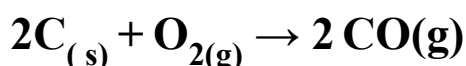
$$x + \frac{8}{4} = 11$$

$$\therefore x = 9$$

$\therefore$ Hydrocarbon will be $= C_9H_8$

Question69

Assume carbon burns according to following equation :



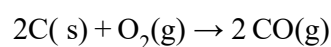
When 12g carbon is burnt in 48g of oxygen, the volume of carbon monoxide produced is _____ $\times 10^{-1}$ L at STP [nearest integer]

[Given : Assume CO as ideal gas, Mass of C is 12gmol^{-1} , Mass of O is 16gmol^{-1} and molar volume of an ideal gas at STP is 22.7Lmol^{-1}]

[31-Jan-2023 Shift 2]

Answer: 227

Solution:



1 mol 1.5 mol

Limiting reagent is carbon. One mole carbon produces one mole CO. Hence, volume at STP is 227×10^{-1} litre

Question70

The density of 3M solution of NaCl is 1.0gmL^{-1} .

Molality of the solution is _____ $\times 10^{-2}$ m
(Nearest integer).

Given: Molar mass of Na and Cl is 23 and 35.5g mol⁻¹ respectively.

[1-Feb-2023 Shift 1]

Answer: 364

Solution:

$$\begin{aligned} m &= \frac{1000 \times M}{1000 \times d - M \times M.W. \text{ of solute}} \\ &= \frac{1000 \times 3}{1000 \times 1 - (3 \times 58.5)} = 3.64 \\ &= 364 \times 10^{-2} \end{aligned}$$

Question 71

The molality of a 10%(v/v) solution of di-bromine solution in CCl₄ (carbon tetrachloride) is ' x '. x = _____ × 10⁻²M.

(Nearest integer)

[Given : molar mass of Br₂ = 160g mol⁻¹

atomic mass of C = 12g mol⁻¹

atomic mass of Cl = 35.5g mol⁻¹

density of dibromine = 3.2g cm⁻³

density of CCl₄ = 1.6g cm⁻³]

[1-Feb-2023 Shift 2]

Answer: 139

Solution:

(10 ml solute in 90 ml solvent
mass of solute = 10 × 3.2 = 32g

mass of solvent = $90 \times 1.6\text{g}$

$$m = \frac{32 \times 1000}{160 \times 90 \times 1.6} = 1.388$$

$$m = 138.8 \times 10^{-2} = 139$$

Question72

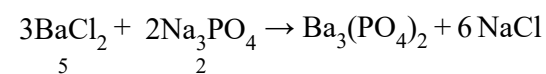
If 5 moles of BaCl_2 is mixed with 2 moles of Na_3PO_4 , the maximum number of moles of $\text{Ba}_3(\text{PO}_4)_2$ formed is

(Nearest integer)

[6-Apr-2023 shift 1]

Answer: 1

Solution:



Na_3PO_4 is limiting reagent.

2 mole Na_3PO_4 gives 1 mole of $\text{Ba}_3(\text{PO}_4)_2$

Question73

0.5g of an organic compound (X) with 60% carbon will produce _____ $\times 10^{-1}\text{g}$ of CO_2 on complete combustion.

[8-Apr-2023 shift 1]

Answer: 11

Solution:

$$\text{Moles of carbon} = \frac{0.5 \times 0.6}{12}$$

$$\text{Moles of CO}_2 = \frac{0.5 \times 0.6}{12}$$

$$\text{Mass of CO}_2 = \frac{0.5 \times 0.6}{12} \times 44 = 11 \times 10^{-1} \text{ gram}$$

Question74

Which of the following have same number of significant figures?

A. 0.00253

B. 1. 0003

C. 15.0

D. 163

Choose the correct answer from the options given below

[8-Apr-2023 shift 2]

Options:

A. B and C only

B. A, B and C only

C. A, C and D only

D. C and D only

Answer: C

Solution:

0.00253, 15.0, 163

All have three significant figures.

Question75

The number of molecules and moles in 2.8375 litres of O₂ at STP are respectively

[10-Apr-2023 shift 1]

Options:

A. 7.527×10^{22} and 0.125 mol

B. 1.505×10^{23} and 0.250 mol

C. 7.527×10^{23} and 0.125 mol

D. 7.527×10^{22} and 0.250 mol

Answer: A

Solution:

$$\text{Moles of O}_2(n_{\text{O}_2}) = \frac{\text{Volume of O}_2}{22.7} = 0.125 \text{ moles}$$

$$\text{Molecules of O}_2 = \text{moles} \times N_A$$

$$= 0.125 \times 6.022 \times 10^{23}$$

$$= 7.527 \times 10^{22} \text{ molecules}$$

Ans (1) 7.527×10^{22} and 0.125 mole

Question 76

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : 3.1500g of hydrated oxalic acid dissolved in water to make 250.0 mL solution will result in 0.1M oxalic acid solution.

Reason R : Molar mass of hydrated oxalic acid is 126gmol^{-1} In the light of the above statements, choose the correct answer from the options given below:

[10-Apr-2023 shift 2]

Options:

A. A is false but R is true

B. A is true but R is false

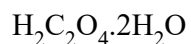
C. Both A and R are true but R is NOT the correct explanation of A

D. Both A and R are true and R is the correct explanation of A

Answer: D

Solution:

Assertion is correct.



$$M = \frac{3.15 \times 1000}{126 \times 250}$$
$$= \frac{12.6}{126} = 0.1$$

Reason is correct. It is used as a fact in explanation of assertion.

Question77

Match List I with List II

LIST I	LIST II
A 16g of $\text{CH}_4(\text{g})$	I. Weight 28g
B 1g of $\text{H}_2(\text{g})$	II 60.2×10^{23} electrons
C 1 mole of $\text{N}_2(\text{g})$	III Weight 32g
D 0.5 mol of $\text{SO}_2(\text{g})$	IV Occupies 11.4L volume at STP

Choose the correct answer from the options given below:
[10-Apr-2023 shift 2]

Options:

A. A-II, B-IV, C-I, D-III

B. A-II, B-IV, C-III, D-I

C. A-II, B-III, C-IV, D-I

D. A-I, B-III, C-II, D-IV

Answer: A

Solution:

$$16\text{gCH}_4 = \text{mole} = 1$$

$$\text{e}^- = 60.0 \times 10^{23}$$

$$19 \text{ Hz} = 0.5 \text{ mole} = 11.4(\text{L}) \text{ STP}$$

1 mole $\text{N}_2 = 2 \text{ rg}$

0.5 mol $\text{SO}_2 = \text{ weights } 32\text{g}.$

Question78

25 mL of silver nitrate solution (1M) is added dropwise to 25 mL of potassium iodide (1.05M) solution. The ion(s) present in very small quantity in the solution is/are
[11-Apr-2023 shift 1]

Options:

A. NO_3^- only

B. Ag^+ and I^- both

C. K^+ only

D. I only

Answer: B

Solution:

On adding AgNO_3 into KI, AgI will form and solubility of AgI is very low.

So, $[\text{Ag}^+]$ and $[\text{I}^-]$ will be present in very small quantity.

Question79

A solution of sugar is obtained by mixing 200g of its 25% solution and 500g of its 40% solution (both by mass). The mass percentage of the resulting sugar solution is _____ (Nearest integer)

[11-Apr-2023 shift 1]

Answer: 36

Solution:

$$\text{Solution (I)} \rightarrow \text{Mass of sugar} = 200 \times \frac{25}{100} = 50 \text{ gm}$$

$$\text{Mass of solution} = 200 \text{ gm}$$

$$\text{Solution (II)} \rightarrow \text{Mass of solution} = 500 \text{ gm}$$

$$\text{Mass of sugar} = \frac{40}{100} \times 500 = 200 \text{ gm}$$

$$\text{Final \%w/w} = \frac{\text{Total mass of sugar}}{\text{Total mass of solution}} \times 100$$

$$= \frac{50 + 200}{200 + 500} \times 100 = \frac{250}{7}$$

$$= 35.71\% \approx 36$$

Question80

A solution is prepared by adding 2g of " X " to 1 mole of water.
Mass percent of " X " in the solution is
[11-Apr-2023 shift 2]

Options:

- A. 5%
- B. 20%
- C. 2%
- D. 10%

Answer: D

Solution:

$$\text{Solute (X)} = 2\text{g}$$

$$\text{Solvent (H}_2\text{O)} = 1 \text{ mole} = 18\text{g}$$

$$\text{Total mass} = 2 + 18 = 20\text{g}$$

$$\% \text{ mass of X} = \frac{2}{20} \times 100 = 10\%$$

Question81

The volume of hydrogen liberated at STP by treating 2.4g of magnesium with excess of hydrochloric acid is _____

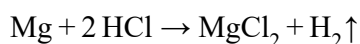
$\times 10^{-2} \text{L}$.

Given: Molar volume of gas is 22.4L at STP. Molar mass of magnesium is 24g mol^{-1}

[11-Apr-2023 shift 2]

Answer: 224

Solution:



$$w = 2.4 \text{g}$$

$$N = \frac{2.4}{24} = 0.1 \text{ mole}$$

$$1 \text{ mole of gas at STP} \Rightarrow 22.4 \text{ lit.}$$

$$\therefore 0.1 \text{ mole of gas} = 0.1 \times 22.4$$

$$= 2.24 \text{ lit.} = 224 \times 10^{-2} \text{ litre}$$

Question82

A metal chloride contains 55.0% of chlorine by weight . 100 mL vapours of the metal chloride at STP weigh 0.57g. The molecular formula of the metal chloride is (Given: Atomic mass of chlorine is 35.5u)

[12-Apr-2023 shift 1]

Options:

A. MCl

B. MCl_3

C. MCl_2

D. MCl_4

Answer: C

Solution:

Molecular weight of metal chloride

$$= \frac{0.57}{100} \times 22700$$

$$= 129.39$$

$$\text{weight of Cl} = 129.39 \times 0.55$$

$$= 71.1645$$

$$\therefore \text{Mole of ClCl} = \frac{71.1645}{35.5} \cong 2$$

Hence MCl_2

Question83

An organic compound gives 0.220g of CO_2 and 0.126g of H_2O on complete combustion. If the % of carbon is 24 then the % of hydrogen is _____ $\times 10^{-1}$. (Nearest integer)
[13-Apr-2023 shift 1]

Answer: 56

Solution:

$$\% \text{ of carbon} = \frac{\frac{0.220}{44} \times 12}{x} \times 100$$

(x = mass of organic compound)

$$24 = \frac{6}{x}$$

$$x = 0.25 \text{ gm}$$

$$\% \text{ of H} = \frac{\frac{0.126}{18} \times 2 \times 1}{0.25} \times 100$$

$$= 5.6 = 56 \times 10^{-1}$$

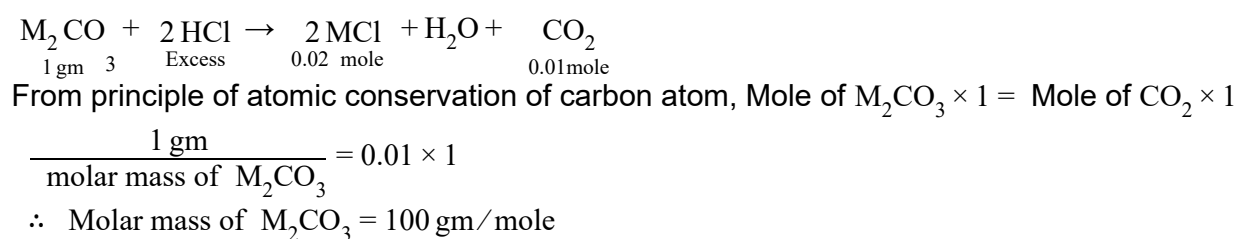
Question84

1g of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01 mol of CO_2 . The molar mass of M_2CO_3 is _____

gmol⁻¹. (Nearest integer)
[13-Apr-2023 shift 2]

Answer: 100

Solution:



Question85

Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is,
Given : Atomic masses of C, H and N are 12,1 and 14 amu respectively.

The molar mass of the compound A is 162gmol⁻¹.
[28-Jun-2022-Shift-2]

Options:

- A. $\text{C}_4\text{H}_6\text{N}_2$
- B. $\text{C}_2\text{H}_3\text{N}$
- C. $\text{C}_5\text{H}_7\text{N}$
- D. $\text{C}_{10}\text{H}_{14}\text{N}_2$

Answer: D

Solution:

Mole ratio of H, C and N

$$= \frac{8.7}{1} : \frac{74}{12} : \frac{17.3}{14}$$

$$= 8.7 : 6.167 : 1.23$$

$$= \frac{8.7}{1.23} : \frac{6.167}{1.23} : \frac{1.23}{1.23}$$

$$= 7 : 5 : 1$$

∴ Empirical formula = C_5H_7N

∴ Molecular formula = $(C_5H_7N)_n$

Given molecular mass = 162

Molecular mass of $(C_5H_7N)_n$

$$= (5 \times 12 + 7 \times 1 + 14) \times n$$

$$= (81) \times n$$

$$\therefore 81 \times n = 162$$

$$\Rightarrow n = 2$$

∴ Molecular formula = $C_{10}H_{14}N_2$

Question86

The complete combustion of 0.492g of an organic compound containing ' C ', ' H ' and ' O ' gives 0.793g of CO_2 and 0.442g of H_2O . The percentage of oxygen composition in the organic compound is _____ (nearest integer)
[28-Jun-2022-Shift-2]

Answer: 46

Solution:

Total organic compound = 0.492 gm

Produced CO_2 = 0.793 gm

$$\therefore \text{Moles of } CO_2 = \frac{0.793}{44}$$

$$\therefore \text{Moles of C atoms} = \frac{0.793}{44}$$

$$\therefore \text{Weight of C atoms} = \frac{0.793}{44} \times 12 = 0.216g$$

Produced H_2O = 0.442 gm

$$\therefore \text{Moles of } H_2O = \frac{0.442}{18}$$

$$\therefore \text{Moles of H atoms} = \frac{0.442}{18}$$

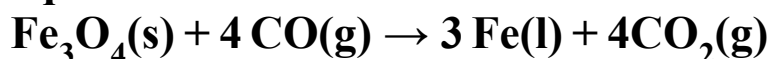
$$\therefore \text{Weight of H atoms} = \frac{0.442}{18} \times 2 = 0.05\text{g}$$

$$\begin{aligned}\therefore \text{Weight of O atoms} \\ &= 0.492 - (0.216 + 0.05) \\ &= 0.226 \text{ gm}\end{aligned}$$

$$\begin{aligned}\% \text{ by mass of oxygen in compound} \\ &= \frac{0.226}{0.492} \times 100 = 46\%\end{aligned}$$

Question87

Production of iron in blast furnace follows the following equation



**when 4.640 kg of Fe_3O_4 and 2.520 kg of CO are allowed to react then the amount of iron (in g) produced is:
[29-Jun-2022-Shift-1]**

Options:

A. 1400

B. 2200

C. 3360

D. 4200

Answer: C

Solution:

Given,

$$\text{Mass of Fe}_3\text{O}_4 = 4.640 \text{ kg} = 4640 \text{ gm}$$

$$\text{Molar mass of Fe}_3\text{O}_4 = 56 \times 3 + 16 \times 4 = 232\text{g}$$

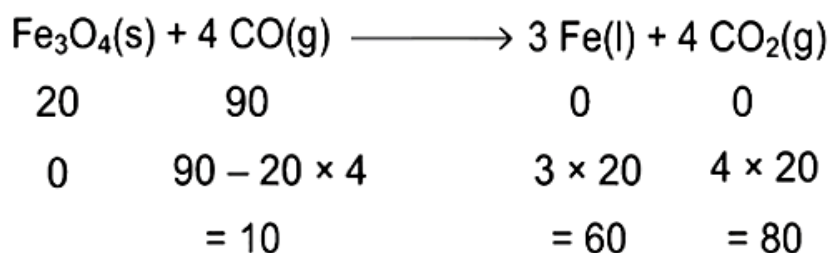
$$\therefore \text{Moles of Fe}_3\text{O}_4 = \frac{4640}{232} = 20$$

Also, given

$$\text{Mass of CO} = 2.520 \text{ kg} = 2520 \text{ gm}$$

$$\text{Molar mass of CO} = 12 + 16 = 28 \text{ gm}$$

$$\therefore \text{Molar of CO} = \frac{2520}{28} = 90$$



Here Fe_3O_4 is limiting reagent as to find limiting reagent, divide the given moles of reactants with their respective stoichiometric coefficient and reactant for which this ratio is minimum will be limiting reagent

$$\text{For } \text{Fe}_3\text{O}_4, \frac{\text{moles}}{\text{stoichiometric coefficient}} = \frac{20}{1}$$

$$\text{For } \text{CO}, \frac{\text{moles}}{\text{stoichiometric coefficient}} = \frac{90}{4} = 22.5$$

$\therefore \text{Fe}_3\text{O}_4$ is limiting reagent.

Now produced $\text{Fe} = 20 \times 3 = 60 \text{ mol}$

$\therefore \text{Weight of Fe} = 60 \times 56 = 3360 \text{g}$

Question88

Geraniol, a volatile organic compound, is a component of rose oil. The density of the vapour is 0.46gL^{-1} at 257°C and 100 mm Hg. The molar mass of geraniol is _____ gmol^{-1} . (Nearest Integer)

[. Given : $R = 0.082 \text{L atm K}^{-1} \text{mol}^{-1}$]

[29-Jun-2022-Shift-1]

Answer: 152

Solution:

From ideal gas equation we know

$$PV = nRT$$

$$\Rightarrow PV = \frac{W}{M}RT$$

$$\Rightarrow P = \frac{W}{V} \cdot \frac{RT}{M}$$

$$\Rightarrow P = d \cdot \frac{RT}{M} \left[\because d = \frac{W}{V} \right]$$

We know, 760 mm of Hg = 1 atm

$$\therefore 100 \text{ mm of Hg} = \frac{100}{760} \text{ atm}$$

$$\therefore \text{Pressure (P)} = \frac{100}{760} \text{ atm}$$

$$\text{Density (d)} = 0.46$$

$$R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

$$T = (257 + 273) \text{ K} = 530 \text{ K}$$

Putting the values in above equation, we get

$$\frac{100}{760} = \frac{0.46 \times 0.082 \times 530}{M}$$

$$\Rightarrow M = 152$$

Question89

Using the rules for significant figures, the correct answer for the expression $\frac{0.02858 \times 0.112}{0.5702}$ will be

[29-Jun-2022-Shift-2]

Options:

A. 0.005613

B. 0.00561

C. 0.0056

D. 0.006

Answer: B

Solution:

$$\frac{0.02858 \times 0.112}{0.5702} = 0.00561$$

Reported answer should not be more precise than least precise term (0.112 is the least precise term with three significant figures) in calculations, so there should be three significant figures in reported answer.

Question90

If a rocket runs on a fuel ($\text{C}_{15}\text{H}_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are :

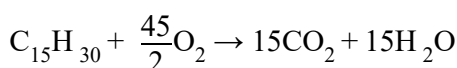
(Given : density of the fuel is 0.756g/mL)
[24-Jun-2022-Shift-1]

Options:

- A. 1188g and 1296g
- B. 2376g and 2592g
- C. 2592g and 2376g
- D. 3429g and 3142g

Answer: C

Solution:



Given, volume of fuel = 1L = 1000ml

And density of fuel = 0.756g/ml

We know,

$$d = \frac{w}{v}$$

$$\Rightarrow 0.756 = \frac{w}{1000}$$

$$\Rightarrow w = 756\text{gm}$$

$$\therefore \text{weight of fuel} = 756\text{gm}$$

$$\text{Molar mass of } \text{C}_{15}\text{H}_{30} = 15 \times 12 + 30 = 210$$

$$\therefore \text{Moles of } \text{C}_{15}\text{H}_{30} = \frac{756}{210}$$

From equation you can see,

$$1 \text{ mole of } \text{C}_{15}\text{H}_{30} \text{ react with } \frac{45}{2} \text{ mole of } \text{O}_2$$

$$\therefore \frac{756}{210} \text{ moles of } \text{C}_{15}\text{H}_{30} \text{ react with } \frac{45}{2} \times \frac{756}{210} \text{ moles of } \text{O}_2$$

$$\therefore \text{Moles of } \text{O}_2 \text{ required} = \frac{45}{2} \times \frac{756}{210}$$

$$\therefore \text{Mass of } \text{O}_2 \text{ required} = \frac{45}{2} \times \frac{756}{210} \times 32 = 2592\text{g}$$

Also,

From 1 mole of $\text{C}_{15}\text{H}_{30}$ 15 moles of CO_2 formed

$$\therefore \text{From } \frac{756}{210} \text{ moles of } \text{C}_{15}\text{H}_{30} 15 \times \frac{756}{210} \text{ moles of } \text{CO}_2 \text{ formed}$$

$$\therefore \text{Moles of } \text{CO}_2 \text{ formed} = 15 \times \frac{756}{210}$$

$$\therefore \text{Mass of } \text{CO}_2 \text{ formed} = 15 \times \frac{756}{210} \times 44 =$$

$$2376\text{g}$$

Question91

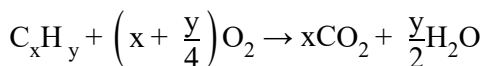
120g of an organic compound that contains only carbon and hydrogen gives 330g of CO₂ and 270g of water on complete combustion. The percentage of carbon and hydrogen, respectively are
[24-Jun-2022-Shift-2]

Options:

- A. 25 and 75
- B. 40 and 60
- C. 60 and 40
- D. 75 and 25

Answer: D

Solution:



From the reaction,

Produced CO₂ = x mol

and produced H₂O = $\frac{y}{2}$ mol

Given produced CO₂ = 330g

$$\therefore \text{moles of CO}_2 = \frac{330}{44} = \frac{30}{4} = x$$

Also given produced H₂O = 270 gm

$$\therefore \text{Moles of H}_2\text{O} = \frac{270}{18} = 15 = \frac{y}{2}$$

$$\Rightarrow y = 30$$

$$\therefore x : y = \frac{30}{4} : 30 = 1 : 4$$

Formula of the compound = (CH₄)_n

$\therefore$ Weight of C in (CH₄)_n = 12n

Weight of H in (CH₄)_n = 4n

$\therefore$ Weight ratio of C and H

$$= 12n : 4n$$

$$= 3 : 1$$

$$\therefore \% \text{ of C} = \frac{3}{4} \times 100 = 75$$

$$\text{and \% of H} = \frac{1}{4} \times 100 = 25$$

Question92

The number of N atoms in 681g of $C_7H_5N_3O_6$ is $x \times 10^{21}$. The value of x is ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)
(Nearest Integer)
[25-Jun-2022-Shift-1]

Answer: 5418

Solution:

Molar mass of $C_7H_5N_3O_6$

$$= 12 \times 7 + 5 + 14 \times 3 + 16 \times 6$$

$$= 227 \text{ gm}$$

Given mass of $C_7H_5N_3O_6 = 681 \text{ gm}$

$$\therefore \text{Moles of } C_7H_5N_3O_6 = \frac{681}{227} = 3$$

In one molecule of $C_7H_5N_3O_6$, 3N atoms present.

$\therefore$ In 1 mole $C_7H_5N_3O_6$, 3 moles of N atoms presents.

$\therefore$ In 3 moles of $C_7H_5N_3O_6$, $3 \times 3 = 9$ moles of N atoms presents.

We know, 1 mole of N atoms = 6.02×10^{23} N atoms.

$$\therefore 9 \text{ moles of N atoms} = 9 \times 6.02 \times 10^{23} = 54.18 \times 10^{23} = 5418 \times 10^{21}$$

Question93

A protein ' A ' contains 0.30% of glycine (molecular weight 75).
The minimum molar mass of the protein ' A ' is $\text{____} \times 10^3 \text{ gmol}^{-1}$
[nearest integer]
[25-Jun-2022-Shift-2]

Answer: 25

Solution:

Let, molar mass of protein A = x

Protein A contains 0.30% glycine

$$\therefore \frac{x \times 0.3}{100} = 75$$

$$\Rightarrow x = 25000 = 25 \times 10^3$$

Question94

A commercially sold conc. HCl is 35% HCl by mass. If the density of this commercial acid is 1.46g/mL, the molarity of this solution is:

(Atomic mass : Cl = 35.5 amu, H = 1 amu)

[26-Jun-2022-Shift-1]

Options:

A. 10.2M

B. 12.5M

C. 14.0M

D. 18.2M

Answer: C

Solution:

35% HCl by mass means in 100 gm HCl solution 35 gm HCl present.

Now, volume of 100 gm HCl solution

$$= \frac{100}{1.46} \text{ ml}$$

$$= \frac{100}{1.46} \text{ L}$$

$$\text{Moles of HCl} = \frac{35}{36.5}$$

$$\text{Now, molarity} = \frac{\text{moles of solute}}{\text{volume of solution (in L)}}$$

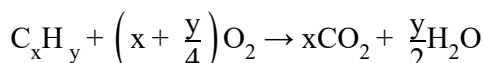
$$= \frac{\frac{35}{36.5}}{\frac{100}{1.46}} = 14$$

Question95

On complete combustion 0.30g of an organic compound gave 0.20g of carbon dioxide and 0.10g of water. The percentage of carbon in the given organic compound is ____ (Nearest integer)
[26-Jun-2022-Shift-1]

Answer: 18

Solution:



Given organic compound $C_xH_y = 0.3 \text{ gm}$

Produced carbon dioxide (CO_2) = 0.2 gm

Produced water (H_2O) = 0.1 gm

$$\text{Moles of } CO_2 = \frac{0.2}{44}$$

$$\therefore \text{Moles of C atom} = \frac{0.2}{44}$$

$$\therefore \text{Mass of C atom} = \frac{0.2}{44} \times 12 = 0.0545$$

$$\text{Moles of } H_2O = \frac{0.1}{18}$$

$$\therefore \text{Moles of H atoms} = \frac{0.1}{18} \times 2$$

$$\therefore \text{Mass of H atoms} = \frac{0.1 \times 2}{18} \times 1 = 0.0111$$

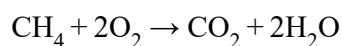
$$\therefore \% \text{ of C atom} = \frac{0.0545}{0.3} \times 100 = 18\%$$

Question96

The moles of methane required to produce 81g of water after complete combustion is ____ $\times 10^{-2}$ mol. [nearest integer]
[26-Jun-2022-Shift-2]

Answer: 225

Solution:



POAC on H atom

$$n_{\text{CH}_4} \times 4 = n_{\text{H}_2\text{O}} \times 2$$

$$n_{\text{CH}_4} = \frac{81}{18} \times 2 \times \frac{1}{4} = \frac{81}{36}$$

$$n_{\text{CH}_4} = 2.25$$

$$= 225 \times 10^{-2}$$

Nearest Integers = 225

Question97

116g of a substance upon dissociation reaction, yields 7.5g of hydrogen, 60g of oxygen and 48.5g of carbon. Given that the atomic masses of H, O and C are 1, 16 and 12, respectively. The data agrees with how many formulae of the following?

A. CH_3COOH ,

B. HCHO ,

C. CH_3OOCH_3 ,

D. CH_3CHO

[27-Jun-2022-Shift-2]

Answer: 2

Solution:

$$\% \text{H} = \frac{7.5}{116} \times 100 = 6.5$$

$$\% \text{O} = \frac{60}{116} \times 100 = 51.7$$

$$\% \text{C} = \frac{48.5}{116} \times 100 = 41.8$$

Relative atomicities = H $\Rightarrow$ 6.5

$$\text{O} \Rightarrow \frac{51.7}{16} = 3.25$$

$$\text{C} \Rightarrow \frac{41.8}{12} = 3.5$$

Empirically formula is approx.. CH_2O

(A) $\text{C}_2\text{H}_4\text{O}_2$ (B) CH_2O relate to this formula.

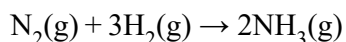
Question98

56.0 L of nitrogen gas is mixed with excess hydrogen gas and it is found that 20L of ammonia gas is produced. The volume of unused nitrogen gas is found to be _____ L.

[25-Jul-2022-Shift-2]

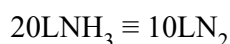
Answer: 46

Solution:



Since H_2 is in excess and 20L of ammonia gas is produced.

Hence, 2 moles $\text{NH}_3 \equiv 1$ mole N_2 ($v \propto n$)



Volume of N_2 left = 56 – 10

= 46L

Question99

Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)

(Given : Atomic mass of Mg is 24g mol^{-1} ; $N_A = 6.02 \times 10^{23}\text{ mol}^{-1}$)

[26-Jul-2022-Shift-1]

Answer: 24

Solution:

In 2L $\rightarrow$ 96 mg of Mg

$$\text{Number of atoms of Mg} = \frac{96 \times 10^{-3}}{24} \times N_A$$

$$= 4 \times 10^{-3} \times 6 \times 10^{23}$$

$$= 24 \times 10^{20}$$

$$x = 24$$

Question100

When 800 mL of 0.5M nitric acid is heated in a beaker, its volume is reduced to half and 11.5g of nitric acid is evaporated.

The molarity of the remaining nitric acid solution is $x \times 10^{-2}$ M. (Nearest integer)

(Molar mass of nitric acid is 63 g mol^{-1})

[26-Jul-2022-Shift-1]

Answer: 54

Solution:

$$m \text{ moles of HNO}_3 = 800 \times 0.5$$

$$\text{Moles of HNO}_3 = 400 \times 10^{-3} = 0.4 \text{ moles}$$

$$\text{Weight of HNO}_3 = 0.4 \times 63 \text{ g} = 25.2 \text{ g}$$

$$\text{Remaining acid} = 25.2 - 11.5 = 13.7 \text{ g}$$

$$M = \frac{13.7 \times 1000}{400 \times 63}$$

$$= \frac{137}{252} = 0.54$$

$$= 54 \times 10^{-2}$$

Question101

Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3g of hemoglobin is (Given: Atomic mass of Fe is

56u, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.)

[26-Jul-2022-Shift-2]

Options:

- A. 1.21×10^5
- B. 12.0×10^{16}
- C. 1.21×10^{20}
- D. 3.4×10^{22}

Answer: C

Solution:

According to the question,

100g of hemoglobin contains 0.34g of iron

3.3g of hemoglobin contains $\frac{0.34}{100} \times 3.3$ g of iron

$$\text{moles of Fe} = \frac{0.34 \times 3.3}{100 \times 56} = \frac{N}{N_A}$$

$$N = \frac{0.34 \times 3.3 \times 6.022 \times 10^{23}}{100 \times 56}$$
$$= 1.21 \times 10^{20}$$

Question102

250g solution of D-glucose in water contains 10.8% of carbon by weight. The molality of the solution is nearest to

(Given: Atomic Weights are, H, 1u; C, 12u; O, 16u)

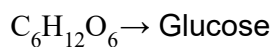
[27-Jul-2022-Shift-1]

Options:

- A. 1.03
- B. 2.06
- C. 3.09
- D. 5.40

Answer: B

Solution:



$$\text{We know: } \frac{\text{mass of C}}{\text{mass of glucose}} = \frac{72}{180}$$

$$\text{Given: } \%C = 10.8 = \frac{\text{mass of C}}{\text{mass of solution}} \times 100$$

$$\frac{10.8 \times 250}{100} = \text{mass of C} \Rightarrow \text{Mass of C} = 27 \text{ gm}$$

$$\therefore \text{mass of glucose} = 67.5 \text{ gm}$$

$$\therefore \text{moles of glucose} = 0.375 \text{ moles}$$

$$\text{Mass of solvent} = 250 - 67.5 \text{ gm} = 182.5 \text{ gm}$$

$$\therefore \text{Molality} = \frac{0.375}{0.1825} = 2.055 \approx 2.06$$

Question103

The normality of H_2SO_4 in the solution obtained on mixing 100 mL of 0.1M H_2SO_4 with 50 mL of 0.1M NaOH _____ is $\times 10^{-1}\text{N}$. (Nearest Integer)
[27-Jul-2022-Shift-2]

Answer: 1

Solution:

$$\text{No. of equivalents of } \text{H}_2\text{SO}_4 = 100 \times 0.1 \times 2 = 20$$

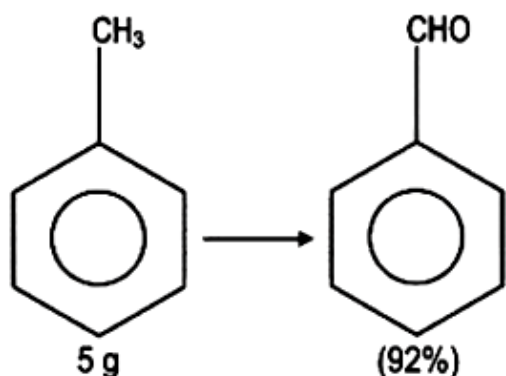
$$\text{No. of equivalents of NaOH} = 50 \times 0.1 = 5$$

$$\text{No. of equivalents of } \text{H}_2\text{SO}_4 \text{ left} = 20 - 5 = 15$$

$$\Rightarrow 150 \times x = 15$$

$$x = \frac{1}{10} = 0.1\text{N} = 1 \times 10^{-1}\text{N}$$

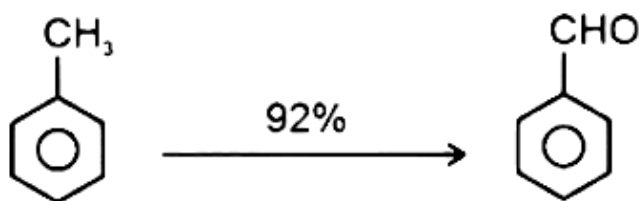
Question104



In the above reaction, 5g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is _____ $\times 10^{-2}$ g. (Nearest integer)
 [27-Jul-2022-Shift-2]

Answer: 530

Solution:



$$\text{Moles} = \frac{5}{92}$$

$$\text{Moles of benzaldehyde produced} = \frac{5}{92} \times 0.92 = 0.05$$

$\therefore$ Mass of benzaldehyde formed

$$= 0.05 \times 106$$

$$= 5.3\text{g}$$

$$= 530 \times 10^{-2}$$

Question105

In the given reaction,

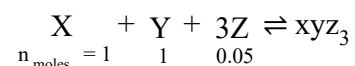


if one mole of each of X and Y with 0.05 mol of Z gives compound X Y Z_3 - (Given : Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively.) The yield of X Y Z_3 is

_____g. (Nearest integer)
[28-Jul-2022-Shift-1]

Answer: 2

Solution:



Limiting reagent is Z = $\frac{0.05}{3} = .016$

3 moles of Z $\rightarrow$ 1 mole of XYZ₃

0.05 mole of Z $\rightarrow \frac{1}{3} \times 0.05$ mole of XYZ₃

M.wt. of XYZ₃ = 10 + 20 + 90
= 120 amu

Wt. of XYZ₃ = $\frac{.05}{3} \times 120$
= 2g

Question106

On complete combustion of 0.492g of an organic compound containing C, H and O, 0.7938g of CO₂ and 0.4428g of H₂O was produced. The % composition of oxygen in the compound is

_____.
[28-Jul-2022-Shift-1]

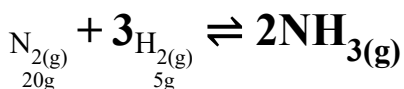
Answer: 46

Solution:

$$\begin{aligned} \% \text{ of H} &= \frac{2}{18} \times \frac{\text{wt. of H}_2\text{O}}{\text{wt. of organic compound}} \times 100 \\ &= \frac{2}{18} \times \frac{0.4428}{0.492} \times 100 \\ &= 0.11 \times 0.9 \times 100 \end{aligned}$$

$$\begin{aligned}
 &= 0.099 \times 100 = 9.9 \\
 \% \text{ of C} &= \frac{12}{44} \times \frac{0.7938}{0.492} \times 100 \\
 &= 0.27 \times 1.61 \times 100 \\
 &= 43.47 \\
 \% \text{ Oxygen} &= 100 - (43.47 + 9.9) \\
 &= 100 - 53.37 \approx 46
 \end{aligned}$$

Question107



Consider the above reaction, the limiting reagent of the reaction and number of moles of NH_3 formed respectively are :

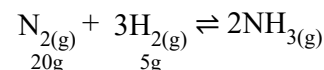
[29-Jul-2022-Shift-1]

Options:

- A. H_2 , 1.42 moles
- B. H_2 , 0.71 moles
- C. N_2 , 1.42 moles
- D. N_2 , 0.71 moles

Answer: C

Solution:



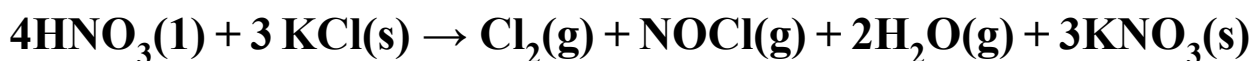
Ideally 28g N_2 reacts with 6g H_2 limiting reagent is N_2

$\therefore$ Amount of NH_3 formed on reacting 20g N is,

$$\begin{aligned}
 &= \frac{34 \times 20}{28} = 24.28\text{g} \\
 &= 1.42 \text{ moles}
 \end{aligned}$$

Question108

Consider the reaction



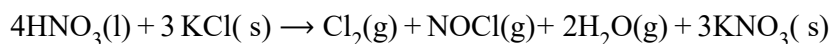
The amount of HNO_3 required to produce 110.0g of KNO_3 is
(Given: Atomic masses of H, O, N and K are 1, 16, 14 and 39 ,
respectively.)
[29-Jul-2022-Shift-2]

Options:

- A. 32.2g
- B. 69.4g
- C. 91.5g
- D. 162.5g

Answer: C

Solution:



$$\therefore 110\text{g of KNO}_3 \Rightarrow \text{moles of KNO}_3 = \frac{110}{101}$$

$$= 1.089 \text{ mol}$$

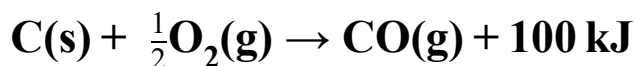
As, 4 mole of HNO_3 produces 3 mol of KNO_3 .

Hence, the moles of HNO_3 required to produce

$$1.089 \text{ moles of KNO}_3 = \frac{4}{3} \times 1.089 = 1.452 \text{ mol}$$

Hence, mass of HNO_3 required is $1.452 \times 63 = 91.5\text{g}$

Question109



When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into ' CO ' and the remaining is converted into ' CO_2 '. The heat generated when 0.6 kg of coal is burnt is _____.

[29-Jul-2022-Shift-2]

Options:

- A. 1600 kJ

B. 3200 kJ

C. 4400 kJ

D. 6600 kJ

Answer: D

Solution:

Weight of coal = 0.6 kg = 600 gm

∴ 60% of it is carbon

So weight of carbon = $600 \times \frac{60}{100} = 360\text{g}$

∴ moles of carbon = $\frac{360}{12} = 30$ moles

∴ Heat generated = $12 \times 400 + 18 \times 100 = 6600\text{ kJ}$

Question110

A 1.84 mg sample of polyhydric alcoholic compound 'X' of molar mass 92.0g/mol gave 1.344 mL of H₂ gas at STP. The number of alcoholic hydrogens present in compound 'X' is

[29-Jul-2022-Shift-2]

Answer: 3

Solution:

Volume of H₂ gas = 1.344 mL

Mole of H₂ gas = $\frac{1.344}{22400} = 6 \times 10^{-5}$

No of H atoms per molecule of H₂ = 2.

No. of moles of organic compound = $\frac{1.84 \times 10^{-3}}{92} = 2 \times 10^{-5}$

No. of -OH (hydroxyl group in one molecule) = $\frac{6 \times 10^{-5}}{2 \times 10^{-5}} = 3$

Question111

The number of significant figures in 50000.020×10^{-3} is
[26 Feb 2021 Shift 1]

Answer: 7

Solution:

Non-zero digits are always significant. Any zeros between two significant digits are significant.

$\therefore$ Zero's between 5 and 2 are all significant.

(Number of significant figures = 7)

Question112

The NaNO_3 weighed out to make 50mL of an aqueous solution containing 70.0mgNa^+ per mL is g. (Rounded off to the nearest integer) [Given : Atomic weight in gmol^{-1} , $\text{Na} : 23$; $\text{N} : 14$; $\text{O} : 16$].
[26 Feb 2021 Shift 2]

Answer: 13

Solution:

70mgNa^+ is present in 1mL of NaNO_3 solution.

$\therefore$ 50mL of NaNO_3 will contain = $70 \times 50\text{mg}$

$$= \frac{70 \times 50}{1000} = 3.5\text{gm}$$

Moles of Na^+ in solution = Moles of NaNO_3 in solution

$[\because \text{NaNO}_3 \rightarrow \text{Na}^+ + \text{NO}_3^-]$

$$= \frac{3.5}{23}\text{mol} \quad [\because \text{Molar mass of } \text{Na}^+ = 23\text{gmol}^{-1}]$$

Mass of $\text{NaNO}_3 = \text{mole} \times \text{molar mass}$

[$\because$ molar mass of $\text{NaNO}_3 = 85 \text{ g mol}^{-1}$]

$$= \frac{3.5}{23} \times 85 = 12.934 \text{ g} \sim 13 \text{ g}$$

Question 113

Complete combustion of 1.80g of an oxygen containing compound ($\text{C}_x\text{H}_y\text{O}_z$) gave 2.64g of CO_2 and 1.08g of H_2O . The percentage of oxygen in the organic compound is
[25 Feb 2021 Shift 1]

Options:

A. 50.33

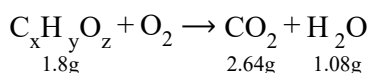
B. 53.33

C. 63.53

D. 51.63

Answer: B

Solution:



$$n_{\text{C}} = \frac{\text{CO}}{(\text{Moles})_2} = \frac{2.64 \text{ g (Given mass)}}{44 \text{ (Molecular mass)}} = 0.06$$

$$n_{\text{H}} = 2 \times n_{\text{H}_2\text{O}} = \frac{1.08}{18} \times 2 = 0.12$$

Weight of oxygen in $\text{C}_x\text{H}_y\text{O}_z$

$$= 1.80 - 12 \times \frac{2.64}{44} - \frac{1.08}{18} \times 2$$

$$= 1.80 - 0.72 - 0.12 = 0.96 \text{ g}$$

$$\% \text{ of oxygen by weight} = \frac{0.96}{1.80} \times 100 = 53.33\%$$

Question 114

The formula of a gaseous hydrocarbon, which requires 6 times of its own volume of O_2 for complete oxidation and produces 4

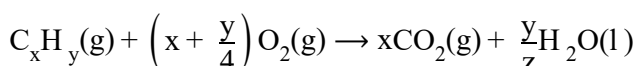
times its own volume of CO_2 is C_xH_y . The value of y is

[24 Feb 2021 Shift 2]

Answer: 8

Solution:

Combustion reaction:



Suppose, volume of C_xH_y is V and volume of O_2 is 6 times greater than $\text{C}_x\text{H}_y = 6V$

then volume of $x\text{CO}_2 \Rightarrow V_x = 4V$

$$x = 4$$

$$\text{Since, } V_{\text{O}_2} = 6 \times V_{\text{C}_x\text{H}_y}$$

$$V \left(x + \frac{y}{4}\right) = 6V$$

$$\left(x + \frac{y}{4}\right) = 6$$

Put value of $x = 4$ in Eq. (i) We get,

$$4 + \frac{y}{4} = 6 \Rightarrow y = 8$$

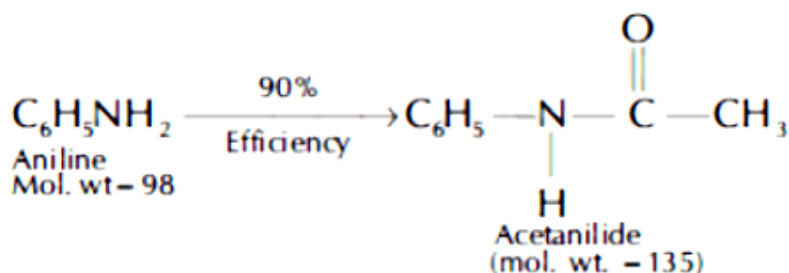
Question115

1.86g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is $\times 10^{-2}$.

[24 Feb 2021 Shift 2]

Answer: 243

Solution:



Given, weight = 1.86g

Here, 1 mole of aniline gives 1 mole of acetanilide

$\therefore$ mole of aniline = mole of acetanilide

$$\Rightarrow \frac{1.86}{98} = \frac{W_{\text{Acetanilide}}}{135}$$

$$W_{\text{Acetanilide}} = \frac{1.86 \times 135}{98} \text{g} = 2.70 \text{g}$$

But efficiency of reaction is 90% only.

Hence, mass of acetanilide produced

$$= 2.70 \times \frac{90}{100} \text{g} = 2.43 \text{g} = 243 \times 10^{-2} \text{g}$$

$$x = 243$$

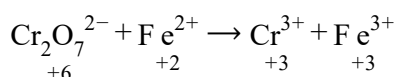
Question 116

15mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20mL of 0.03M aqueous $\text{Cr}_2\text{O}_7^{2-}$. The molarity of the Fe^{2+} solution is $\times 10^{-2}\text{M}$ (Round off to the nearest integer).

[17 Mar 2021 Shift 1]

Answer: 24

Solution:



Dichromate ($\text{Cr}_2\text{O}_7^{2-}$) oxidised Fe^{2+} to Fe^{3+} and itself get reduced to Cr^{3+} .

Valency factor of $\text{Cr}_2\text{O}_7^{2-}$ = 6 as Cr gets reduced from +6 to +3

From law of equivalence,

Milliequivalent of $\text{Cr}_2\text{O}_7^{2-}$ = Milliequivalent of Fe^{2+}

$$M_1 V_1 n_1 = M_2 V_2 n_2$$

$$(M_1 \times V_1) \times 6 = (M_2 \times V_2) \times 1$$

$$(0.03 \times 20) \times 6 = (M_2 \times 15) \times 1$$

$$\Rightarrow M_2 = 0.24 \text{ molar}$$

$$= 24 \times 10^{-2} \text{ molar}$$

Question117

The mole fraction of a solute in a 100 molal aqueous solution $\times 10^{-2}$ (Round off to the nearest integer).

[Given, atomic masses H : 1.0u, O : 16.0u]

[17 Mar 2021 Shift 1]

Answer: 64

Solution:

Given, molality = 100

$\Rightarrow$ 100 moles of solute in 1kg of solvent

Mole fraction of solute (χ_{solute}) = $\frac{\text{Moles of solute}}{\text{Total moles}}$

$$= \frac{n_{\text{Solute}}}{n_{\text{Solute}} + n_{\text{Solvent}}}$$

$$n_{\text{solvent}} = \frac{1000}{18} = 55.5$$

$$\chi_{\text{solute}} = \frac{100}{100 + 55.5} = \frac{100}{155.5} = 0.643$$

$$\chi_{\text{solute}} = 64.3 \times 10^{-2} \Rightarrow 64 \times 10^{-2}$$

Question118

10.0mL of Na_2CO_3 solution is titrated against 0.2M HCl solution. The following titre values were obtained in 5 readings. 4.8mL, 4.9mL, 5.0mL, 5.0mL and 5.0mL based on these readings and convention of titrimetric estimation of concentration of Na_2CO_3 solution is mM (Round off to the nearest integer).

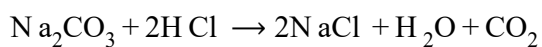
[18 Mar 2021 Shift 2]

Answer: 50

Solution:

10.0mL of Na_2CO_3 solution is titrated against 0.2M HCl

Volume of Na_2CO_3 solution (mL)	Volume of 0.2M HCl solution (mL)	Mean volume of HCl (ml)
10	4.8	
10	4.9	
10	5.0	5.0
10	5.0	
10	5.0	



10mL 0.2M_{HCl}

$M_{\text{Na}_2\text{CO}_3} = ?$ 5mL

M_{eq} of $\text{Na}_2\text{CO}_3 = M_{\text{eq}}$ of HCl

$$M_{\text{Na}_2\text{CO}_3} \times 10 \times 2 = 0.2 \times 5 \times 1$$

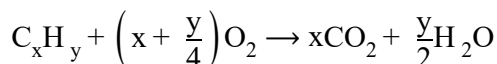
$$M_{\text{Na}_2\text{CO}_3} = 5 \times 10^{-2} \text{M} = 50 \times 10^{-3} \text{M} = 50 \text{mM}$$

Question119

Complete combustion of 750g of an organic compound provides 420g of CO_2 and 210g of H_2O . The percentage composition of carbon and hydrogen in organic compound is 15.3 and respectively (Round off to the nearest integer).
[16 Mar 2021 Shift 1]

Answer: 3

Solution:



Any hydrocarbon, on combustion gives CO_2 and H_2O . This is Liebig's method for estimation of 'C' and 'H' percentage.

Mass of water formed = 210g

18g of H_2O contains = 2g of hydrogen

210g of H_2O contains = $\left(\frac{2}{18} \times 210 = \frac{70}{3}\right)$ g of hydrogen.

Given, mass of organic compound = 750g

Percentage of hydrogen

$$= \frac{\text{Mass of hydrogen}}{\text{Mass of organic compound}} \times 100$$

$$= \frac{70}{3 \times 750} \times 100$$

$$= 3.11\%$$

Nearest integer = 3

Question120

10.0 mL of 0.05 M $KMnO_4$ solution was consumed in a titration with 10.0 mL of given oxalic acid dihydrate solution. The strength of given oxalic acid solution is $\times 10^{-2}$ g/L. (Round off to the nearest integer)
[27 Jul 2021 Shift 2]

Answer: 1575

Solution:



$$\text{or, } \frac{10 \times 0.05}{1000} \times 5 = \frac{10 \times M}{1000} \times 2$$

$$\therefore \text{Conc. of oxalic acid solution} = 0.125M$$

$$= 0.125 \times 126 \text{ g/L} = 15.75 \text{ g/L}$$

$$= 1575 \times 10^{-2} \text{ g/L}$$

Question121

When 10mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15mL of 0.02M solution of $\text{K}_2\text{Cr}_2\text{O}_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}\text{M}$. The value of x is _____.(Nearest integer)
[25 Jul 2021 Shift 1]

Answer: 18

Solution:

$$\begin{aligned}\text{milli-equivalents of Fe}^{2+} &= \text{milli-equivalents of K}_2\text{Cr}_2\text{O}_7 \\ M \times 10 \times 1 &= 0.02 \times 15 \times 6 \\ M &= 0.18 = 18 \times 10^{-2}\text{M}\end{aligned}$$

Question122

If the concentration of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in blood is 0.72gL^{-1} , the molarity of glucose in blood is _____ $\times 10^{-3}\text{M}$. (Nearest integer)
[Given: Atomic mass of C = 12, H = 1, O = 16u]
[22 Jul 2021 Shift 2]

Answer: 4

Solution:

$$[\text{Glucose}] = \frac{C(\text{gm/l})}{M(\text{gm/mol})} = \frac{0.72}{180} = 4 \times 10^{-3}\text{M}$$

Question123

The number of significant figures in 0.00340 is _____.
[25 Jul 2021 Shift 2]

Answer: 3

Solution:

Number of significant figures = 3

Question124

4g equimolar mixture of NaOH and Na_2CO_3 contains xg of NaOH and y g of Na_2CO_3 . The value of x is _____ g. (Nearest integer)
[20 Jul 2021 Shift 2]

Answer: 1

Solution:

Total mass = 4g

Now

$$\text{NaOH} : a \text{ mol} \quad W_{\text{NaOH}} + W_{\text{Na}_2\text{CO}_3} = 4$$

$$\text{Na}_2\text{CO}_3 : 'a' \text{ mol} \quad \Rightarrow 40a + 106a = 4$$

$$\Rightarrow a = \frac{4}{146} \text{ mol}$$

$$\Rightarrow \text{therefore mass of NaOH is : } \frac{4}{146} \times 40 \text{ g}$$

$$= 1.095 \approx 1$$

Question125

250mL of 0.5M NaOH was added to 500mL of 1M HCl. The number of unreacted HCl molecules in the solution after complete reaction is _____ $\times 10^{21}$.

(Nearest integer)

($N_A = 6.022 \times 10^{23}$)

[20 Jul 2021 Shift 1]

Answer: 226

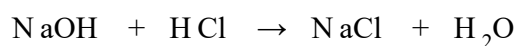
Solution:

We know that no. of moles = $V_{\text{litre}} \times \text{Molarity}$ & No. of millimoles = $V_{\text{ml}} \times \text{Molarity}$

so millimoles of NaOH = $250 \times 0.5 = 125$

Millimoles of HCl = $500 \times 1 = 500$

Now reaction is



t = 0	125	500	0	0
-------	-----	-----	---	---

t = t	0	375	125	125
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so millimoles of HCl left = 375

Moles of HCl = 375×10^{-3}

No. of HCl molecules = $6.022 \times 10^{23} \times 375 \times 10^{-3}$

= 225.8×10^{21}

$\approx 226 \times 10^{21} = 226$

Question126

Sodium oxide reacts with water to produce sodium hydroxide.

20.0g of sodium oxide is dissolved in 500 mL of water.

Neglecting the change in volume, the concentration of the resulting NaOH solution is $\times 10^{-1}$ M.(Nearest integer)

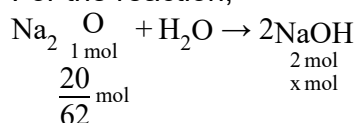
[Atomic mass: Na = 23.0, O = 16.0, H = 1.0]

[31 Aug 2021 Shift 2]

Answer: 13

Solution:

For the reaction,



$$\text{Moles of NaOH formed} \Rightarrow x = \frac{20}{62} \times 2$$

$$\text{Concentration of NaOH} = \frac{\text{Moles of NaOH}}{\text{Volume of solution (in litre)}}$$

$$= \frac{\frac{20}{62} \times 2}{\frac{500}{1000}} = 1.29\text{M} = 13 \times 10^{-1}\text{M}$$

Question 127

100g of propane is completely reacted with 1000g of oxygen. The mole fraction of carbon dioxide in the resulting mixture is

$x \times 10^{-2}$. The value of x is (Nearest integer)

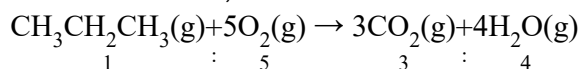
[Atomic weight : H = 1.008, C = 12.00, O = 16.00]

[27 Aug 2021 Shift 2]

Answer: 19

Solution:

For the reaction,



1 mole of propane reacts completely with 5 moles of oxygen to form 3 moles of carbon dioxide and 4 moles of steam.

44 g of propane = 1 mole of propane

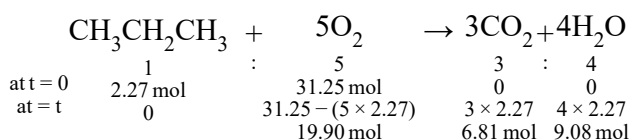
$$100\text{g of propane} = \frac{1}{44} \times 100 = 2.27 \text{ mol}$$

32g of O_2 = 1 mole of O_2

$$1000\text{g of } \text{O}_2 = \frac{1}{32} \times 1000 = 31.25 \text{ moles}$$

$\therefore$ 2.27 moles of propane requires $5 \times 2.27 = 11.35$ moles of O_2

moles of CO_2 formed = $3 \times 2.7 = 6.681$ mol of CO_2



When reaction is completed 19.90 moles of O_2 ,

6.81 moles of CO_2 and 9.08 moles of steam are left in the flask.

$$\text{Mole fraction of } \text{CO}_2 = \frac{\text{Moles of } \text{CO}_2(\text{g})}{\text{Total number of moles}}$$

$$= \frac{6.81}{19.90 + 6.81 + 9.08} = 0.19$$

$$x \times 10^{-2} = 0.19$$

$$x = 19$$

Question128

A chloro compound A, (i) Forms aldehydes on ozonolysis followed by the hydrolysis.

(ii) When vaporised completely, 1.53g of A gives 448 mL of vapour at STP.

The number of carbon atoms in a molecule of compound A is

..... .

[26 Aug 2021 Shift 2]

Answer: 3

Solution:

Given, 448 mL of A gives 1.53 g of vapours.

$$\therefore 22400 \text{ mL of A gives } = \frac{1.53}{448} \times 22400 = 76.50 \text{ g of A}$$

$\therefore$ Molecular mass is 76.5

$\therefore$ The possible compound is $\text{CH}_3 - \text{CH} = \text{CH} - \text{Cl}$

On ozonolysis followed by hydrolysis, it gives aldehyde as follows.

The compound (A) $\text{CH}_3 - \text{CH} = \text{CH} - \text{Cl}$ (chloropropene) has 3 carbon atoms.

Question129

The number of atoms in 8g of sodium is $x \times 10^{23}$. The value of x is (Nearest integer) [Given : $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ Atomic

mass of Na = 23.0u]
[1 Sep 2021 Shift 2]

Answer: 2

Solution:

Given, mass of Na = 8g

Molar mass of Na = 23g mol^{-1}

$$\frac{\text{Weight of sodium atom}}{\text{Molecular mass of sodium atom}} = \frac{\text{Number of atoms}}{\text{Avogadro's number}}$$

$$\frac{8\text{g}}{23\text{g}} = \frac{\text{Number of atoms}}{6.022 \times 10^{23}}$$

$$\text{Number of atoms} = \frac{8 \times 6.022}{23} \times 10^{23}$$

$$\text{Number of atoms} = 2.09 \times 10^{23}$$

$$x \approx 2$$

Hence, answer is 2.

Question130

4.5g of compound A(M W = 90) was used to make 250mL of its aqueous solution. The molarity of the solution is M is $x \times 10^{-1}$. The value of x is (Rounded off to the nearest integer) [2020]

Answer: 2

Solution:

$$M = \frac{4.5/90}{250/1000} = 0.2 = 2 \times 10^{-1}$$

Question131

The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is _____. (Molecular Weight of $\text{HNO}_3 = 63$)

[NV, Jan. 09, 2020(I)]

Answer: 14

Solution:

Mass percent of $\text{HNO}_3 = 63$

Thus, 100g of nitric acid solution contains 63g of nitric acid by mass.

$$\text{No. of moles} = \frac{63\text{g}}{63\text{g mol}^{-1}} = 1$$

$$\text{Volume of } 100\text{g of nitric acid solution} = \frac{\text{Mass}}{\text{Density}} = \frac{100\text{g}}{1.4\text{g/mL}} = 71.4\text{mL}$$

$$\text{Molarity} = \frac{\text{No. of moles}}{\text{volume(mL)}} \times 1000 = \frac{1}{71.4} \times 1000 = 14\text{M}$$

Question132

Amongst the following statements, that which was not proposed by Dalton was:

[Jan. 07,2020 (I)]

Options:

A. Chemical reactions involve reorganization of atoms. These are neither created nor destroyed in a chemical reaction.

B. All the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.

C. When gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.

D. Matter consists of indivisible atoms.

Answer: C

Solution:

Except (3) all postulates was given by the Dalton.

Question133

10.30mg of O_2 is dissolved into a liter of sea water of density 1.03g/mL. The concentration of O_2 in ppm is _____.

[NV, Jan. 09, 2020 (II)]

Answer: 10

Solution:

$$\text{ppm} = \frac{10.3 \times 10^{-3}}{1.03 \times 1000} \times 10^6 = 10$$

Question134

Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100kg of wheat is _____.

Atomic weight: Fe = 55.85; S = 32.00; O = 16.00

[NV, Jan. 08 , 2020 (I)]

Answer: 4.96

Solution:

$$10\text{ppm} = \frac{\text{Mass of Fe (in g)}}{100 \times 1000} \times 10^6$$

$$\Rightarrow \text{Mass of Fe} = 1\text{g}$$

$$\text{Molar mass of FeSO}_4 \cdot 7\text{H}_2\text{O} = 278$$

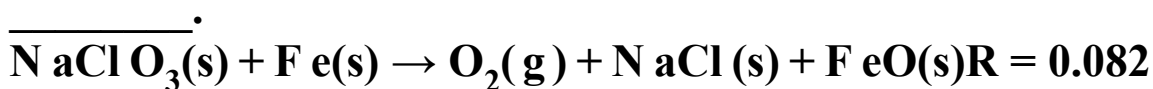
56g of iron present in 1mole of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

1g of Fe present $\frac{278}{56}$ g in of salt = 4.96g

Question135

NaClO_3 is used, even in spacecrafts, to produce O_2 . The daily consumption of pure O_2 by a person is 492L at 1 atm, 300K .

How much amount of NaClO_3 , in grams, is required to produce O_2 for the daily consumption of a person at 1atm, 300K ?

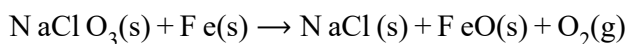


$\text{Latm mol}^{-1}\text{K}^{-1}$

[NV, Jan. 08,2020 (II)]

Answer: 2130

Solution:



Moles of NaClO_3 = Moles of O_2

$$\text{Moles of } \text{O}_2 = \frac{PV}{RT} = \frac{1 \times 492}{0.082 \times 300} = 20\text{mol}$$

$$\text{Mass of } \text{NaClO}_3 = 20 \times 106.5 = 2130\text{g}$$

Question136

The ammonia (NH_3) released on quantitative reaction of 0.6g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by:

[Jan. 07, 2020 (II)]

Options:

A. 200mL of 0.4N HCl

- B. 200mL of 0.2N HCl
- C. 100mL of 0.2N HCl
- D. 100mL of 0.1N HCl

Answer: C

Solution:

1 mol of urea = 2 mol of NH_3

60g of urea = 2mol of NH_3

0.6g of urea = $\frac{2}{60} \times 0.6\text{mol} = 0.02\text{mol}$ of NH_3

mol of NH_3 = mol of HCl

$\therefore$ mol of HCl = 0.02mol

$\Rightarrow$ Normality of HCl = 0.2N

Volume of HCl = 100mL

Question137

A solution of two components containing n_1 moles of the 1st component and n_2 moles of the 2nd component is prepared. M_1 and M_2 are the molecular weights of component 1 and 2 respectively. If d is the density of the solution in gmL^{-1} , C_2 is the molarity and x_2 is the mole fraction of the 2nd component, then C_2 can be expressed as:

[Sep. 06,2020(I)]

Options:

A. $C_2 = \frac{1000x_2}{M_1 + x_2(M_2 - M_1)}$

B. $C_2 = \frac{d x_2}{M_2 + x_2(M_2 - M_1)}$

C. $C_2 = \frac{1000d x_2}{M_1 + x_2(M_2 - M_1)}$

D. $C_2 = \frac{d x_1}{M_2 + x_2(M_2 - M_1)}$

Answer: C

Solution:

	1st component	2nd component
mole	n_1	n_2
m.w	M_1	M_2
mass	$n_1 M_1$	$n_2 M_2$

[$\because$ mass = mole $\times$ m.w.]

Mass of solution = $n_1 M_1 + n_2 M_2$

Mole fraction of the 2nd component (x_2) = $\frac{n_2}{n_1 + n_2}$

$$\Rightarrow n_1 = \frac{n_2(1 - x_2)}{x_2}$$

Mass of solution = $n_1 M_1 + n_2 M_2$

$$= \frac{n_2 M_1(1 - x_2)}{x_2} + n_2 M_2$$

$$= \frac{n_2}{x_2} [M_2 x_2 - x_2 M_1 + M_1]$$

$$\text{Volume of solution} = \frac{n_2 [M_2 x_2 - x_2 M_1 + M_1]}{1000 d x_2}$$

$$C_2 = \frac{1000 n_2 d x_2}{n_2 [M_2 x_2 - x_2 M_1 + M_1]}$$

$$\Rightarrow C_2 = \frac{1000 d x_2}{M_1 + x_2 (M_2 - M_1)}$$

Question138

The average molar mass of chlorine is 35.5 gmol^{-1} . The ratio of ^{35}Cl to ^{37}Cl in naturally occurring chlorine is close to:
[Sep. 06, 2020 (II)]

Options:

A. 4 : 1

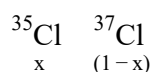
B. 3 : 1

C. 2 : 1

D. 1 : 1

Answer: B

Solution:



$$\begin{aligned} M_{\text{avg}} &= 35 \times x + 37(1-x) = 35.5 \\ &= 35x + 37(1-x) = 35.5 \\ \Rightarrow 2x &= 1.5 \end{aligned}$$

$$x = \frac{3}{4}$$

$$\text{So, ratio of } {}^{35}\text{Cl} : {}^{37}\text{Cl} = \frac{3}{4} / \frac{1}{4} = 3 : 1.$$

Question139

The ratio of the mass percentages of 'C&H' and 'C&O' of a saturated acyclic organic compound 'X' are 4 : 1 and 3 : 4 respectively. Then, the moles of oxygen gas required for complete combustion of two moles of organic compound 'X' is

_____.
[NV, Sep. 02,2020 (II)]

Answer: 5

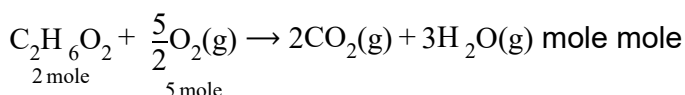
Solution:

Mass ratio of C : H is 4 : 1 $\Rightarrow$ 12 : 3 and
C : O is 3 : 4 $\Rightarrow$ 12 : 16
So,

	mass	mole	mole ratio
C	12	1	1
H	3	3	3
O	16	1	1

Empirical formula $\Rightarrow \text{CH}_3\text{O}$

As compound is saturated acyclic, so molecular formula is $\text{C}_2\text{H}_6\text{O}_2$



$\therefore$ Number of moles of O_2 required to oxidise 2 moles of (X) = 5.

Question 140

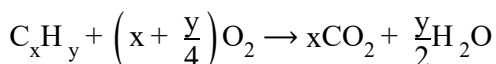
The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is

 .
[NV, Sep. 05, 2020(I)]

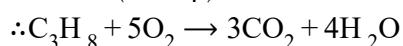
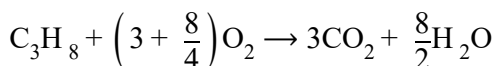
Answer: 18

Solution:

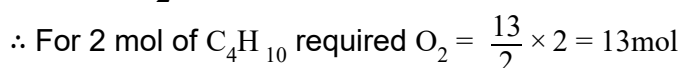
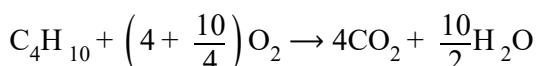
Complete combustion of hydrocarbons can be represented by the following reaction.



For propane combustion reaction is



Similarly, for butane is



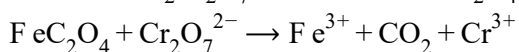
Question141

The volume, in mL, of 0.02M $K_2Cr_2O_7$ solution required to react with 0.288g of ferrous oxalate in acidic medium is _____. (Molar mass of Fe = 56gmol^{-1})
[NV, Sep. 05,2020(II)]

Answer: 50

Solution:

M. eq. of $K_2Cr_2O_7$ = M. eq. of FeC_2O_4



$$V \times 0.02 \times 6 = \frac{0.288}{144} \times 3 \times 1000$$

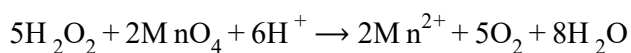
$$V = 50\text{mL}$$

Question142

A 20.0mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316g of $KMnO_4$ in acid solution. The purity of H_2O_2 (in %) is _____. (mol. wt. of H_2O_2 = 34; mol. wt. of $KMnO_4$ = 158)
[NV, Sep. 04,2020(I)]

Answer: 85

Solution:



$$\text{Moles of } KMnO_4 = \frac{0.316}{158} = 2 \times 10^{-3}$$

Equivalents of $\text{H}_2\text{O}_2 = \text{Equivalents of K MnO}_4$

Equivalents of $\text{K MnO}_4 = 2 \times 10^{-3} \times 5 = 0.01$

Moles of $\text{H}_2\text{O}_2 = \frac{0.01}{2} = 0.005$

Mass of pure $\text{H}_2\text{O}_2 = 0.005 \times 34 = 0.170\text{g}$

Percentage purity = $\frac{0.17}{0.2} \times 100 = 85\%$.

Question143

The mass of ammonia in grams produced when 2.8kg of dinitrogen quantitatively reacts with 1kg of dihydrogen is

_____.
[NV, Sep. 04,2020(I)]

Answer: 3400

Solution:

Mole of $\text{N}_2 = \frac{2800}{28} = 100$ and Mole of $\text{H}_2 = \frac{1000}{2} = 500$

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

Mass of NH_3 formed = $200 \times 17 = 3400\text{g}$

Question144

A 100mL solution was made by adding 1.43g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$. The normality of the solution is 0.1N .

The value of x is _____.

(The atomic mass of Na is 23g/mol)

[NV, Sep. 04, 2020 (II)]

Answer: 10

Solution:

$$\text{Normality} = \frac{\text{No. of equivalents of solute}}{\text{Volume of solution (in L)}}$$
$$0.1 = \frac{1.43}{\frac{(106 + 18x)}{2} \times 0.1} \Rightarrow \frac{106 + 18x}{2} = 143$$
$$\Rightarrow 18x = 286 - 106 = 180 \Rightarrow x = 10.$$

Question145

The mole fraction of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest integer, is _____ .
[NV, Sep. 03, 2020 (I)]

Answer: 47

Solution:

Let total mole of solution = 1

So, mole of glucose = 0.1

Mole of H_2O = 0.9

$$\%(\text{w/w}) \text{ of } \text{H}_2\text{O} = \left[\frac{0.9 \times 18}{0.9 \times 18 + 0.1 \times 180} \right] \times 100$$
$$= 47.368 = 47.37.$$

Question146

6.023×10^{22} molecules are present in 10g of a substance 'x'. The molarity of a solution containing 5g of substance x' in 2 L solution is _____ $\times 10^{-3}$
[NV, Sep. 03, 2020 (II)]

Answer: 25

Solution:

$$\text{Number of mole of } x = \frac{6.022 \times 10^{22}}{6.022 \times 10^{23}} = \frac{10}{\text{Molar mass of } x}$$

So molar mass of $x = 100\text{g}$

$$\text{Molarity} = \frac{5}{100 \times 2} = 0.025\text{M}.$$

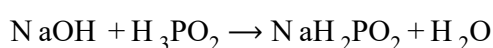
Question147

**The volume (in mL) of 0.1N N aOH required to neutralise 10mL of 0.1N phosphinic acid is _____.
[NV, Sep. 03, 2020 (II)]**

Answer: 10

Solution:

Phosphinic acid is hypophosphorous acid (H_3PO_2).



For neutrization,

$$(N_1 V_1)_{\text{add}} = (N_2 V_2)_{\text{base}}$$

$$0.1 \times 10 = 0.1 \times (V_{\text{mL}})_{\text{NaOH}}$$

$$V_{\text{NaOH}} = 10\text{mL}$$

Question148

**8g of NaOH is dissolved in 18g of H_2O . Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solution respectively are :
[Jan. 12, 2019 (II)]**

Options:

A. 0.2,22.20

- B. 0.2,11.11
- C. 0.167,11.11
- D. 0.167,22.20

Answer: C

Solution:

$$\text{No. of moles of H}_2\text{O}(n_1) = \frac{18}{18} = 1$$

$$\text{No. of moles of NaOH}(n_2) = \frac{8}{40} = \frac{1}{5}$$

$$\text{Mole fraction of NaOH} = \frac{n_2}{n_2 + n_1} = \frac{\frac{1}{5}}{\frac{1}{5} + 1} = 0.167$$

$$\text{Molality} = \frac{\text{No. of moles of solute}}{\text{Mass of solvent (kg)}} = \frac{1}{5} \times \frac{1000}{18} = 11.11\text{m}$$

Question149

**The amount of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) required to prepare 2 L of its 0.1M aqueous solution is:
[Jan. 10 2019(II)]**

Options:

- A. 136.8g
- B. 17.1g
- C. 68.4g
- D. 34.2g

Answer: C

Solution:

As we know,

$$\text{Molarity} = \frac{\text{No. of moles of sugar}}{\text{Volume of solution (in L)}}$$

$$\Rightarrow 0.1 = \frac{\text{No. of moles of sugar}}{2\text{L}}$$

So, no. of moles of sugar = 0.2 mole

∴ Mass of sugar = No. of moles of sugar × Molar mass of sugar = $0.2 \times 342 = 68.4\text{g}$

Question150

50mL of 0.5M oxalic acid is needed to neutralize 25mL of sodium hydroxide solution. The amount of NaOH in 50mL of the given sodium hydroxide solution is:

[Jan. 12,2019 (I)]

Options:

A. 40g

B. 10g

C. 20g

D. 80g

E. None of Above

Answer: E

Solution:

Oxalic acid Sodium hydroxide

$$N_1 V_1 = N_2 V_2$$

$$(2 \times M_1) V_1 = M_2 V_2$$

$$2 \times 0.5 \times 50 = M_2 \times 25$$

$$M_2 = 2$$

$$\text{Molarity} = \frac{\text{No.of moles}}{\text{Vol.(L)}}$$

$$2 = \frac{\text{No.of moles}}{50/1000}$$

$$\text{No. of moles} = \frac{1}{10} = 0.1$$

$$\text{No. of moles} = \frac{W}{\text{Molar mass}}$$

$$W = 0.1 \times 40 = 4\text{g}$$

No option is correct.

Question151

A 10mg effervescent tablet containing sodium bicarbonate and oxalic acid releases 0.25mL of CO_2 at $T = 298.15\text{K}$ and $P = 1$ bar. If molar volume of CO_2 is 25.0L under such condition, what is the percentage of sodium bicarbonate in each tablet?

[Molar mass of $\text{NaHCO}_3 = 84\text{gmol}^{-1}$]

[Jan. 11, 2019 (I)]

Options:

A. 0.84

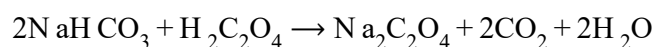
B. 33.6

C. 16.8

D. 8.4

Answer: D

Solution:



$$\text{Moles of } \text{CO}_2 \text{ evolved} = \frac{0.25}{25 \times 10^3} = 10^{-5}$$

$$\therefore \text{Moles of } \text{NaHCO}_3 = 10^{-5}$$

$$\therefore \text{Mass of } \text{NaHCO}_3 = 84 \times 10^{-5} \text{g}$$

$$= 0.84 \times 10^{-3} \text{g} = 0.84 \text{mg}$$

$$\therefore \% \text{ by weight} = \times 100 = 8.4\%$$

Question152

25mL of the given HCl solution requires 30mL of 0.1M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30mL of 0.2M aqueous NaOH solution?

[Jan.11,2019 (II)]

Options:

A. 25mL

B. 75mL

C. 50mL

D. 12.5mL

Answer: A

Solution:

25mL of HCl solution requires 30mL of 0.1M Na_2CO_3 solution.

$$\therefore N_1 V_1 = N_2 V_2$$

$$\therefore 25 \times N_1 = 30 \times 0.2 (0.1\text{M Na}_2\text{CO}_3 = 0.2\text{N Na}_2\text{CO}_3)$$

$$N_1 = \frac{6}{25} = 0.24\text{N}$$

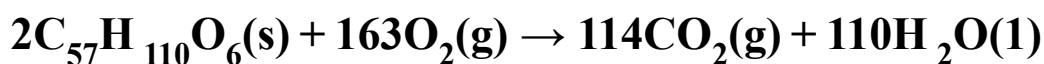
Now, HCl solution is titrated with NaOH solution.

$$M_1 V_1 = M_2 V_2; 0.24\text{N HCl} = 0.24\text{M HCl}$$

$$\therefore V \times 0.24 \times 1 = 30 \times 0.2 \times 1 \Rightarrow V = 25\text{mL}$$

Question153

For the following reaction the mass of water produced from 445g of $\text{C}_{57}\text{H}_{110}\text{O}_6$ is:



[Jan. 9, 2019 (II)]

Options:

A. 490g

B. 445g

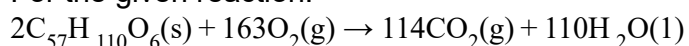
C. 495g

D. 890g

Answer: C

Solution:

For the given reaction:



$$\text{Moles of } \text{C}_{57}\text{H}_{110}\text{O}_6 = \frac{445}{890} = 0.5$$

Now, moles of water = $\frac{110}{2} \times 0.5 = 27.5$

$\therefore 1 = 27.5 \times 18 = 495\text{g}$

Question154

**The percentage composition of carbon by mole in methane is :
[April 8, 2019(II)]**

Options:

A. 75%

B. 80%

C. 25%

D. 20%

Answer: D

Solution:

CH_4 has one atom of carbon among 5 atoms (1C + 4H)

$\therefore \text{Mole \% of C} = \frac{1}{5} \times 100 = 20\%$

Question155

**5 moles of AB_2 weigh $125 \times 10^{-3}\text{kg}$ and 10 moles of A_2B_2 weigh $300 \times 10^{-3}\text{kg}$. The molar mass of A(M_A) and molar mass of B(M_B) in kgmol^{-1} are:
[April 12, 2019(I)]**

Options:

A. $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

B. $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$

C. $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$

D. $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$

Answer: D

Solution:

5mol AB_2 weighs 125g

$$\therefore AB_2 = 25\text{g/mol}$$

10mol A_2B_2 weighs 300g

$$\therefore A_2B_2 = 30\text{g/mol}$$

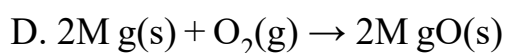
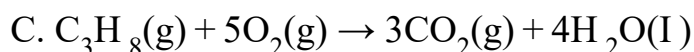
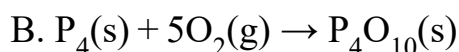
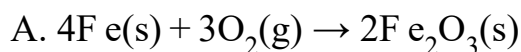
$$\therefore \text{Molar mass of A}(M_A) = 5\text{g or } 5 \times 10^{-3}\text{kg}$$

$$\text{Molar mass of B}(M_B) = 10\text{g or } 10 \times 10^{-3}\text{kg}$$

Question156

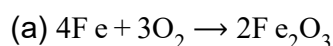
The minimum amount of $O_2(g)$ consumed per gram of reactant is for the reaction : (Given atomic mass: Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)
[April 10, 2019 (II)]

Options:

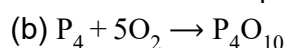


Answer: A

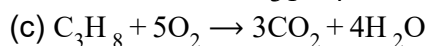
Solution:



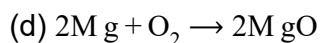
$$1\text{g Fe requires} = \frac{3 \times 32}{4 \times 56} = 0.43\text{g of oxygen}$$



1g of P requires $= \frac{5 \times 32}{31 \times 4} = 1.3\text{g}$ of oxygen



1g of C_3H_8 requires $= \frac{5 \times 32}{44} = 3.6\text{g}$ of O_2



1 g Mg requires $= \frac{32}{2 \times 24}\text{g} = 0.66\text{g}$ of O_2

Question 157

For a reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$; identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures.

[April 9, 2019 (I)]

Options:

A. 56g of N_2 + 10g of H_2

B. 35g of N_2 + 8g of H_2

C. 28g of N_2 + 6g of H_2

D. 14g of N_2 + 4g of H_2

Answer: A

Solution:

According to the stoichiometry of balanced equation, 28g N_2 reacts with 6g H_2

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
1 mole 28 g 3 mole 6 g
 $\therefore$ For 56g of N_2 , 12g of H_2 is required.

Question 158

0.27g of a long chain fatty acid was dissolved in 100cm^3 of hexane. 10mL of this solution was added dropwise to the surface of water in a round watch glass. Hexane evaporates and a monolayer is formed. The distance from edge to centre of the

watch glass is 10cm. What is the height of the monolayer?

[Density of fatty acid = 0.9gcm^{-3} ; $\pi = 3$]

[April 8, 2019 (II)]

Options:

A.

$$10^{-8}\text{ m}$$

B.

$$10^{-6}\text{ m}$$

C.

$$10^{-4}\text{ m}$$

D.

$$10^{-2}\text{ m}$$

Answer: B

Solution:

Given: 0.27g is present in 100cm^3 of hexane

$\therefore$ 10mL of aqueous solution contains only 0.027 g acid.

$$\text{Volume of } 0.027\text{g acid} = \frac{0.027}{0.9}\text{mL}$$

$$\therefore \pi r^2 h = \frac{0.027}{0.9} \left(\text{given } r = 10\text{cm}, \pi = 3 \right)$$

$$\therefore h = 10^{-4}\text{cm} = 10^{-6}\text{m}$$

Question159

An unknown chlorohydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only, chlorine atoms present in 1g of chlorohydrocarbon are:

(Atomic wt. of Cl = 35.5u; Avogadro constant
= $6.023 \times 10^{23}\text{mol}^{-1}$)

[Online April 16,2018]

Options:

- A. 6.023×10^9
- B. 6.023×10^{23}
- C. 6.023×10^{21}
- D. 6.023×10^{20}

Answer: D

Solution:

Given percentage of chlorine in an hydrocarbon = 3.55%

i.e.,

100g of chlorohydrocarbon has 3.55g of chlorine.

1 g of chlorohydrocarbon will have $\frac{3.55}{100} = 0.0355$ g of chlorine.

Atomic wt. of Cl = 35.5g/mol

Number of moles of Cl = $\frac{0.0355\text{g}}{35.5\text{g/mol}} = 0.001\text{mol}$

Number of atoms of Cl = $0.001\text{mol} \times 6.023 \times 10^{23}\text{mol}^{-1}$
 $= 6.023 \times 10^{20}$

Question160

The ratio of mass percent of C and H of an organic compound ($\text{C}_x\text{H}_y\text{O}_z$) is 6 : 1. If one molecule of the above compound ($\text{C}_x\text{H}_y\text{O}_2$) contains half as much oxygen as required to burn one molecule of compound C_xH_y completely to CO_2 and H_2O .

The empirical formula of compound $\text{C}_x\text{H}_y\text{O}_z$ is:

[2018]

Options:

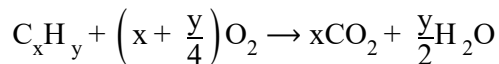
- A. $\text{C}_3\text{H}_6\text{O}_3$
- B. $\text{C}_2\text{H}_4\text{O}$
- C. $\text{C}_3\text{H}_4\text{O}_2$
- D. $\text{C}_2\text{H}_4\text{O}_3$

Answer: D

Solution:

So, $x = 1, y = 2$

Equation for combustion of C_xH_y



$$\text{Oxygen atoms required} = 2\left(x + \frac{y}{4}\right)$$

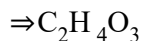
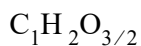
As mentioned,

$$2\left(x + \frac{y}{4}\right) = 2z; \left(x + \frac{y}{4}\right) = z$$

Now putting the values of x and y

$$\Rightarrow \left(1 + \frac{2}{4}\right) = z \Rightarrow z = 1.5$$

$\therefore$ Molecule ($C_xH_yO_z$) can be written as



Question 161

A sample of $NaClO_3$ is converted by heat to $NaCl$ with a loss of 0.16 g of oxygen. The residue is dissolved in water and precipitated as $AgCl$. The mass of $AgCl$ (in g) obtained will be: (Given: Molar mass of $AgCl = 143.5 \text{ g mol}^{-1}$) [Online April 15, 2018 (I)]

Options:

A. 0.35

B. 0.54

C. 0.41

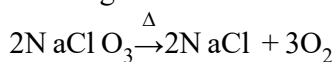
D. 0.48

Answer: D

Solution:

No. of moles of oxygen in 0.16g of oxygen molecule

$$= \frac{0.16 \text{ g}}{32 \text{ g/mol}} = 0.005 \text{ mol}$$



According to the reaction,
 $3 \text{ moles of } \text{O}_2 = 2 \text{ moles of } \text{NaCl} = 2 \text{ moles of } \text{AgCl}$
Molar mass of $\text{AgCl} = 143.5 \text{ g/mol}$
 $0.005 \text{ moles of } \text{O}_2 \text{ will ppt.} = 0.005 \times \frac{2}{3} \text{ moles } \text{AgCl}$
 $= 0.0033 \text{ moles of } \text{AgCl}$
 $\therefore \text{Mass of } \text{AgCl} \text{ (in g) obtained will be}$
 $= 143.5 \text{ g/mol} \times 0.0033 \text{ moles} = 0.48 \text{ g.}$

Question162

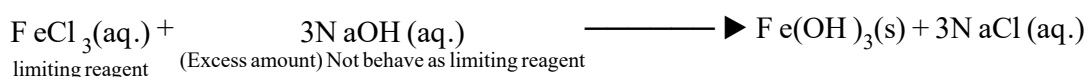
Excess of NaOH (aq) was added to 100 mL of $\text{FeCl}_3(\text{aq})$ resulting into 2.14 g of Fe(OH)_3 . The molarity of $\text{FeCl}_3(\text{aq})$ is:
(Given molar mass of $\text{Fe} = 56 \text{ g mol}^{-1}$ and molar mass of $\text{Cl} = 35.5 \text{ g mol}^{-1}$)
[Online April 8, 2017]

Options:

- A. 0.2 M
- B. 0.3 M
- C. 0.6 M
- D. 1.8 M

Answer: A

Solution:



$$\text{Moles of } \text{Fe(OH)}_3 = \frac{\text{weight in g}}{\text{Mol. mass of } \text{Fe(OH)}_3}$$

$$= \frac{2.14 \text{ g}}{107 \text{ g/mol}} = 0.02 \text{ mol}$$

$1.0 \text{ mole of } \text{Fe(OH)}_3 \text{ is obtained from } = 1.0 \text{ mole of } \text{FeCl}_3$

$0.02 \text{ mole of } \text{Fe(OH)}_3 \text{ will be obtained from } = 0.02 \text{ mole } \text{FeCl}_3 \text{ Molarity}$

$$= \frac{\text{No. of moles}}{\text{Volume in L}} = \frac{0.02 \text{ mole}}{0.1 \text{ L}} = 0.2 \text{ M}$$

Question163

The most abundant elements by mass in the body of a healthy human adult are: [2017] Oxygen (61.4%); Carbon (22.9%), Hydrogen (10.0%); and Nitrogen (2.6%). The weight which a 75kg person would gain if all ^1H atoms are replaced by ^2H atoms is [2017]

Options:

- A. 15kg
- B. 37.5kg
- C. 7.5kg
- D. 10kg

Answer: C

Solution:

Percentage (by mass) of elements given in the body of a healthy human adult is :

Oxygen = 61.4%, Carbon = 22.9%

Hydrogen = 10.0% and Nitrogen = 2.6%

$\therefore$ Total weight of person = 75kg

$\therefore$ Mass due to ^1H is $= 75 \times \frac{10}{100} = 7.5\text{kg}$

If ^1H atoms are replaced by ^2H atoms.

Mass gain by person would be = 7.5kg

Question164

What quantity (in mL) of a 45% acid solution of a monoprotic strong acid must be mixed with a 20% solution of the same acid to produce 800mL of a 29.875% acid solution?

[Online April 9,2017]

Options:

- A. 320
- B. 325
- C. 316

D. 330

Answer: C

Solution:

$$\frac{V \times 45}{100} + \frac{(800 - V) \times 20}{100} = \frac{800 \times 29.875}{100}$$
$$\Rightarrow \frac{9V}{20} + 160 - \frac{V}{5} = 239 \Rightarrow V = 316$$

Question165

5L of an alkane requires 25L of oxygen for its complete combustion. If all volumes are measured at constant temperature and pressure, the alkane is:
[Online April 9, 2016]

Options:

A. Isobutane

B. Ethane

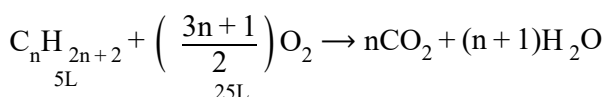
C. Butane

D. Propane

Answer: D

Solution:

Since the compound undergoing combustion is an alkane. Hence the combustion reaction can be written as



Since volumes are measured at constant T & P, hence according to Avogadro's law
Volume $\propto$ mole

1 Lalkane requires $\frac{3n+1}{2}$ L of O_2

5 Lalkane requires 25L of O_2

$$\frac{1}{5} = \frac{\frac{3n+1}{2}}{25}$$

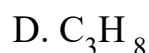
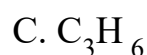
$$\therefore n = 3$$

Hence alkane is propane (C_3H_8)

Question 166

At 300K and 1atm, 15mL of a gaseous hydrocarbon requires 375mL air containing 20%O₂ by volume for complete combustion. After combustion the gases occupy 330mL. Assuming that the water formed is in liquid form and the volumes were measured at the same temperature and pressure, the formula of the hydrocarbon is:
[2016]

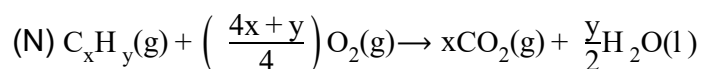
Options:



E. None of above

Answer: E

Solution:



$$\text{Volume of } O_2 \text{ used} = \frac{20}{100} \times 375 = 75\text{mL}$$

$$\text{Volume of air} = 375 - 75 = 300\text{mL}$$

$$\begin{aligned} \text{Total volume of gases after combustion} \\ = \text{vol. of } CO_2 + \text{vol. of air} = 330\text{mL} \end{aligned}$$

$$\text{Volume of } CO_2 = 330 - 300 = 30\text{mL}$$

$$15\text{mL } C_xH_y \text{ gives } = 30\text{mL } CO_2$$

$$1\text{mL } C_xH_y \text{ gives } = \frac{30}{15} = 2\text{mL } CO_2$$

At constant T and P; Volume $\propto$ mole

$$\therefore 1\text{mol } C_xH_y = 2\text{mol } CO_2$$

$$x = 2$$

$$\left(\frac{4x+y}{4} \right) = \frac{75}{15}$$

$$4x + y = 20$$

$$y = 20 - 4 \times 2 = 12$$

Hence, formula of the hydrocarbon is C_2H_{12} .

Question167

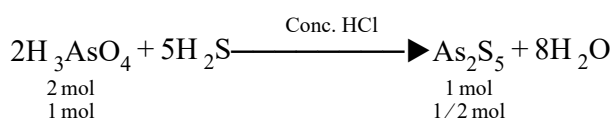
The amount of arsenic pentasulphide that can be obtained when 35.5g arsenic acid is treated with excess H_2S in the presence of conc. HCl (assuming 100% conversion) is:
[Online April 9, 2016]

Options:

- A. 0.25mol
- B. 0.50mol
- C. 0.333mol
- D. 0.125mol

Answer: D

Solution:



∴ The molar mass of H_3AsO_4 is $3 \times 1 + 79 + 4 \times 16$
= 142g/mol

∴ Number of moles of $H_3AsO_4 = \frac{35.5}{142} = 0.25\text{mol}$

∴ Number of moles of $As_2S_5 = \frac{0.25}{2} = 0.125\text{mol}$.

Question168

The volume of 0.1N dibasic acid sufficient to neutralize 1 g of a base that furnishes 0.04 mole of OH^- in aqueous solution is :
[Online April 10,2016]

Options:

- A. 400mL

B. 600mL

C. 200mL

D. 800mL

Answer: A

Solution:

Applying law of equivalence

Equivalence of acid = Equivalence of base

Equivalent of acid = Normality $\times$ volume = $0.1 \times V$

Another formula of equivalence = n factor $\times$ number of moles

$\therefore$ Equivalent of base = n factor of $\text{OH}^- \times$ moles of OH

= 1×0.04

$0.1 \times V = 1 \times 0.04$

$V = 0.4\text{L} = 0.4 \times 1000 = 400\text{mL}.$

Question169

The molecular formula of a commercial resin used for exchanging ions in water softening is $\text{C}_8\text{H}_7\text{SO}_3 - \text{Na}^+$ (Mol. wt.

206 . What would be the maximum uptake of Ca^{2+} ions by the resin when expressed in mole per gram resin?

[2015]

Options:

A. $\frac{2}{309}$

B. $\frac{1}{412}$

C. $\frac{1}{103}$

D. $\frac{1}{206}$

Answer: B

Solution:

2 mole of water softner require 1 mole of Ca^{2+} ion

So, 1 mole of water softner require $\frac{1}{2}$ mole of Ca^{2+} ion

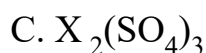
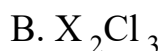
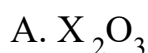
Thus, $\frac{1}{2 \times 206} = \frac{1}{412} \text{mol/g}$ will be maximum uptake.

Question170

Choose the incorrect formula out of the four compounds for an element X below:

[Online April 11, 2015]

Options:



Answer: C

Solution:

Mass of substance = 250mg = 0.250g

Mass of AgBr = 141mg = 0.141g

1 mole of AgBr = 1g atom of Br

188g of AgBr = 80g of Br

$\therefore$ fof AgBr contain bromine = 80g

0.141g of AgBr contain bromine

$$= \frac{80}{188} \times 0.141 = 0.06\text{g}$$

0.06g of bromine is present in 0.250g of organic compound

$$\therefore \% \text{ of bromine} = \frac{0.06}{0.250} \times 100 = 24\%$$

Question171

3 g of activated charcoal was added to 50mL of acetic acid solution (0.06N) in a flask. After an hour it was filtered and the strength of the filtrate was found to be 0.042N . The amount of

**acetic acid adsorbed (per gram of charcoal) is:
[2015]**

Options:

A. 42mg

B. 54mg

C. 18mg

D. 36mg

Answer: C

Solution:

Let the weight of acetic acid initially be w_1 in 50mL of 0.060N solution.

$$N = \frac{w_1 \times 1000}{M \cdot wt. \times 50} \quad \left(\text{Normality} = 0.06N \right)$$

$$0.06 = \frac{w_1 \times 1000}{60 \times 50}$$

$$\Rightarrow w_1 = \frac{0.06 \times 60 \times 50}{1000} = 0.18g = 180mg$$

After an hour, the strength of acetic acid = 0.042N

so, let the weight of acetic acid be w_2

$$N = \frac{w_2 \times 1000}{60 \times 50}$$

$$0.042 = \frac{w_2 \times 1000}{3000}$$

$$\Rightarrow w_2 = 0.126g = 126mg$$

So amount of acetic acid adsorbed per 3g

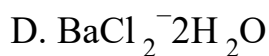
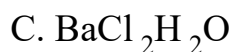
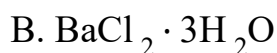
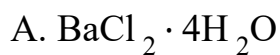
$$= 180 - 126mg = 54mg$$

$$\therefore \text{amount of acetic acid absorbed per g} = 54/3 = 18mg$$

Question172

**A sample of a hydrate of barium chloride weighing 61g was heated until all the water of hydration is removed. The dried sample weighed 52g. The formula of the hydrated salt is :
(atomic mass, Ba = 137amu, Cl = 35.5 amu)
[Online April 10, 2015]**

Options:



Answer: D

Solution:

Weight of hydrated $\text{BaCl}_2 = 61\text{g}$

Weight of anhydrous $\text{BaCl}_2 = 52\text{g}$

Loss in mass = 9g

Assuming $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ as hydrate

Mass of $\text{H}_2\text{O} = 9\text{g}$

Moles of $\text{H}_2\text{O} = \frac{9}{18} = 0.5\text{mol}$

Gross molecular wt. of $\text{BaCl}_2 = 208\text{g/mol}$.

% of H_2O in this hydrated $\text{BaCl}_2 = \frac{9}{61} \times 100 = 14.75\%$

% of H_2O in $\text{BaCl}_2 \cdot x\text{H}_2\text{O} = \frac{18x}{208 + 18x} \times 100$

Thus, $\frac{18x}{208 + 18x} \times 100 = 14.75$

On solving $x = 2$

Hence, the formula of hydrated salt is $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$

Question173

**$\text{A} + 2\text{B} + 3\text{C} \rightleftharpoons \text{AB}_2\text{C}_3$ Reaction of 6.0g of A, 6.0×10^{23} atoms of B, and 0.036 mol of C yields 4.8g of compound AB_2C_3 . If the atomic mass of A and C are 60 and 80amu , respectively, the atomic mass of B is (Avogadro no. = 6×10^{23}) :
[Online April 11, 2015]**

Options:

A. 50amu

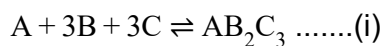
B. 60amu

C. 70amu

D. 40amu

Answer: A

Solution:



$$\text{No. of moles of A} = \frac{6.0\text{g}}{60\text{g/mol}} = 0.1\text{mol}$$

$$\text{No. of moles of B} = \frac{6.00 \times 10^{23}}{6.000 \times 10^{23}} = 1\text{mol}$$

$$\text{No. of moles of C} = 0.036$$

Therefore, C is the limiting reagent,

The number of moles of product formed

$$= \frac{0.036}{3} = 0.012\text{mol}$$

The expression for the molar mass is Molar mass

$$= \frac{\text{Given mass}}{\text{mole of product}}$$

$$60 + (2 \times x) + (3 \times 80) = \frac{4.8}{0.012}$$

$$x = 50\text{amu}$$

Hence, atomic mass of B is 50 amu

Question174

In Carius method of estimation of halogens, 250mg of an organic compound gave 141mg of AgBr. The percentage of bromine in the compound is :

(at. mass Ag = 108; Br = 80)

[2015]

Options:

A. 48

B. 60

C. 24

D. 36

Answer: C

Solution:

Mass of substance = 250mg = 0.250g

Mass of AgBr = 141mg = 0.141g

1 mole of AgBr = 1g atom of Br

188g of AgBr = 80g of Br

∴ 188g of AgBr contain bromine = 80g

0.141 g of AgBr contain bromine

$$= \frac{80}{188} \times 0.141 = 0.06\text{g}$$

0.06g of bromine is present in 0.250g of organic compound

$$\therefore \% \text{ of bromine} = \frac{0.06}{0.250} \times 100 = 24\%$$

Question175

Dissolving 120g of a compound of (mol. wt. 60) in 1000g of water gave a solution of density 1.12g/mL. The molarity of the solution is:

[Online April 9,2014]

Options:

A. 1.00M

B. 2.00M

C. 2.50M

D. 4.00M

Answer: B

Solution:

Given

mass of solute (w) = 120g

mass of solvent (w) = 1000g

Mol. mass of solute = 60g

density of solution = 1.12g/mL

From the given data,

Mass of solution = 1000 + 120 = 1120g

$$\therefore d = \frac{\text{Mol. mass}}{V} \text{ or } V = \frac{\text{Mol. mass}}{d}$$

$$\text{Volume of solution } V = \frac{1120}{1.12} = 1000\text{mL or } = 1 \text{ litre}$$

$$\text{Now molarity (M)} = \frac{w}{\text{Mol. mass} \times V(\text{L})} = \frac{120}{60 \times 1} = 2\text{M}$$

Question176

**The amount of oxygen in 3.6 moles of water is:
[Online April 9, 2014]**

Options:

- A. 115.2g
- B. 57.6g
- C. 28.8g
- D. 18.4g

Answer: B

Solution:

1 mole of water contains = 16 g of oxygen
∴ 3.6 mole of water contains
= $16 \times 3.6 = 57.6\text{g}$

Question177

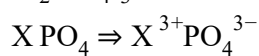
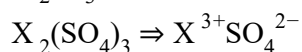
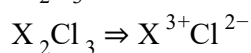
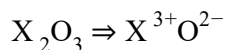
**A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is:
[Online April 11, 2014]**

Options:

- A. NH_2
- B. N_3H
- C. NH_3
- D. N_2H_4
- E. None of above.

Answer: D

Solution:



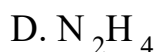
Since Cl^{2-} does not exist. So, X_2Cl_3 is incorrect.

The correct formula should be XCl_3

Question178

A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is:
[Online April 11,2014]

Options:



Answer: D

Solution:

In an unknown compounds containing N and H given % of H = 12.5%
 $\therefore$ % of N = 100 – 12.5 = 87.5%

Element	Percentage	Atomic ratio	Simple ratio
H	12.5%	$\frac{12.5}{1} = 12.5$	$\frac{12.5}{6.25} = 2$
N	87.5	$\frac{87.5}{14} = 6.25$	$\frac{6.25}{6.25} = 1$

Empirical formula = NH_2

Mol. wt = $2 \times \text{vapour density} = 16 \times 2 = 32$.

Molecular formula = $n \times \text{empirical formula mass}$

$$n = \frac{32}{16} = 2$$

$\therefore$ Molecular formula of the compound will be = $(\text{NH}_2)_2$



Question179

The amount of BaSO_4 formed upon mixing 100mL of 20.8% BaCl_2 solution with 50mL of 9.8% H_2SO_4 solution will be:
(Ba = 137, Cl = 35.5, S = 32, H = 1 and O = 16)
[Online April 12, 2014]

Options:

A. 23.3g

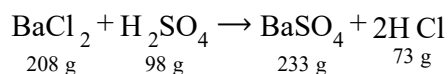
B. 11.65g

C. 30.6g

D. 33.2g

Answer: B

Solution:



$$\text{Mass of BaCl}_2 \text{ in solution} = 100 \times \frac{20.8}{100} = 20.8\text{g}$$

$$\text{Mass of H}_2\text{SO}_4 \text{ in solution} = 50 \times \frac{9.8}{100} \times 4.9 = 4.9\text{g}$$

98g of H_2SO_4 reacts with 208g BaCl_2

$$4.9\text{g H}_2\text{SO}_4 \text{ will react with } \frac{208}{98} \times 4.9 = 10.4\text{g BaCl}_2$$

H_2SO_4 reacts as a limiting reagent because BaCl_2 is given in excess

98g H_2SO_4 produces 233g BaSO_4

$$4.9\text{g H}_2\text{SO}_4 \text{ will produce } \frac{233}{98} \times 4.9 = 11.65\text{g BaSO}_4$$

Question180

The density of 3M solution of sodium chloride is 1.252 g mL^{-1} .

The molality of the solution will be:

(molar mass, $\text{NaCl} = 58.5\text{ g mol}^{-1}$)

[Online April 22, 2013]

Options:

A. 260m

B. 2.18m

C. 2.79m

D. 3.00m

Answer: C

Solution:

The relation between molarity (M) and molality (m) is $d = M \left(\frac{1}{m} + \frac{M_2}{1000} \right)$, $M_2 =$ Mol. mass of solute

On putting value $1.252 = 3 \left(\frac{1}{m} + \frac{58.5}{1000} \right)$

On solving $m = 2.79$

Question181

Number of atoms in the following samples of substances is the largest in:

[Online April 23, 2013]

Options:

A. 4.0g of hydrogen

B. 71.0g of chlorine

C. 127.0g of iodine

D. 48.0g of magnesium

Answer: A

Solution:

4g of hydrogen = 4 mole of hydroge

$$n = 4 \times 6.023 \times 10^{23} \text{ atoms}$$

$$71.0\text{g of chlorine} = \frac{71.0}{71.0} = 1 \text{ moles of chlorine}$$

$$= 6.023 \times 10^{23} \text{ atoms}$$

$$127\text{g of iodine} = \frac{127}{254} \text{mol}$$

$$= 6.023 \times 10^{23} \times \frac{1}{2} \text{ atoms}$$

$$48.0\text{g of magnesium} = \frac{48.0}{24.0} \text{mol}$$

$$= 2 \times 6.023 \times 10^{23} \text{ atoms}$$

$\therefore 4.0\text{gH}_2$ has largest number of atoms.

Question182

10mL of 2(M)NaOH solution is added to 200mL of 0.5 (M) of NaOH solution. What is the final concentration ?
[Online April 25, 2013]

Options:

A. 0.57(M)

B. 5.7(M)

C. 11.4(M)

D. 1.14(M)

Answer: A

Solution:

From molarity equation

$$M_1V_1 + M_2V_2 = M V_{(\text{total})}$$
$$M = \frac{M_1V_1 + M_2V_2}{\text{Total}} = \frac{2 \times 10 + 0.5 \times 200}{210}$$

$$M = \frac{120}{210} = 0.57\text{M}$$

Question183

6 litres of an alkene require 27 litres of oxygen at constant temperature and pressure for complete combustion. The alkene is:

[Online April 25, 2013]

Options:

A. Ethene

B. Propene

C. 1 -Butene

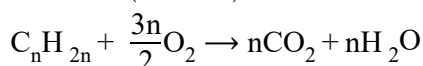
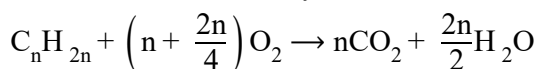
D. 2 -Butene

Answer: B

Solution:

General combustion reaction for hydrocarbons is $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$

For alkane, $x = n$ and $y = 2n$



$\therefore$ 1 mole alkene reacts with $\frac{3n}{2}$ mole of O_2

moles $\propto$ volume (at constant temp & spress.)

1 L alkene requires $\frac{3n}{2}$ L of O_2

6L alkene requires 27L of O_2

$$\frac{1}{6} = \frac{\frac{3n}{2}}{27}$$

$$3n = \frac{54}{6}$$

$$n = 3^6$$

Hence alkene is propene (C_3H_6)

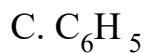
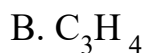
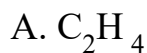
Question184

A gaseous hydrocarbon gives upon combustion 0.72g of water and 3.08g. of CO_2 . The empirical formula of the hydrocarbon is

:

[2013]

Options:



Answer: D

Solution:

$$\therefore 18g, H_2O \text{ contains } = 2gH$$

$$\therefore 0.72gH_2O \text{ contains}$$

$$= \frac{2}{18} \times 0.72g = 0.08gH$$

$$\therefore 44gCO_2 \text{ contains } = 12gC$$

$$\therefore 3.08gCO_2 \text{ contains } = \frac{12}{44} \times 3.08 = 0.84gC$$

$$\therefore C : H = \frac{0.84}{12} : \frac{0.08}{1}; 0.07 : 0.08 = 7 : 8$$

$$\therefore \text{Empirical formula} = C_7H_8$$

Question185

**1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186mol e of CO_2 . The molar mass of M_2CO_3 in $gmol^{-1}$ is:
[2013]**

Options:

A. 1186

B. 84.3

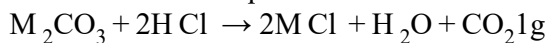
C. 118.6

D. 11.86

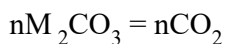
Answer: B

Solution:

Given chemical eqⁿ



0.01186 mol From the above chemical eqⁿ.



$$\frac{1}{\text{Molar mass of M}_2\text{CO}_3} = 0.01186$$

$$\therefore \text{Molar mass of M}_2\text{CO}_3 = \frac{1}{0.01186}$$

$$\text{Molar mass} = 84.3 \text{ g/mol}$$

Question 186

The concentrated sulphuric acid that is peddled commercial is 95% H_2SO_4 by weight. If the density of this commercial acid is 1.834 g cm^{-3} , the molarity of this solution is [Online May 7, 2012]

Options:

A. 17.8M

B. 12.0M

C. 10.5M

D. 15.7M

Answer: A

Solution:

95% H_2SO_4 by weight means 100g H_2SO_4 solution contains 95g H_2SO_4 by mass.

$$\text{Molar mass of H}_2\text{SO}_4 = 98 \text{ g mol}^{-1}$$

$$\text{Moles in 95g H}_2\text{SO}_4 = \frac{95}{98} = 0.969 \text{ mol}$$

Volume of 100g H_2SO_4 solution

$$= \frac{\text{mass}}{\text{density}} = \frac{100 \text{ g}}{1.834 \text{ g cm}^{-3}}$$

$$= 54.52 \text{ cm}^3 = 54.52 \times 10^{-3} \text{ L}$$

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in L}}$$

$$= \frac{0.969}{54.52 \times 10^{-3}} = 17.8\text{M}$$

Question187

The ratio of number of oxygen atoms (O) in 16.0 g ozone (O_3), 28.0g carbon monoxide (CO) and 16.0 oxygen (O_2) is (Atomic mass: C = 12, O = 16 and Avogadro's constant $N_A = 6.0 \times 10^{23} \text{mol}^{-1}$)
[Online May 7, 2012]

Options:

- A. 3 : 1 : 2
- B. 1 : 1 : 2
- C. 3 : 1 : 1
- D. 1 : 1 : 1

Answer: D

Solution:

O_3 molecular weight = $16 + 16 + 16 = 48\text{g/mol}$

It means weight of 1mol of O_3 is 48g and in 1mol of O_3 we have 3 atoms of Oxygen

In 48g of O_3 , number of atoms of oxygen = 3

so in 16g of O_3 , number of atoms of oxygen = $(3/48) \times 16 = 1$

CO molecular weight = $12 + 16 = 28\text{g/mol}$

It means weight of 1 mol of CO is 28g and in 1 mol of CO we have 1 atom of Oxygen

so in 28g of CO, number of atoms of oxygen = 1

O_2 molecular weight = $16 + 16 = 32\text{g/mol}$

It means weight of 1 mol of O_2 is 32g and in 1mol of O_2 we have 2 atoms of Oxygen

In 32g of O_2 , number of atoms of oxygen = 2

so in 16g of O_2 , number of atoms of oxygen = $(2/32) \times 16 = 1$

So answer is 1:1:1

Question188

**The ppm level of F^- in a 500g sample of a tooth paste containing $0.2gF^-$ is
[Online May 12, 2012]**

Options:

- A. 400
- B. 1000
- C. 250
- D. 200

Answer: A

Solution:

$$\begin{aligned}\text{ppm} &= \frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 10^6 \\ &= \frac{0.2}{500} \times 10^6 = 400 \text{ ppm}\end{aligned}$$

Question189

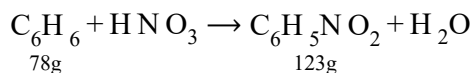
**5g of benzene on nitration gave 6.6g of nitrobenzene. The theoretical yild of the nitrobenzene will be
[Online May 12, 2012]**

Options:

- A. 4.5g
- B. 5.6g
- C. 8, 09g
- D. 6.6g

Answer: C

Solution:



Now since 78g of benzene on nitration give = 123g nitrobenzene
hence 5g of benzene on nitration give

$$= \frac{123}{78} \times 5 = 7.88\text{g}$$

The nearest answer is (c) i.e. theoretical yield = 7.88g

Question190

A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be [Online May 26, 2012]

Options:



Answer: C

Solution:

74.75% of chlorine means 74.75g chlorine is present in 100g of metal chloride.

$$\begin{aligned}\text{Weight of metal} &= 100 - 74.75 \\ &= 25.25\text{g}\end{aligned}$$

$$\begin{aligned}\text{Equivalent weight} &= \frac{\text{weight of metal}}{\text{weight of chlorine}} \times 35.5 \\ &= \frac{25.25}{74.75} \times 35.5 = 12\end{aligned}$$

Valency of metal

$$\begin{aligned}&= \frac{2 \times \text{V.D.}}{\text{Equivalent wt. of metal} + 35.5} \\ &= \frac{2 \times 94.8}{12 + 35.5} = 4\end{aligned}$$

∴ Formula of metal chloride is MCl_4

Alternate method:

$$\begin{aligned}\text{Mol. wt} &= 2 \times \text{vapour density} \\ &= 2 \times 94.8 = 189.6\text{g}\end{aligned}$$

Since 74.75% is chlorine therefore,
189.6 metal chloride contains

$$= \frac{74.75}{100} \times 189.6 = 141.72 \text{ g chloride}$$

$$\text{Number of atoms of chloride} = \frac{141.72}{35.5} = 3.99 \approx 4$$

Hence, formula of metal chloride is MCl_4

Question191

An aqueous solution of oxalic acid dihydrate contains its 6.3 g in 250mL. The volume of 0.1N N aOH required to completely neutralize 10mL of this solution

[Online May 12, 2012]

Options:

A. 4mL

B. 20mL

C. 2mL

D. 40mL

Answer: D

Solution:

Normality of oxalic acid solution

$$= \frac{6.3 \times 1000}{63 \times 250} = 0.4\text{N}$$

Now from

$$N_1 V_1 = N_2 V_2$$

$$0.4 \times 10 = 0.1 \times V_2$$

$$V_2 = 40\text{mL}$$

Question192

The molality of a urea solution in which 0.0100g of urea, $[(\text{NH}_2)_2\text{CO}]$ is added to 0.3000d m^3 of water at STP is:

[2011RS]

Options:

A. $5.55 \times 10^{-4} \text{m}$

B. 33.3 m

C. $3.33 \times 10^{-2} \text{m}$

D. 0.555m

Answer: A

Solution:

Molality = Moles of solute/ Mass of solvent in kg

$$\begin{aligned}\text{Molality} &= \frac{0.01/60}{0.3} = \frac{0.01}{60 \times 0.3}; \\ &= 5.55 \times 10^{-4} \text{m}\end{aligned}$$

Question193

The density (in gmL^{-1}) of a 3.60M sulphuric acid solution that is 29% H_2SO_4 (molar mass = 98gmol^{-1}) by mass will be [2007]

Options:

A. 1.45

B. 1.64

C. 1.88

D. 1.22

Answer: D

Solution:

Since molarity of solution is 3.60M . It means 3.6 moles of H_2SO_4 is present in its 1 litre solution.

Mass of 3.6 moles of H_2SO_4

$$= \text{Moles} \times \text{Molecular mass}$$

$$= 3.6 \times 98\text{g} = 352.8\text{g}$$

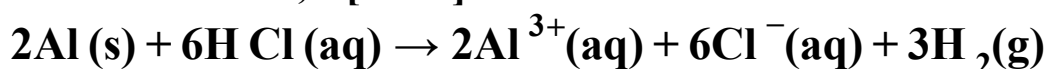
$\therefore$ 1000mL solution has 352.8g of H_2SO_4

Given that 29g of H_2SO_4 is present in

$$\begin{aligned}
 &= 100\text{g of solution} \\
 \therefore 352.8\text{g of H}_2\text{SO}_4 &\text{ is present in} \\
 &= \frac{100}{29} \times 352.8\text{g of solution} \\
 &= 1216\text{g of solution} \\
 \text{Density} &= \frac{\text{Mass}}{\text{Volume}} = \frac{1216}{1000} \\
 &= 1.216\text{g/mL} = 1.22\text{g/mL}
 \end{aligned}$$

Question194

In the reaction, [2007]



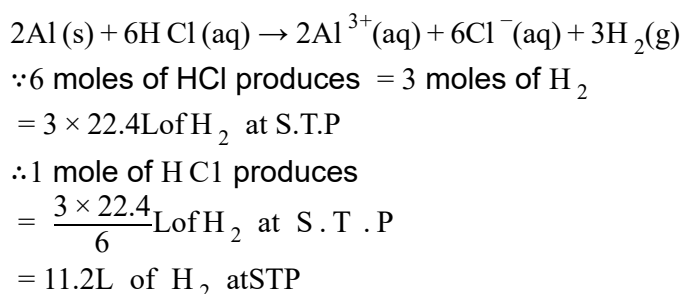
[2007]

Options:

- A. 11.2LH₂(g) at STP is produced for every mole of HCl(aq) consumed
- B. 6LHCl(aq) is consumed for every 3L of H₂(g) produced
- C. 33.6LH₂(g) is produced regardless of temperature and pressure for every mole of Al that reacts
- D. 67.2H₂(g) at STP is produced for every mole of Al that reacts.

Answer: A

Solution:



Question195

How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms?
[2006]

Options:

- A. 1.25×10^{-2}
- B. 2.5×10^{-2}
- C. 0.02
- D. 3.125×10^{-2}

Answer: D

Solution:

1 Mole of $\text{Mg}_3(\text{PO}_4)_2$ contains 8 moles of oxygen atoms

$\therefore$ 8 mole of oxygen atoms $\equiv$ 1 mole of $\text{Mg}_3(\text{PO}_4)_2$

0.25 mole of oxygen atom $\equiv \frac{1}{8} \times 0.25$ mole of $\text{Mg}_3(\text{PO}_4)_2$
 $= 3.125 \times 10^{-2}$ mole of $\text{Mg}_3(\text{PO}_4)_2$

Question196

Density of a 2.05M solution of acetic acid in water is 1.02 g/mL. The molality of the solution is
[2006]

Options:

- A. 2.28 mol kg^{-1}
- B. 0.44 mol kg^{-1}
- C. 1.14 mol kg^{-1}
- D. 3.28 mol kg^{-1}

Answer: A

Solution:

Apply the formula $d = M \left(\frac{1}{m} + \frac{M_2}{1000} \right)$

$$\therefore 1.02 = 2.05 \left(\frac{1}{m} + \frac{60}{1000} \right)$$

On solving we get, $m = 2.288 \text{ mol / kg}$

Question197

Two solutions of a substance (non electrolyte) are mixed in the following manner: 480mL of 1.5M first solution + 520mL of 1.2M second solution. What is the molarity of the final mixture? [2005]

Options:

A. 2.70M

B. 1.344M

C. 1.50M

D. 1.20M

Answer: B

Solution:

From the molarity equation

$$M_1 V_1 + M_2 V_2 = M V$$

Let M be the molarity of final mixture,

$$M = \frac{M_1 V_1 + M_2 V_2}{V} \text{ where } V = V_1 + V_2$$

$$M = \frac{480 \times 1.5 + 520 \times 1.2}{480 + 520} = 1.344M$$

Question198

If we consider that $1/6$, in place of $1/12$, mass of carbon atom as the relative atomic mass unit, the mass of one mole of the substance will [2005]

Options:

- A. be a function of the molecular mass of the substance
- B. remain unchanged
- C. increase two fold
- D. decrease twice

Answer: D

Solution:

$$\text{Relative atomic mass} = \frac{\text{Mass of one atom of the element}}{\frac{1}{12}^{\text{th}} \text{ part of the mass of one atom of carbon} - 12}$$

$$\text{or } \frac{\text{Mass of one atom of the element}}{\text{Mass of one atom of the C - 12}} \times 12$$

Now if we use $\frac{1}{6}$ in place of $\frac{1}{12}$ the formula becomes Relative atomic mass

$$= \frac{\text{Mass of one atom of element}}{\text{Mass of one atom of carbon}} \times 6$$

$\therefore$ Relative atomic mass decrease twice.

Question199

6.02×10^{20} molecules of urea are present in 100ml of its solution. The concentration of urea solution is (Avogadro constant, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) [2004]

Options:

- A. 0.02M
- B. 0.01M
- C. 0.001M
- D. 0.1M

Answer: B

Solution:

Moles of urea present in 100mL of sol.

$$= \frac{6.02 \times 10^{20}}{6.02 \times 10^{23}} \text{mol}$$

$$\therefore M = \frac{6.02 \times 10^{20} \times 1000}{6.02 \times 10^{23} \times 100} = 0.01M$$

[$\therefore M$ = Moles of solute present in 1L of solution]

Question200

To neutralise completely 20mL of 0.1M aqueous solution of phosphorous acid (H_3PO_3), the value of 0.1M aqueous KOH solution required is [2004]

Options:

A. 40mL

B. 20mL

C. 10mL

D. 60mL

Answer: A

Solution:

The neutralization reaction is $\text{H}_3\text{PO}_3 + 2 \text{KOH} \rightarrow \text{K}_2\text{HPO}_3 + 2\text{H}_2\text{O}$.

Phosphorous acid is diprotic acid as it has two ionizable hydrogens. Thus, 1 mole of phosphorous acid will neutralize 2 moles of KOH.

The number of moles of phosphorous acid present in 20 mL of 0.1M aqueous solution is

$$0.1 \times 20 \times \frac{1}{1000} = 0.002 \text{ mol.}$$

They will neutralize $2 \times 0.002 \text{ mol} = 0.004$ moles of KOH.

The molarity of KOH solution is 0.1M.

The volume of KOH solution required will be $\frac{0.004}{0.1} = 0.04\text{L} = 40 \text{ ml.}$

Question201

25mL of a solution of barium hydroxide on titration with a 0.1 molar solution of hydrochloric acid gave a titre value of 35mL.

**The molarity of barium hydroxide solution was
[2003]**

Options:

A. 0.14

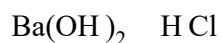
B. 0.28

C. 0.35

D. 0.07

Answer: D

Solution:



$$N_1 V_1 = N_2 V_2$$

$$N_1 \times 25 = 0.1 \times 35$$

$$N_1 = 0.14$$

Since, Ba(OH)_2 is diacid base

$$\text{Hence } N = M \times 2 \text{ or } M = \frac{N}{2}$$

$$M = 0.07\text{M}$$

Question202

**What volume of hydrogen gas, at 273K and 1atm. pressure will be consumed in obtaining 21.6g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen ?
[2003]**

Options:

A. 67.2L

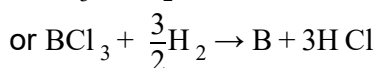
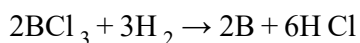
B. 44.8L

C. 22.4L

D. 89.6 L

Answer: A

Solution:



Now, since 10.8g boron requires hydrogen

$$= \frac{3}{2} \times 22.4\text{L at S.T.P.}$$

Hence 21.6 boron requires hydrogen

$$= \frac{3}{2} \times \frac{22.4}{10.8} \times 21.6 = 67.2\text{L at S.T.P.}$$

Question203

**With increase of temperature, which of these changes?
[2002]**

Options:

- A. molality
- B. weight fraction of solute
- C. molarity
- D. mole fraction.

Answer: C

Solution:

Among all the given options, molarity changes with temperature because the term molarity involves volume which increases on increasing temperature.

Question204

**Number of atoms in 558.5 gram Fe (at. wt. of Fe = 55.85gmol⁻¹) is
[2002]**

Options:

- A. twice that in 60g carbon

B. 6.023×10^{22}

C. half that in 8g He

D. $558.5 \times 6.023 \times 10^{23}$

Answer: A

Solution:

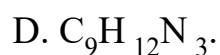
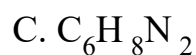
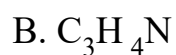
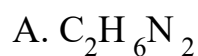
$$\text{Fe (No. of moles)} = \frac{558.5}{55.85} = 10\text{mol}$$

$$\text{C (No. of moles) in 60g of C} = 60/12 = 5\text{mol}.$$

Question 205

In a compound C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular weight of compound is 108 . Molecular formula of compound is [2002]

Options:



Answer: C

Solution:

Element	%	Relative no. of atoms	Simple ratio of atoms
C	9	$\frac{9}{12} = \frac{3}{4}$	3
H	1	$\frac{1}{1} = 1$	4
N	3.5	$\frac{3.5}{14} = \frac{1}{4}$	1

Empirical formula = C_3H_4N

$$(C_3H_4N)_n = 108$$

$$(12 \times 3 + 4 \times 1 + 14)_n = 108$$

$$(54)_n = 108 \quad n = \frac{108}{54} = 2$$

$\therefore$ Molecular formula = $C_6H_8N_2$
